Sensitivity and Locality Lower Bounds via Low-Soundness Label Cover

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Abstract

We study the sensitivity of approximation algorithms, which measures how much their output distributions change under a single input edit. Can weaker approximation guarantees make their solutions more stable? In this work, we show that, for Set Cover, Dominating Set, Min-Rep, and minimum unweighted directed 3-spanner, even a $\log^\beta n$ approximation guarantee requires sensitivity $n^{\Omega(1)}$, for some absolute constant $\beta > 0$ and all sufficiently large n. The lower bounds hold even for randomized algorithms with unrestricted computation. Here n is the universe size for Set Cover and the number of vertices for the graph problems.

On bounded-degree graphs, our polynomial sensitivity lower bounds imply logarithmic locality lower bounds in the randomized LOCAL model and its non-signaling generalization. For Dominating Set, fix any $\varepsilon \in (0, 1)$ and a sufficiently large constant maximum-degree bound Δ. A $(1+\varepsilon)\ln \Delta$ approximation can be achieved with sensitivity and LOCAL round complexity independent of n. We show that improving the ratio to $(1 - \varepsilon)\ln \Delta$ requires sensitivity $n^{\Omega_{\varepsilon,\Delta}(1)}$, and hence locality $\Omega_{\varepsilon,\Delta}(\log n)$. For minimum unweighted directed 3-spanner, we show that every constant approximation factor requires logarithmic locality, even on DAGs of constant maximum total degree.

The key ingredient is a polynomial sensitivity lower bound for a variant of low-soundness Label Cover. Fleming and Yoshida established such a lower bound when the required fraction of satisfied constraints on satisfiable instances is close to one. We extend this to inverse-polylogarithmic fractions, enabling the reductions to the problems above. The transformations used to lower this fraction can spread one input edit across many constraints, making the transferred lower bound vacuous. We overcome this obstacle by adapting the Impagliazzo–Kabanets–Wigderson direct-product construction and Dinur–Harsha iterated composition to control the propagation of input edits, and by designing recovery procedures that are stable under changes to both the supplied labeling and the source instance.

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1 Introduction

A desirable property of an optimization algorithm is that similar inputs lead to similar outputs [36]. Unfortunately, in many traditional optimization problems, a single input edit can force optimal outputs to differ in many coordinates. A natural approach to improving stability is to allow both approximation and randomization, giving the algorithm more freedom in choosing its output.[1] Can this freedom prevent a single input edit from forcing many output changes, even when computation is unrestricted?

We measure stability by *sensitivity* [41]: the maximum earth mover's distance (EMD) between output distributions on inputs differing by one allowed edit. EMD is the minimum expected Hamming distance over all couplings, where the Hamming distance counts changed labels or changes in set membership.[2] Thus a sensitivity lower bound S means that some single input edit forces at least S output changes in expectation under every coupling.

Sensitivity captures robustness of the solution itself, beyond its objective value. Such robustness matters when inputs are incomplete or perturbed, especially when the output is used in subsequent computation [36, 41]. Sensitivity is also relevant to exact data deletion, where the output after a deletion must have the same distribution as recomputation on the remaining input [21, 34, 35], and in distributed computing [18], where LOCAL algorithms on bounded-degree graphs are automatically low-sensitivity.

Fleming and Yoshida [18] used probabilistically checkable proof (PCP) constructions to prove polynomial sensitivity lower bounds for approximation algorithms for problems including Minimum Vertex Cover and Max 3SAT. A key step is a lower bound for Label Cover: for some constant $\varepsilon \in (0, 1)$, Fleming and Yoshida show that satisfying at least a $1 - \varepsilon$ fraction of constraints on every satisfiable instance requires sensitivity $n^{\Omega(1)}$, where n is the number of vertices.

However, classical hardness-of-approximation reductions to further problems, such as Set Cover [17], start from 'low-soundness' Label Cover. In particular, recovering a Label Cover solution from an approximate Set Cover solution may guarantee only a small (even sub-constant) fraction of satisfied constraints, whereas Fleming and Yoshida's sensitivity lower bound requires a fraction close to one. This leads to the central question of this paper, also raised by Fleming and Yoshida:

> *Can we prove polynomial sensitivity lower bounds for low-soundness Label Cover, where an algorithm need only satisfy a tiny fraction of constraints on satisfiable instances?*

The usual parallel-repetition argument [33] does not establish this directly, because one source edit may change so many repeated constraints that the resulting sensitivity lower bound becomes vacuous.

1.1 Our Results

We answer this question affirmatively, even when the required fraction is inverse-polylogarithmic in the instance size. As applications, we obtain new sensitivity lower bounds for Set Cover, Dominating Set, MinRep, and minimum unweighted directed 3-spanner, as well as new locality lower bounds in the LOCAL and non-signaling models for Dominating Set and directed 3-spanner.

We give two constructions of low soundness Label Cover, each serving a different purpose. The first construction, based on Impagliazzo–Kabanets–Wigderson (IKW) direct-product amplification [24], keeps the left label alphabet small enough to obtain Dominating Set sensitivity lower bounds for approximation factors near $\ln \Delta$, where Δ is the maximum-degree bound. The second construction, based on Dinur–Harsha

[1] Unless stated otherwise, approximation and value guarantees are in expectation; see Section 2 for the formal definitions.

[2] The allowed edits are specified for each problem.

(DH) iterated composition [15], has larger alphabet but improved size, resulting in stronger sensitivity lower bounds even for polylogarithmic approximation ratios. We first state the Label Cover guarantees that make these results possible, then give their consequences for covering and network-design problems.

1.1.1 Low-Soundness Label Cover

A Label Cover instance consists of a bipartite (multi)graph (U, V, E), finite label alphabets Σ_U, Σ_V, and a projection constraint $f_e : \Sigma_U \to \Sigma_V$ for each edge $e = (u, v)$. Formally, we will use *left-predicate Label Cover*, which also specifies a predicate $P_u : \Sigma_U \to \{0, 1\}$ at each left vertex. A labeling of the vertices satisfies the edge $e = (u, v)$ when its left label passes P_u and projects to its right label under f_e. Its *value* is the fraction of satisfied edges, and the value of the instance is the maximum value across all possible assignments. The instance is called satisfiable if some labeling has value 1. An input edit changes one projection or one left predicate, and output distance counts changed vertex labels.

Our constructions transform the 'source' binary constraint systems (2-CSPs) used in Fleming and Yoshida's lower bound [18] into 'target' Label Cover instances in a way that preserves satisfiability and admits a randomized recovery procedure. In particular, on a satisfiable source instance, the recovery procedure uses (randomized) solutions to the transformed target instance with expected value above some small specified threshold to produce back a source assignment whose expected value is close to one. Following the PCP literature, we call such constructions *low-soundness* because the required target value is small.

Fleming and Yoshida's source lower bound applies to algorithms attaining value close to one on every satisfiable member of their family. A low-sensitivity algorithm attaining the required target value, combined with sufficiently stable recovery, would violate that lower bound. We therefore control both the number of target edits caused by a source edit and the sensitivity of recovery. Section 1.2 explains the transfer in detail.

We first prove a sensitivity lower bound using the IKW construction [24]. Its small left alphabet lets us obtain Dominating Set sensitivity lower bounds for approximation factors proportional to the logarithm of the maximum degree. The parameter k below controls the soundness threshold, how small expected value we need for recovery to the initial source instance.

Theorem 1.1 (Sensitivity lower bound via IKW; informal version of Theorem 6.1)**.** For every sufficiently large N and sufficiently large integer $k = o(\log N)$, there is a family of instances with $n = N^{\Theta(k)}$ vertices such that any randomized algorithm for left-predicate Label Cover that attains expected value at least $\exp\left(-C\sqrt{k}\right)$ on every satisfiable input has sensitivity $n^{\Omega(1/k)}$. Here $C > 0$ is an absolute constant. The instances are biregular, with degrees and alphabet sizes bounded by $2^{O(k)}$.

As k grows, the required value decreases, but so does the exponent in the IKW sensitivity lower bound. To obtain an exponent independent of the required value, we instead adapt the DH construction [15]. This leads to a truly polynomial sensitivity lower bound even for inverse-polylogarithmic value guarantees, at the cost of a larger left alphabet.

Theorem 1.2 (Sensitivity lower bound via Dinur–Harsha; informal version of Theorem 11.1)**.** There are absolute constants $B, \delta > 0$ such that, for every sufficiently large n and every $1 < g \le \log^B n$, any randomized algorithm for left-predicate Label Cover that attains expected value at least $1/g$ on every satisfiable input has sensitivity $\Omega(n^\delta)$. This holds on instances with n left vertices and at most $n \log^{O(1)} n$ right vertices, with left and right degrees and right alphabet size bounded by $g^{O(1)}$, and left alphabet size bounded by $\exp\left(g^{O(1)}\right)$.

Table 1 compares the two constructions at the same required value.

Table 1: Sensitivity lower bounds for left-predicate Label Cover. For IKW, n counts all vertices and $g = \exp\left(C\sqrt{k}\right)$; the bound holds at the sizes and in the parameter range of Theorem 6.1. For DH, n counts left vertices, and the bound holds for all sufficiently large n and $1 < g \le \log^B n$ as in Theorem 11.1. The DH exponent $\delta > 0$ is independent of g.

Construction	Value guarantee	Sensitivity	Degrees and alphabet sizes
IKW	$1/g$	$n^{\Omega(1/\log^2 g)}$	Degrees and alphabets $\exp\big(O(\log^2 g)\big)$
Dinur–Harsha	$1/g$	$\Omega(n^{\delta})$	Degrees and right alphabet $g^{O(1)}$; left alphabet $\exp\big(g^{O(1)}\big)$

1.1.2 Covering and Network Design

For Set Cover, Dominating Set, Min-Rep, and minimum unweighted directed 3-spanner, we prove that even a $\log^{\beta} n$ approximation requires sensitivity $n^{\Omega(1)}$, for some absolute constant $\beta > 0$. The bounds apply to randomized algorithms with unrestricted computation and hold for all sufficiently large n, where n is the universe size for Set Cover and the number of vertices for the graph problems.

For Dominating Set and directed 3-spanner, our sensitivity lower bounds also yield lower bounds for distributed approximation. By the sensitivity-to-locality bound of Fleming and Yoshida [18], polynomial sensitivity implies logarithmic locality on graphs of constant maximum degree (c.f., Section 1.2). In the LOCAL model, vertices communicate with their neighbors in synchronous rounds, with unrestricted message size and local computation; the number of rounds is the locality. Our locality lower bounds hold even in the more general *non-signaling* model, where the joint output distribution on every vertex set depends only on that set's radius-t input neighborhood at locality t (Section 2.4). We highlight two applications.

Dominating Set: stability and locality near $\ln \Delta$. Dominating Set asks for a smallest vertex set such that every vertex either belongs to the set or has a neighbor in it. An input edit inserts or deletes one edge, and output distance counts changes in vertex membership. Fix any $\varepsilon \in (0, 1)$ and a sufficiently large constant maximum-degree bound Δ. A $(1 + \varepsilon) \ln \Delta$ approximation can be attained with sensitivity and round complexity independent of n in the CONGEST model, which restricts LOCAL messages to $O(\log n)$ bits. This follows from the algorithm of Kuhn, Moscibroda, and Wattenhofer [26] combined with randomized rounding (Section 13.2). In contrast, we show that improving the ratio to $(1 - \varepsilon) \ln \Delta$ requires sensitivity $n^{\Omega_{\varepsilon,\Delta}(1)}$ and locality $\Omega_{\varepsilon,\Delta}(\log n)$, with the locality lower bound holding even in the non-signaling model (Theorem 13.3 and corollary 13.4). Together, these bounds identify $\ln \Delta$, to leading order, as the approximation threshold for achieving sensitivity and round complexity independent of n at constant maximum degree.

To compare with previous locality lower bounds, let α denote the approximation ratio and t the locality, which in LOCAL is the number of communication rounds. Chang and Li [7] prove the LOCAL lower bound

$$\alpha = 1 + \eta \quad \Longrightarrow \quad t = \Omega\left(\frac{\log n}{\eta}\right)$$

for sufficiently small constants $\eta > 0$. For every constant $c \ge 1$, the KMW LOCAL lower bound [11, 27] gives

$$\alpha = \log^{c} \Delta \quad \Longrightarrow \quad t = \Omega\left(\min\left\{\sqrt{\frac{\log n}{c \log\log n}}, \frac{\log \Delta}{c \log\log \Delta}\right\}\right).$$

At constant Δ, the expression inside this bound is independent of n for all sufficiently large n. Our result retains logarithmic dependence on n while allowing approximation ratios approaching $\ln \Delta$ from below:

$$\alpha = (1 - \varepsilon) \ln \Delta \quad \Longrightarrow \quad t = \Omega_{\varepsilon,\Delta}(\log n),$$

for every fixed $\varepsilon \in (0, 1)$ and sufficiently large constant Δ, even in the non-signaling model. Section 13 gives the full degree-dependent bounds and comparisons.

Directed 3-spanner: locality of constant-factor approximation. A directed 3-spanner retains a subset of the input arcs such that every input arc has a directed replacement path of length at most three. This feasibility condition can be checked within a constant-radius neighborhood. We show that approximating the minimum number of retained arcs within any fixed constant factor nevertheless requires information from logarithmic distances, even with unrestricted message size and local computation.

We measure sensitivity under insertion or deletion of one arc, with output distance counting changed arc memberships. In the distributed formulation, communication is bidirectional on the underlying undirected graph, and each vertex selects incident input arcs whose union is the output spanner. For every constant approximation factor $\alpha \geq 1$, our sensitivity lower bounds imply

$$t = \Omega_{\alpha}(\log n),$$

already on weakly connected DAGs of constant maximum total degree depending on α (Corollary 16.4). This lower bound on t holds in LOCAL and even in the non-signaling model.

Together with the $(1+\varepsilon)$-approximation algorithm of Censor-Hillel and Dory [6], our lower bounds place the LOCAL complexity of every constant approximation factor greater than one between logarithmic and polylogarithmic in n. Their lower bounds concern directed unweighted k-spanner for $k \geq 5$ in CONGEST, and weighted 2-spanner in LOCAL. These do not give a LOCAL lower bound for unweighted directed 3-spanner. Sections 15 and 16 give the full parameter-dependent bounds and comparisons.

Set Cover and Min-Rep. The reductions to Dominating Set and directed 3-spanner pass through Set Cover and Min-Rep, respectively, and also yield sensitivity lower bounds for these two problems. Set Cover asks for a minimum number of sets covering a given universe. A Min-Rep instance is a bipartite graph whose vertices on each side are partitioned into groups; the goal is to select as few vertices as possible so that every pair of groups joined by an edge has an edge with both endpoints selected. An input edit changes one set–element incidence for Set Cover, with the universe and set indices fixed, or one edge for Min-Rep, with the vertices and groups fixed. Output distance counts changes in selected sets or vertices, respectively.

For both problems, our DH construction gives

$$\log^{\beta} n\text{-approximation} \quad \Longrightarrow \quad n^{\Omega(1)} \text{ sensitivity}$$

for some absolute constant $\beta > 0$ and all sufficiently large n, where n is the universe size for Set Cover and the number of vertices for Min-Rep (Theorem 12.6 and corollary 14.5). The Set Cover bound holds even with $O(n)$ sets. Section 12 also gives IKW-based bounds when set sizes and element frequencies are bounded.

1.2 Technical Overview

All our constructions start from the same 2-CSP family of Fleming and Yoshida [18]. A 2-CSP has one variable at each vertex and a binary constraint on each edge. At each sufficiently large size N, the family uses a

common constant-degree regular graph on N vertices and a common constant-size alphabet; only the constraint relations vary. For absolute constants $\eta_{\mathrm{src}}, c_{\mathrm{src}} > 0$, producing assignments of expected value at least $1 - \eta_{\mathrm{src}}$ on its satisfiable members requires sensitivity $N^{c_{\mathrm{src}}}$ (Lemma 2.3). To prove Theorems 6.1 and 11.1, we recover source assignments with this value guarantee from Label Cover labelings whose expected value exceeds a small threshold.

We must control how recovery responds to changes in both the supplied labeling and the source instance. All value guarantees below concern satisfiable source instances.

1.2.1 Transferring Sensitivity through Recovery

Each of our reductions consists of a transformation $\mathcal{T}$ from source instances I to target instances $\mathcal{T}(I)$. To control how the sensitivity changes through these reductions, we equip each instance I with a recovery procedure as follows. Given an algorithm A for the target problem, the recovery procedure may use several independent outputs of A on $\mathcal{T}(I)$, together with additional randomness and the source instance I. We therefore view the recovery procedure as a map $\mathcal{R}_I$ from distributions over target outputs to distributions over source outputs. Write $\mu_I = \mathrm{Law}(A(\mathcal{T}(I)))$ for the target output distribution, where Law denotes the distribution of a random variable. Applying the recovery procedure to outputs of A defines a source algorithm B satisfying

$$\mathrm{Law}(B(I)) = \mathcal{R}_I(\mu_I).$$

We bound the sensitivity of B in terms of four quantities:

- S_A is the sensitivity of the target algorithm A: the maximum EMD between its output distributions on target instances differing by one allowed edit.
- $C_{\mathcal{T}}$ bounds the number of target edits caused by one source edit.
- $C_{\mathcal{R}}$ bounds the factor by which recovery can increase the EMD between two target output distributions, with the source instance I unchanged. For every source instance I and all distributions μ, ν over its target range,
$$\mathsf{EMD}(\mathcal{R}_I(\mu), \mathcal{R}_I(\nu)) \le C_{\mathcal{R}} \cdot \mathsf{EMD}(\mu, \nu).$$
- $D_{\mathcal{R}}$ bounds the EMD caused by replacing I by a neighboring source instance while supplying the same output distribution. For every neighboring pair $I, \widetilde{I}$ and every distribution μ over their common target range,
$$\mathsf{EMD}(\mathcal{R}_I(\mu), \mathcal{R}_{\widetilde{I}}(\mu)) \le D_{\mathcal{R}}.$$

For neighboring source instances $I, \widetilde{I}$, the target instances $\mathcal{T}(I), \mathcal{T}(\widetilde{I})$ can be joined by a sequence of at most $C_{\mathcal{T}}$ single target edits. Coupling the outputs of A along this sequence gives $\mathsf{EMD}(\mu_I, \mu_{\widetilde{I}}) \le C_{\mathcal{T}} S_A$.

Compare the recovered distributions in two steps. First keep the recovery map $\mathcal{R}_I$ fixed and change the supplied distribution from μ_I to $\mu_{\widetilde{I}}$. Then keep $\mu_{\widetilde{I}}$ fixed and replace $\mathcal{R}_I$ by $\mathcal{R}_{\widetilde{I}}$. The triangle inequality therefore bounds the sensitivity $\mathrm{Sens}(B)$ of B:

$$\mathrm{Sens}(B) \le C_{\mathcal{T}} C_{\mathcal{R}} S_A + D_{\mathcal{R}}.$$

If recovery gives B the approximation guarantee required by a source sensitivity lower bound L, then $C_{\mathcal{T}} C_{\mathcal{R}} S_A \ge L - D_{\mathcal{R}}$. The multiplicative loss is the product $C_{\mathcal{T}} C_{\mathcal{R}}$: critically $C_{\mathcal{T}}$ will often be quite large and we will require recovery $C_{\mathcal{R}} < 1$ that actually contracts EMD to compensate for the transform causing many target edits. We also need $D_{\mathcal{R}} < L$ for the transferred lower bound to be positive.

Theorem 3.2 states the formal transfer bound; Lemma 3.4 bounds the losses under composition.

1.2.2 Sensitivity Transfer through Direct-Product Amplification (Theorem 6.1)

Let Φ be the source 2-CSP, with variable set X and constraint set E, and put $k' = \lfloor \sqrt{k} \rfloor$. The encoding map $\mathcal{T}^k_{\mathrm{IKW}}$ follows the IKW direct-product construction [24] and sends Φ to a Label Cover instance with left vertices E^k and right vertices $X^{k'}$. A label at a left vertex $(e_1, \ldots, e_k)$ assigns values to the $2k$ endpoint positions. Its left predicate requires that repeated occurrences of a variable receive the same value and that all k source constraints are satisfied. A label at a right vertex assigns values to its k' variable positions. Each edge checks that the left label projected onto selected endpoint positions agrees with the right label. Using the IKW agreement analysis, we construct a randomized recovery procedure that can find a nearly satisfying source assignment even when the supplied labeling has value only $\varepsilon = \exp\Big(-\Theta(\sqrt{k})\Big)$. Increasing k thus lets us handle weaker value guarantees. Our contribution here is to make this recovery stable under changes to both the supplied labeling and the source instance.

The obstacle for sensitivity is that one source constraint appears in many tuples. Changing its relation affects only the left predicates at tuples containing it; the graph and edge projections stay unchanged. The edit bound $C_{\mathcal{T}}$ can nevertheless be very large. We therefore need a small $C_{\mathcal{R}}$ to keep the product $C_{\mathcal{T}} C_{\mathcal{R}}$ small.

Bounding changes in the decoded assignment. To apply the agreement analysis on satisfiable source instances, we first replace each violating left label by a label satisfying its predicate, using only the label and predicate at that vertex. This cannot decrease value and, for a fixed source instance, cannot increase Hamming distance between labelings.

The core decoding step chooses a right vertex uniformly at random and uses its label on the variables in that tuple. To recover each source variable outside that tuple, it samples edges incident to the chosen right vertex whose left tuples contain the variable, and reads its value from a left label whose projection agrees with the chosen right label, if one is found. The agreement analysis shows that a sufficient fraction of right vertices lead to good source assignments. For sensitivity, we arrange the sampling so that, averaged over the recovered variable and the chosen right vertex, the left endpoint of each sampled edge is uniform among all left vertices. Two executions using the same randomness can then disagree at a variable only if they read a changed label.

To control changes to right labels as well, we use the standard right-degree reduction and its recovery rule [14, Theorem 5.4], replacing each right vertex with one copy per incident edge and using an expander to redistribute the edge incidences among these copies (Algorithm 1). The recovery rule chooses each original right label from a uniformly chosen copy. A change to one copy's label can affect a decoding trial only when that copy supplies the label of the chosen right vertex. Writing $N = |X|$ for the number of source variables, our coupling analysis, together with the right-vertex copying, gives a factor $N/|E|^k$ in the bound on $C_{\mathcal{R}}$, with additional factors from sampling and candidate selection. On the transformation side, one source edit changes $O(k|E|^k/N)$ left predicates, giving $C_{\mathcal{T}} \le O(k|E|^k/N)$. Thus the factors $|E|^k$ and N cancel in $C_{\mathcal{T}} C_{\mathcal{R}}$.

We must also bound $D_{\mathcal{R}}$, the effect of changing the source instance while keeping the supplied labeling fixed. At the decoding stage, a source edit affects recovery only through the replacement of violating left labels. The replacement can change only at left tuples containing the edited constraint, so the same sampling argument bounds this contribution to $D_{\mathcal{R}}$.

Selecting a good candidate without introducing instability. Suppose the target algorithm guarantees expected value at least 4ε. To achieve the required expected source value of $1 - \eta_{\mathrm{src}}$, we perform $O(\varepsilon^{-2})$

independent trials so that a nearly satisfying candidate exists with sufficiently high constant probability. Selecting the candidate with the best value could be unstable: two assignments may differ on many variables while their values are almost tied. We instead choose a threshold uniformly from a short interval near one and return the first candidate whose value exceeds it. Under a shared threshold, a small change in value changes a candidate's eligibility only with small probability. This also controls the effect of changing a source constraint, since such an edit changes the value of any fixed assignment by at most $1/|E| = O(1/N)$.

We show that, after including these recovery steps, both the multiplicative loss $C_{\mathcal{T}}C_{\mathcal{R}}$ and the additive loss $D_{\mathcal{R}}$ are $N^{o(1)}$ when $k = o(\log N)$ (Proposition 7.4). The source lower bound therefore remains $N^{\Omega(1)}$. The limitation is the size of the amplified instance: it has $n_{\mathrm{LC}} = N^{\Theta(k)}$ vertices, so this becomes $n_{\mathrm{LC}}^{\Omega(1/k)}$. Thus stronger amplification weakens the sensitivity exponent in the final size. On the other hand, labels describe only $O(k)$ source symbols, giving the small alphabets that benefit our Dominating Set application.

1.2.3 Sensitivity Transfer through Iterated Composition (Theorem 11.1)

Our adaptation of the DH construction gives a polynomial sensitivity lower bound for a Label Cover value guarantee of $1/g$, with an exponent independent of g, even when g is polylogarithmic in the instance size. Our construction uses standard ingredients from algebraic PCPs, including polynomial encodings and low-degree testing. We first reduce the source 2-CSP to a Label Cover instance with polynomially many vertices. This reduction already permits recovery from labelings of inverse-polylogarithmic value. However, the degrees and alphabet sizes of the resulting Label Cover instance still depend on the source size. In our subsequent reductions, we create sets or representatives for each label, so large alphabets produce large instances and greater sensitivity loss. To remove this dependence on the source size, we use the Dinur–Harsha composition framework [15], together with alphabet and degree reduction, to obtain degree and alphabet-size bounds depending only on g. Each round replaces the current local constraints by tests on auxiliary proofs, then applies alphabet and degree reduction to prepare the resulting instance for the next round. Throughout the iteration, we maintain decreasing upper bounds on the query complexity and the size of each local checking circuit. Each composition step raises the value needed for recovery, so we start with a threshold sufficiently below $1/g$ to allow repeated composition. For sensitivity transfer, we additionally need to control how changes to the source instance or the supplied labeling affect recovery, and how these losses accumulate under composition.

1. Recovering a good assignment when local answers are ambiguous. The initial reduction maps a source 2-CSP to a Label Cover instance. A satisfying source assignment induces a satisfying labeling by extending the assignment to a polynomial, assigning its restrictions to left vertices, and assigning its point evaluations to right vertices. The constraints check agreement between these labels and satisfaction of groups of source constraints. The groups are chosen so that an assignment violating a constant fraction of source constraints rarely satisfies a whole group.

At a small acceptance rate, the local answers need not identify a unique polynomial, but the Arora–Sudan low-degree test [2] implies there is a short list of global polynomials accounting for most of the accepted comparisons. This list guarantee does not by itself provide a stable recovery procedure: small changes to the supplied labeling may change the list. Our contribution at this stage is a stable recovery procedure that scores every polynomial in a fixed space common to all inputs. Each polynomial's score is the fraction of left answers that agree with its restriction and pass their constraint checks. The list guarantee ensures that some polynomial has a large score whenever enough tests pass. The choice of constraint groups ensures that every polynomial above a suitable score threshold represents a nearly satisfying source assignment.

Each score is an average over the left vertices. If there are A left vertices, changing one left label changes every candidate's score by at most $1/A$; right labels do not enter the scores. Changing one source constraint affects only a small fraction of the terms in each average. To select a polynomial stably, we use a random threshold, as in the IKW recovery, and take the first qualifying polynomial in an order fixed across inputs. Only a bounded number of polynomials can have scores near the threshold. Under a shared threshold, the score perturbation and the bound on the number of near-threshold polynomials control the probability that the selected polynomial changes. We repeat recovery on independent algorithm outputs and return the assignment decoded from the first successful trial. Sufficiently many trials give expected source value at least $1 - \eta_{\mathrm{src}}$ (Theorem 9.1).

2. Simplifying a test without changing where it reads. For composition, we view the constraint at each old left vertex u as a circuit C_u on the labels of its right neighbors, with one input coordinate per incident edge. The circuit accepts a tuple exactly when some left label satisfying P_u projects to its entries along the corresponding edges. We replace this circuit by tests on an auxiliary proof intended to encode an accepted tuple. A decodable PCP [15] provides such a proof as a string over a finite alphabet. Each test reads a few proof symbols and attempts to recover a specified coordinate of the encoded tuple, or rejects. Thus the test for an edge (u, v) attempts to recover the old right label at v.

We use the DH composition rule [15, Section 4] to combine these tests into a new Label Cover instance. Each new left vertex combines tests that decode the label of the same old right vertex, and its predicate requires all of them to return the same label. Each proof position used by the tests becomes a right vertex, whose label specifies the symbol at that position. A new left label specifies proposed symbols for the positions queried by the tests being combined. Projection constraints check that these proposed symbols agree with the corresponding right labels. Thus the number of queries controls the length of the new left labels, without requiring any label to contain an entire proof. The soundness guarantee lets us recover a good labeling of the old instance from one satisfying sufficiently many new constraints. Figure 1(a) illustrates the construction.

Changing the queried positions would therefore change which right vertices a new left vertex is adjacent to. To preserve the graph under a source edit, we modify the decoder to make these positions independent of the circuit C_u, a property called *query-obliviousness*. To achieve this query-obliviousness, we use a fixed universal circuit whose input contains an encoded circuit description and a candidate tuple of right labels. The universal circuit checks that the description is a valid codeword, decodes it, and checks that the described circuit accepts the candidate tuple. We apply a decodable PCP to this fixed universal circuit. To recover a coordinate of the candidate tuple, the decoder also attempts to recover a uniformly sampled symbol of the encoded description. We compare this symbol with the corresponding symbol in the encoding of the description of C_u. This comparison checks that the decoded description corresponds to C_u. In this decoder, only this comparison depends on C_u, so the query locations remain fixed. Distinct descriptions have encodings that agree on few positions, making acceptance under a circuit different from C_u unlikely.

To control the sensitivity of recovery as well, we design the procedure in Figure 1(b) to have local dependencies. For each old left vertex u, assemble the new right labels at (u, z) into a proof p_u. We use the decoder's list of tuples accepted by C_u to recover a label at u: each candidate tuple corresponds to a left label satisfying P_u that projects to it. This step depends only on p_u and the old local constraint at u. Independently, sample the test randomness ρ uniformly. For each old right vertex v, run the local decoders on the symbols in the left label at (v, ρ) in the composed instance. Their common output supplies the recovered label when all decoders accept and agree.

With the same recovery randomness, changing one label in the composed instance affects at most one

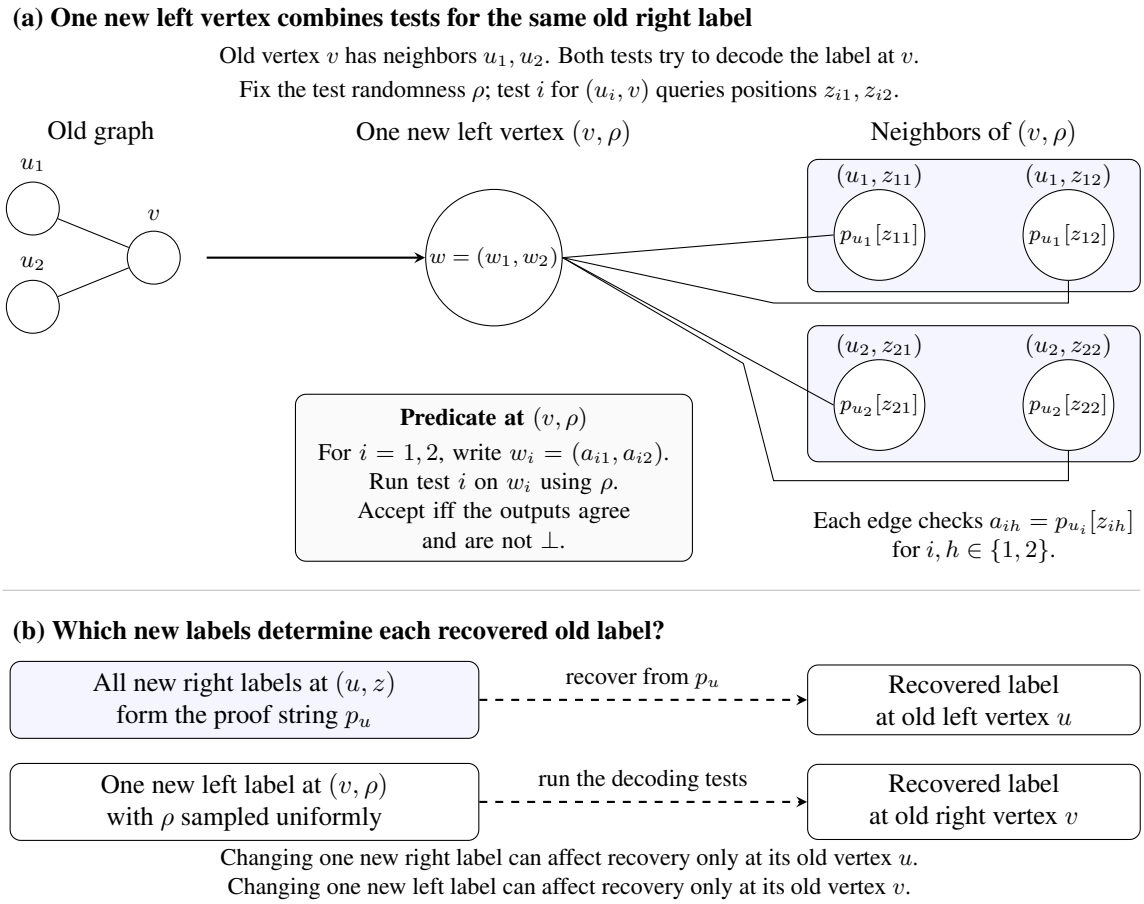


Figure 1: Composition and recovery, illustrated with two old edges $(u_1, v), (u_2, v)$ and two queries per test. In (a), u_1, u_2 are neighbors of v in the old graph; they index the answer strings w_1, w_2 within the single label at (v, ρ). The entry a_{ih} is a proposed answer to query h of test i. Vertex names are outside the circles, and new vertex labels are inside; the shaded boxes on the right group symbols of the same proof. Lines without arrowheads are graph edges; the solid arrow denotes composition. Query positions, and hence the new edges, stay fixed when old constraints change. In (b), dashed arrows show recovery dependencies, with the same random choices used when comparing two labelings. For an old constraint edit at u, keep the new labeling fixed: recovery can change only at u and its old neighbors.

recovered label, giving $C_{\mathcal{R}} \leq 1$ for this composition step. If an old constraint at u changes while the labeling of the composed instance is fixed, only recovery at u and its neighbors can change. Both bounds are independent of proof length (Theorem 10.4).

3. Reducing local circuit size under composition. Reading few proof symbols does not by itself make a test small: the circuit that processes them could still be large. This distinction matters because the next composition step takes that circuit as its input. We analyze the DH algebraic decoder [15, Theorem 6.5] to bound its local circuit size.

At each stage, for input circuits of size at most M, we choose a decoder whose number of queries and local decoding circuit size are both at most M^α, for a sufficiently small absolute constant $\alpha \in (0, 1)$. This

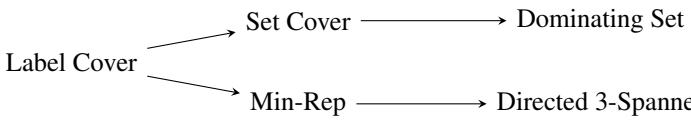
saving is large enough that the local circuit-size bound still decreases after the subsequent alphabet and degree reductions.

4. Keeping the accumulated loss small. For the initial encoding, the stable scoring procedure used in the IKW construction gives polylogarithmic multiplicative and additive sensitivity losses. We now bound the additional losses from iterated composition. Let n_0 be the initial size parameter, which is polynomial in the source size. We show that, at stage i, the factors controlling size growth and sensitivity loss are at most $\log^{O(4^{-i})} n_0$. Since $\sum_{i\geq 0} 4^{-i} = 4/3$, their products remain polylogarithmic in n_0. Thus the final instance size is polynomial in the source size, with an exponent independent of g, and the accumulated transfer parameters satisfy $C_{\mathcal{T}} C_{\mathcal{R}}, D_{\mathcal{R}} \leq \log^{O(1)} n_0$ (Proposition 11.4 and lemma 11.5). Consequently, the sensitivity lower bound remains polynomial in the final instance size.

For a target value $1/g$, we choose a stopping stage at which the local tests have size bounded in terms of g alone, while the recovery threshold remains at most $1/g$ (see the proof of Theorem 11.1).

1.2.4 From Label Cover to Covering Problems

We adapt classical reductions from Label Cover to covering and network-design problems. In the parameter ranges used below, recovery from an α-approximate Set Cover or Min-Rep solution guarantees expected Label Cover value $\Omega(\alpha^{-2})$ (Lemmas 14.2 and 12.3). For growing α, this guarantee does not suffice to apply Fleming and Yoshida's sensitivity lower bound, which requires Label Cover value close to one. In contrast, our low-soundness bounds apply in this range, and stable recovery maps let us transfer sensitivity. The reductions follow two branches, with solutions recovered in the reverse direction:



The covering branch uses the classical reductions from Label Cover to Set Cover [17, 29] and from Set Cover to bounded-degree Dominating Set [8]. We show that these reductions admit stable recovery while preserving the leading logarithmic approximation coefficient.

Start with a biregular Label Cover instance on (U, V, E), with left and right degrees d_L, d_R. Recall that the classical Set Cover construction has one set for each left vertex–label pair and one copy of a universe B for each right vertex. Each right label indexes a partition of B into d_R parts, supplied by compatible left labels on the incident edges; a satisfying labeling gives a cover of size $|U|$.

For our local recovery, we need a right label supported by enough incident edges to be found by local sampling. We use a quantitative strengthening of the partition guarantee in [17, Lemma 3.2]: a small cover must contain a constant fraction of the parts of one partition, whereas the cited lemma guarantees at least two parts. We show that the standard probabilistic construction gives this stronger guarantee for covers of size sufficiently below $d_R \ln |B|$, when $|B|$ is sufficiently large relative to $|\Sigma_V| d_R$ (Lemma 12.1).

Given a cover, for each left vertex u, let L_u contain the labels y whose sets (u, y) are selected. Recovery labels each u uniformly from L_u. Independently, for each right vertex v, it samples a uniform incident edge $e = (u, v)$ and a fresh uniform label $y \in L_u$, then assigns v the right label $f_e(y)$ prescribed by the edge e. The strengthened partition guarantee ensures enough agreement between these independent choices to obtain the expected value bound above.

This sampling is also stable. Adding or removing a set at u changes only L_u. Under a coupling using the same sampled edges and identical label samples wherever the lists agree, recovery can change only at u and

at right vertices whose sampled edge is incident to u. The expected number of changed labels is therefore at most $1 + d_L/d_R$. The recovery does not use the predicates, and a projection swap can change only its right endpoint's recovered label. A source edit changes at most $d_L|\Sigma_U||B|$ incidences (Lemma 12.2).

Each set has size at most $d_L|B|$, and each element belongs to at most $d_R|\Sigma_U|$ sets. To keep both quantities at most a prescribed bound Γ, for any constant $\varepsilon \in (0, 1)$ we choose the IKW parameters so that the degrees and alphabet sizes are $\Gamma^{o(1)}$. This lets us bound both set sizes and element frequencies by Γ while taking $\ln |B| = (1 - o(1)) \ln \Gamma$. A sufficiently small ε-dependent margin then yields the $(1 - \varepsilon) \ln \Gamma$-approximation lower bound (Theorem 12.5).

We then pass from Set Cover to Dominating Set using the classical incidence-graph reduction. Recovering a cover requires only local replacements of selected element-vertices by incident sets, giving constant sensitivity loss (Proposition 13.1).

We choose the set-size and frequency bounds proportional to Δ so that the reduction keeps the maximum degree at most Δ and extends a cover to a dominating set of size at most $(1 + O(\varepsilon))$ times the cover size. The logarithms of these source bounds are $\ln \Delta - O_\varepsilon(1)$, so the reduction preserves the leading coefficient and yields the sensitivity lower bound for $(1 - \varepsilon) \ln \Delta$-approximation (Theorem 13.3). For polylogarithmic approximation ratios, we instead use the DH construction. Taking $g = \Theta(\alpha^2)$ preserves a polynomial sensitivity lower bound with an absolute exponent; the DH degree and alphabet bounds yield approximation factor $\log^\beta \Delta$ for a sufficiently small constant $\beta > 0$ (Theorem 13.5).

1.2.5 From Label Cover through Min-Rep to Directed Spanners

Our spanner application follows the approach of encoding Min-Rep solutions by short paths, as in classical spanner hardness reductions [12]. We adapt this approach to directed 3-spanners with bounded degree. The key difficulties are to encode the Min-Rep constraints using paths of length three and to amplify the cost of selecting representatives while preserving the sensitivity lower bound.

Adapting Label Cover to Min-Rep. In the reduction to Min-Rep, the labels of each source vertex become a group of representatives. Edges join pairs whose labels satisfy the source constraints; we call such pairs compatible. We make as many copies of each group as the degree of its source vertex. To recover a labeling, independently at each vertex choose a uniform group copy and a uniform selected representative in that copy. We show that an α-approximate solution yields expected Label Cover value at least α^{-2} on a satisfiable source instance, and that changing one selected representative affects at most one recovered label (Lemma 14.2).

Obtaining directed 3-spanners with bounded degree. Before introducing repetitions, add a source s_i for each left group A_i and a sink t_j for each right group B_j. Include selection arcs $s_i \to a$ for every $a \in A_i$ and $b \to t_j$ for every $b \in B_j$, and orient each compatibility edge from its left representative to its right representative. For each pair of groups A_i, B_j containing a compatible pair, add an arc $s_i \to t_j$. We call this a *demand arc*: a 3-spanner must either retain it or provide a path $s_i \to a \to b \to t_j$ through compatible representatives $a \in A_i$ and $b \in B_j$. The two selection arcs on this path correspond to selecting a and b.

Every input arc must be spanned, including unused selection arcs. We therefore add one auxiliary vertex h_i per left group and $\bar{h}_j$ per right group, with arcs providing the two-arc routes $s_i \to h_i \to a$ and $b \to \bar{h}_j \to t_j$ for all representatives in the respective groups. These two-arc routes span omitted selection arcs, while any demand path using one has at least four arcs. Using a separate auxiliary vertex for each group avoids a high-degree vertex shared by all groups.

Preserving sensitivity when copying vertices. We next increase the cost of selecting representatives to absorb the cost of the compatibility and auxiliary arcs. To this end, for an integer x, replace each source and sink by copies s_i^p, t_j^p, indexed by $p \in [x]$, while sharing the representatives and auxiliary vertices among the copies. Each copy inherits the selection and auxiliary arcs of the original vertex. For each pair of groups containing a compatible pair, add demand arcs $s_i^p \to t_j^p$ only between copies with the same index p.

Let r be the number of groups. On a satisfiable source instance, choosing one representative per group according to a satisfying labeling uses xr selection arcs across the copies. Adding all compatibility and auxiliary arcs gives a 3-spanner. We choose x large enough that these additional arcs number $O(xr)$ as well, so the spanner has $O(xr)$ arcs. The choice of x is unchanged when an input edge is edited.

Given a spanner H, recovery chooses $p \in [x]$ uniformly and selects every representative incident to a selection arc of H with index p. For each demand arc $s_i^p \to t_j^p$ in H, recovery also selects a compatible pair in $A_i \times B_j$, using only the compatibility edges between these groups to choose the pair. Each selection or demand arc in H is used for only one value of p and contributes at most two representatives, so the expected number selected is at most $2|H|/x$. In particular, an α-approximate spanner yields a Min-Rep solution of expected size $O(\alpha r)$ on a satisfiable source instance.

We must also show that this copying preserves the sensitivity lower bound. A compatibility-edge edit changes at most x demand arcs in addition to the compatibility arc itself, giving $C_{\mathcal{T}} = O(x)$. For a fixed input, couple recovery from two spanners using the same sampled index p. Each changed selection or demand arc then affects at most two representatives, and only with probability $1/x$, giving $C_{\mathcal{R}} = O(1/x)$. Thus $C_{\mathcal{T}} C_{\mathcal{R}} = O(1)$, independently of x. With the supplied spanner held fixed, an input edit changes the chosen compatible pair for at most one pair of groups, contributing an additive loss $D_{\mathcal{R}} = O(1)$.

Combining the reductions, an α-approximate spanner gives a Label Cover value guarantee of $\Omega(\alpha^{-2})$. For DH, the size increase and sensitivity losses are at most $\exp\big(g^{O(1)}\big)$. Setting $g = \Theta(\alpha^2)$ and taking $\alpha = \log^{\beta} n$ for a sufficiently small constant $\beta > 0$ makes these factors $n^{o(1)}$, so a polynomial sensitivity lower bound is preserved. Theorem 16.2 gives the corresponding locality lower bounds.

From sensitivity to locality. Finally, we convert the sensitivity lower bounds into locality lower bounds. Fleming and Yoshida [18] show that, for vertex-set outputs on graphs of maximum degree Δ, a sensitivity lower bound S implies non-signaling locality

$$t \geq \Omega\left(\frac{\log S}{\log(\Delta + 1)}\right) - O(1).$$

Intuitively, non-signaling allows outputs before and after an edge edit to be coupled so that changes are confined to a neighborhood of the edited edge. Bounded degree limits this neighborhood's growth (Section 2.4). For directed spanners, counting changed incident arcs introduces only another degree factor (Lemma 16.1).

1.3 Related Work

Algorithmic sensitivity. Sensitivity has been studied for network centrality [31], graph approximation algorithms [36, 38, 41], clustering [37], and dynamic programming [28]. For every constant $\varepsilon \in (0, 1)$, maximum matching in general graphs admits a randomized $(1 - \varepsilon)$-approximation with polylogarithmic worst-case sensitivity [39]. Recent results for Max Cut include worst-case sensitivity bounds for approximations arbitrarily close to the Goemans–Williamson ratio and algorithms with low average sensitivity on expanders admitting a cut containing nearly all edges [23].

On the lower-bound side, Fleming and Yoshida [18] proved a sensitivity lower bound for Label Cover algorithms attaining value close to one on satisfiable instances. Locally testable codes yield linear sensitivity lower bounds for near-optimal approximation of satisfiable CSPs such as Max E3LIN2, as well as lower bounds for Clique and maximum k-coverage [40]. We extend the Label Cover lower bound to inverse-polylogarithmic value guarantees while retaining a polynomial sensitivity lower bound with a constant exponent, enabling the covering and spanner applications.

Our deletion interpretation uses the output-distribution requirement of exact unlearning [21, 34, 35]: after deletion, the updated output must have the same distribution as a fresh run on the remaining input. The joint distribution of the outputs before and after an update is a coupling, so their expected Hamming distance is at least their EMD, even if the algorithm retains auxiliary state. For single-edge or single-arc edits, take the graph containing the changed edge or arc as the initial input. Symmetry of EMD then gives the same lower bound for deletion. This bounds output changes under exact deletion; it does not by itself give a lower bound on update time.

PCP constructions with low soundness. PCP constructions with low soundness are central in PCP-based hardness of approximation [22]. Constructions include the two-query PCP of Moshkovitz and Raz [30], Dinur–Harsha composition [15], parallel repetition and direct-product testing [24, 33], and derandomized parallel repetition via structured PCPs [16]. More recent constructions obtain low-soundness PCPs using high-dimensional expanders [4, 32]. To transfer sensitivity lower bounds, we bound both the number of target constraints changed by one source edit and the change in recovery when either the supplied labeling or the source instance changes. We prove these bounds for our adaptations of the IKW and DH constructions. We do not know whether constructions based on parallel repetition or high-dimensional expanders admit comparable sensitivity-transfer bounds.

Covering reductions and distributed Dominating Set. The Set Cover reductions originate in classical approximation hardness: Lund and Yannakakis [29] introduced the set-system construction, Feige [17] used it for Set Cover, and Chlebík and Chlebíková [8] treated bounded-degree Dominating Set. We show that these reductions transfer our Label Cover sensitivity lower bounds to Set Cover and Dominating Set.

For distributed Dominating Set, Chang and Li [7] proved an $\Omega(\log n)$ LOCAL lower bound for a constant approximation factor. Kuhn, Moscibroda, and Wattenhofer (KMW) [27] gave degree-dependent upper and lower bounds; Coupette and Lenzen [11] gave a streamlined presentation of the lower-bound construction. The KMW lower bound for the linear-programming relaxation of Dominating Set also holds in the non-signaling model [5, 19].

Spanner approximation. Min-Rep is a standard intermediate problem in spanner hardness reductions; Dinitz, Kortsarz, and Raz [12] use Label Cover with large girth to obtain hardness for undirected spanners. Our directed construction permits explicit control of the three-arc paths, the effect of individual arc edits, and the maximum degree.

Distributed approximation algorithms for directed spanners include those of Censor-Hillel and Dory [6] and Dinitz and Nazari [13].

1.4 Organization of the Paper

Section 2 gives the formal definitions and the common 2-CSP source. Part I develops the sensitivity-transfer framework and the degree and alphabet reductions. Part II proves the IKW Label Cover theorem, Theo-

rem 6.1. Part III proves the DH theorem, Theorem 11.1, through initialization by polynomial restrictions, query-oblivious composition, and iteration. Part IV transfers the bounds to Set Cover and Dominating Set and gives locality consequences for Dominating Set, with a comparison to prior distributed bounds. Part V develops the second application through Min-Rep and directed 3-spanner, then derives the non-signaling and LOCAL lower bounds for directed 3-spanner.

2 Preliminaries

For a positive integer n, let $[n]$ denote the set $\{1, 2, \ldots, n\}$. We use bold symbols such as $\boldsymbol{\pi}$ to denote random assignments. For a random variable X, write $\mathsf{Law}(X)$ for its probability distribution.

For the minimization problems considered later, a solution is *feasible* if it satisfies the constraints of the problem, and an instance is *feasible* if it admits such a solution. For a feasible instance I, write cost_I for its objective function and $\mathsf{opt}(I)$ for the minimum cost of a feasible solution. A randomized algorithm A is an α-approximation if, for every feasible instance I, it outputs a feasible solution almost surely and satisfies

$$\mathbf{E}[\mathrm{cost}_I(A(I))] \le \alpha \cdot \mathsf{opt}(I).$$

2.1 Label Cover

For the Dinur-Harsha route we will work with the following extension of the Label Cover problem which allows for additional predicates on the left vertices. This will allow us to maintain sensitivity. The usual Label Cover problem is recovered by taking each predicate P_u to be identically 1.

Definition 2.1 (Left-predicate Label Cover instance)**.** A *left-predicate Label Cover instance* is a tuple

$$I = (U, V, E, \Sigma_U, \Sigma_V, \mathcal{P}, F), \qquad \mathcal{P} = \{P_u : \Sigma_U \to \{0,1\}\}_{u \in U}, \qquad F = \{f_e : \Sigma_U \to \Sigma_V\}_{e \in E},$$

where $G = (U, V, E)$ is a bipartite multigraph, Σ_U and Σ_V are finite alphabets, each P_u is a unary predicate on left labels. Parallel edges are distinct elements of E. Thus, when we write $e = (u, v)$, the symbol e identifies a particular parallel edge with endpoints u and v, and the $f_e \in F$ are called *projections*. An assignment is a pair $\pi = (\pi_U, \pi_V)$ with $\pi_U : U \to \Sigma_U$ and $\pi_V : V \to \Sigma_V$. An edge $e = (u, v)$ is satisfied by π if

$$P_u(\pi_U(u)) = 1 \qquad \text{and} \qquad f_e(\pi_U(u)) = \pi_V(v).$$

Thus a left label that fails its predicate satisfies none of its incident edges. For an assignment π, set $\mathsf{val}_I(\pi) := 1$ when $E = \varnothing$; when $E \neq \varnothing$, let

$$\mathsf{val}_I(\pi) := \frac{1}{|E|} \cdot \left|\left\{e = (u, v) \in E : P_u(\pi_U(u)) = 1 \text{ and } f_e(\pi_U(u)) = \pi_V(v)\right\}\right|$$

denote the fraction of satisfied edges, and let $\mathsf{val}(I) := \max_\pi \mathsf{val}_I(\pi)$. When the predicates are clear from context, we may abbreviate the tuple by omitting $\mathcal{P}$.

For two assignments $\pi, \tilde{\pi} : X \to \Sigma$ over the same domain X, let $d_{\mathrm{H}}(\pi, \tilde{\pi}) = |\{x \in X : \pi(x) \neq \tilde{\pi}(x)\}|$ denote their Hamming distance. For two probability distributions P, Q on the same space $\mathcal{X}$, the *earth mover's distance* (EMD) between P and Q is

$$\mathsf{EMD}(P, Q) := \min_{\Gamma \in \Pi(P,Q)} \mathop{\mathbf{E}}_{(X,Y)\sim\Gamma} \big[d_H(X, Y)\big],$$

where $\Pi(P, Q)$ denotes the set of couplings of P and Q.

2.2 Sensitivity

Throughout our reductions, we will measure sensitivity for a (left-predicate) Label Cover instances by swapping one projection or one unary predicate, keeping the underlying graph fixed. Let $I = (U, V, E, \Sigma_U, \Sigma_V, \mathcal{P}, F)$ be a left-predicate Label Cover instance. For $e \in E$ and $f : \Sigma_U \to \Sigma_V$, let $I^{e \leftarrow f}$ denote the instance obtained from I by replacing the projection f_e with f. Similarly, for $u \in U$ and a predicate $Q : \Sigma_U \to \{0, 1\}$, $I^{u \leftarrow Q}$ denotes the instance obtained by replacing P_u with Q.

The *swap sensitivity* of a randomized algorithm A on I is

$$\mathsf{SwapSens}(A, I) := \max \left\{ \begin{array}{l} \max\limits_{e \in E} \max\limits_{f:\Sigma_U \to \Sigma_V} \mathsf{EMD}\Big(A(I), A(I^{e \leftarrow f})\Big), \\ \max\limits_{u \in U} \max\limits_{Q:\Sigma_U \to \{0,1\}} \mathsf{EMD}\big(A(I), A(I^{u \leftarrow Q})\big) \end{array} \right\},$$

and for a family of instances $\mathcal{I}$, $\mathsf{SwapSens}(A, \mathcal{I}) := \max_{I \in \mathcal{I}} \mathsf{SwapSens}(A, I)$. This requires A to be defined on $\mathcal{I}$ and on all one-swap neighbors of its members, whether or not those neighbors lie in $\mathcal{I}$. We say that a family of left-predicate Label Cover instances $\mathcal{I}$ is *swap-closed* if for any instance $I \in \mathcal{I}$, $e \in E$, and $u \in U$, every projection swap $I^{e \leftarrow f}$ and every predicate swap $I^{u \leftarrow Q}$ also belongs to $\mathcal{I}$. For a family of instances $\mathcal{I}$, its *swap-closure* $\mathsf{SwapClo}(\mathcal{I})$ is the family of instances obtained by repeatedly swapping projections and/or left predicates in instances from $\mathcal{I}$.

For instances I and $\tilde{I}$ on the same underlying graph and the same alphabets, the *swap distance* $\mathsf{SwapDist}(I, \tilde{I})$ is the minimum number of projection swaps and predicate swaps needed to transform I into $\tilde{I}$. For a swap-closed family $\mathcal{I}$, an algorithm A on $\mathcal{I}$, and $I, \tilde{I} \in \mathcal{I}$, the triangle inequality gives $\mathsf{EMD}(A(I), A(\tilde{I})) \le \mathsf{SwapSens}(A, \mathcal{I}) \cdot \mathsf{SwapDist}(I, \tilde{I})$. The following lemma allows us to restrict attention to neighboring instances when bounding sensitivity.

Lemma 2.2. Let $\mathcal{I}$ be a family of instances equipped and A be an algorithm on $\mathcal{I}$. If $\mathsf{EMD}(A(I), A(\widetilde{I})) \ge \lambda \cdot \mathsf{SwapDist}(I, \widetilde{I})$ for some $I, \widetilde{I} \in \mathcal{I}$ with $\mathsf{SwapDist}(I, \widetilde{I}) = k > 0$, then every shortest swap path

$$I = I^{(0)}, I^{(1)}, \ldots, I^{(k)} = \widetilde{I}, \qquad I^{(t)} \in \mathcal{I} \quad (0 \le t \le k),$$

contains a consecutive pair satisfying $\mathsf{EMD}\big(A(I^{(t)}), A(I^{(t+1)})\big) \ge \lambda$. Such a path exists, in particular, when $\mathcal{I}$ is swap-closed.

Proof. Repeatedly applying the triangle inequality,

$$\mathsf{EMD}(A(I), A(\widetilde{I})) \le \sum_{t=0}^{k-1} \mathsf{EMD}\big(A(I^{(t)}), A(I^{(t+1)})\big).$$

Hence, the average of the k terms is at least $\mathsf{EMD}(A(I), A(\widetilde{I})) \ge \lambda k$, so one has value at least λ. □

Finally, for a family of instances $\mathcal{I}$, let $\mathcal{I}_{\mathrm{sat}} \subseteq \mathcal{I}$ denote the instances that are satisfiable. Let $\mathsf{SwapSens}_s(\mathcal{I})$ be the minimum of $\mathsf{SwapSens}(A, \mathcal{I})$ over all algorithms A such that for every $I \in \mathcal{I}_{\mathrm{sat}}$, if $\boldsymbol{\pi} \sim A(I)$ then $\mathbf{E}[\mathsf{val}_I(\boldsymbol{\pi})] \ge s$.

The Single-edge-edit Model. In the single-edge-edit model, an edit inserts or deletes one edge while leaving the vertex sets, alphabets, and left predicates unchanged. Let A also be defined on instances obtained

by such edits. If $I^{e\setminus}$ denotes the instance obtained by deleting e, then a projection swap $I \to I^{e\leftarrow f}$ is realized by a path of two edge edits, $I \to I^{e\setminus} \to I^{e\leftarrow f}$. The triangle inequality for EMD therefore gives

$$\max_{e\in E}\max_{f:\Sigma_U\to\Sigma_V} \mathsf{EMD}\Big(A(I), A(I^{e\leftarrow f})\Big) \qquad \le 2 \max_{\substack{J\in\{I\}\cup\{I^{e\setminus}:e\in E\};\\ \widetilde{J} \text{ is obtained from } J\\ \text{by one edge insertion or deletion}}} \mathsf{EMD}(A(J), A(\widetilde{J})),$$

where an inserted edge may carry an arbitrary projection. In particular, every swap-sensitivity lower bound that is proved using only projection swaps implies the same lower bound in the single-edge-edit model up to a factor 2.

2.3 The Base 2-CSP

Both our low-soundness constructions start from the bounded-degree 2-CSPs in the proof of Fleming and Yoshida's Label Cover sensitivity lower bound [18]. A 2-CSP consists of a multigraph (X, E), an alphabet Σ, and a relation $R_e \subseteq \Sigma^2$ for each edge e. We fix an ordering of the two endpoint incidences of each edge; parallel edges and self-loops are allowed, and degrees count endpoint incidences. An assignment $a : X \to \Sigma$ satisfies $e = (x, y)$ if $(a(x), a(y)) \in R_e$. Its value is the fraction of satisfied edges, with value 1 when $E = \varnothing$. We use the same swap-sensitivity definitions as above, with a swap replacing one relation on its fixed pair of endpoint incidences. The value guarantee is required only on satisfiable members of the swap-closed family.

The construction in [18, proof of Theorem 5.2], yields a family of regular 2-CSPs with constant degree and alphabet-size, and a polynomial swap-sensitivity lower bound for algorithms achieving expected value at least $1 - \eta$ on satisfiable instances, for some constant $\eta \in (0, 1)$. We adapt it by a padding argument to obtain exactly N variables for every sufficiently large N, and making the degree even by duplicating edges.

Lemma 2.3 (Fleming–Yoshida sensitivity lower bound)**.** There are constants $\eta_{\mathrm{src}}, c_{\mathrm{src}} \in (0, 1)$, an even integer $d_0 \ge 2$, and a constant alphabet Σ_{src} such that, for every sufficiently large N, there is a swap-closed family $\mathcal{F}_N$ of 2-CSPs on a common d_0-regular multigraph with N variables and alphabet Σ_{src} satisfying

$$\mathsf{SwapSens}_{1-\eta_{\mathrm{src}}}(\mathcal{F}_N) \ge N^{c_{\mathrm{src}}}.$$

Proof. Fleming and Yoshida's reductions preserve satisfiability. Within each family they produce, all instances have the same variable set, edge multiset, alphabet, and ordering of the endpoints of each edge; only the constraint relations R_e vary. Their sensitivity argument applies to algorithms defined on the whole swap closure and required to satisfy the value guarantee only on its satisfiable members.

The numbers of variables attained by the construction form an unbounded increasing sequence whose successive ratios are bounded by an absolute constant C. Thus, for every sufficiently large N, we can choose a family with M variables such that $N/C \le M \le N$. Duplicate every edge to make the degree d_0 even; this preserves values and loses at most a factor of two in the sensitivity lower bound. Append $N - M$ variables, each carrying $d_0/2$ always-satisfied self-loops. The graph remains d_0-regular, and restriction to the original variables is 1-Lipschitz in Hamming distance. The expected fraction of violated constraints on the original instance is at most $N/M \le C$ times that on the padded instance. Choosing η_{src} sufficiently small therefore preserves a lower bound of $\Omega(M^c) = \Omega(N^c)$ for some constant $c > 0$. Taking $0 < c_{\mathrm{src}} < c$ proves the lemma for sufficiently large N. □

2.4 Non-signaling Locality

Noah: perhaps move to applications? Also see Max's comments

We compare input graphs on a common vertex set V. For a graph $G = (V, E)$ with vertex and edge inputs, let $N_t^G(S)$ denote the radius-t input neighborhood of $S \subseteq V$, retaining vertex names and all input information initially available at vertices in the neighborhood. For example, in the directed spanner setting, this information includes each vertex's incident arcs and their endpoints. Let $\{A(G)\}_G$ be a family indexed by the input graphs under consideration, where $A(G)$ is a probability distribution over vectors $(y_v)_{v \in V}$ of vertex outputs. For $S \subseteq V$, write $A(G)|_S$ for the joint marginal distribution of $(y_v)_{v \in S}$. We call $\{A(G)\}_G$ a *t-local non-signaling algorithm* if, for every pair of inputs $G, \widetilde{G}$ as above and every $S \subseteq V$,

$$N_t^G(S) = N_t^{\widetilde{G}}(S) \quad \Longrightarrow \quad A(G)|_S = A(\widetilde{G})|_S.$$

In particular, the output distributions of every t-round randomized LOCAL algorithm form a t-local non-signaling family. The model also contains quantum-LOCAL and bounded-dependence algorithms at the same locality [1, 10, 19].

We use the following sensitivity-to-locality consequence of this definition [18]. Suppose two graphs $G, \widetilde{G}$ on a common vertex set V, each of maximum degree at most $\Delta \geq 1$, differ by one input edit supported on a constant-size set $X \subseteq V$ (the endpoints in the case of an edge edit). Let $W \subseteq V$ be the union of the vertex sets of $N_{t+1}^G(X)$ and $N_{t+1}^{\widetilde{G}}(X)$. The choice of radius $t+1$ in the definition of W ensures that the radius-t input neighborhoods of $V \setminus W$ coincide in the two graphs, even when vertices at the boundaries of these neighborhoods initially know all their incident edges. For a t-local non-signaling algorithm, the joint output marginals on $V \setminus W$ are identical, so they admit a coupling that agrees outside W. Consequently, the earth mover's distance between the two output distributions, under Hamming distance, is at most

$$|W| \leq O((\Delta + 1)^{t+1}).$$

Thus a sensitivity lower bound $S > 0$ implies a locality lower bound

$$t \geq c \, \frac{\log S}{\log(\Delta + 1)} - C \tag{1}$$

for absolute constants $c, C > 0$.

2.5 Local Circuits and Paths of Predicate Swaps

Two of the key transformations, alphabet reduction and iterated composition, that underlie our PCPs require bounds on the circuits that test whether neighboring right labels are the projections of a left label satisfying its predicate. In this section we describe the encoding of these circuits, and the measure of complexity that we will use, which we call their local description complexity.

We encode alphabet symbols by fixed binary strings and fix a binary circuit encoding that explicitly lists all inputs and outputs, gates, and connections and includes an end marker. Circuit complexity is the bit length of this encoding. Padding appends ignored zero bits after the end marker to reach a prescribed description length, leaving the computed function and number of inputs unchanged.

Definition 2.4 (Local description complexity)**.** For a left-predicate Label Cover instance I and $u \in U$, put $k_u = \deg_I(u)$ and order the incident edges at u as $e_{u,1}, \ldots, e_{u,k_u}$, with right endpoints $v_{u,j} \in V$. Let $\mathcal{N}_I(u) \subseteq \Sigma_V^{k_u}$ be the set of projection tuples of left labels satisfying P_u:

$$\mathcal{N}_I(u) := \{(f_{e_{u,1}}(a), \ldots, f_{e_{u,k_u}}(a)) : a \in \Sigma_U, \ P_u(a) = 1\}.$$

The instance has *local description complexity at most* S if each set $\mathcal{N}_I(u)$ is decided by a circuit $C_u : \Sigma_V^{k_u} \to \{0,1\}$ of complexity at most S. The circuits are chosen vertexwise: when a predicate or projection at u changes, the chosen circuit at every other left vertex is retained.

We also need the local description complexity bound to hold at intermediate instances when left predicates are changed one at a time. A *path of predicate swaps* is a sequence of instances with fixed graph, alphabets, and projections in which each step replaces one left predicate. If two such instances have local description complexity at most S, replacing their differing predicates one at a time gives a path with the same bound at every instance: at each left vertex u, use the circuit from the endpoint instance whose predicate is used at u.

Part I
Sensitivity Transfer and Reduction Tools

In this section we develop the tools which we will need in order to bound the changes in sensitivity through our PCP reductions. We then develop tools for reducing the alphabet and degree of Label Cover instances.

3 Sensitivity Transfer for Output Distributions

We formalize the mechanism for transferring sensitivity lower bounds through reductions that we outlined in Section 1.2. A reduction $\mathcal{T}$ maps instances of a source problem (such as our base 2-CSP or Label Cover) to a target problem, typically with better parameters. Let $\mathcal{X}$ and $\mathcal{Y}$ be instances of the source and target problems respectively, then $\mathcal{T} : \mathcal{X} \to \mathcal{Y}$. Each problem-instance $I \in \mathcal{X}$ has a domain and a range X_I, and denote by Y_I the range of $\mathcal{T}(I)$. We require that neighboring instances I have the same range so that we can compare them.

In order to control the change in sensitivity through $\mathcal{T}$, we equip each instance $I \in \mathcal{X}$ with a randomized *recovery map* $\mathcal{R}_I$. Since we would like to bound the sensitivity of randomized algorithms solving our instances, our recovery algorithm will map a distribution on Y_I (thought of as the output of some randomized algorithm on Y) to a distribution on X_I.

Definition 3.1. With the setup above, we call $(C_{\mathcal{T}}, C_{\mathcal{R}}, D_{\mathcal{R}})$ *sensitivity-transfer parameters* if the following hold.

1. For all neighboring instances $I, \widetilde{I} \in \mathcal{X}$, the target instances $\mathcal{T}(I), \mathcal{T}(\tilde{I})$ have a common range and
$$\mathsf{SwapDist}\big(\mathcal{T}(I), \mathcal{T}(\widetilde{I})\big) \le C_{\mathcal{T}}.$$
2. For every I, the recovery map $\mathcal{R}_I$ is $C_{\mathcal{R}}$-Lipschitz with respect to EMD: for all $\mu, \nu \in \mathfrak{D}(Y_I)$,
$$\mathsf{EMD}\big(\mathcal{R}_I(\mu), \mathcal{R}_I(\nu)\big) \le C_{\mathcal{R}} \cdot \mathsf{EMD}(\mu, \nu).$$
We will refer to this throughout as the *Lipschitz bound*.
3. For all neighboring instances $I, \widetilde{I}$ and every distribution μ on their common target range,
$$\mathsf{EMD}\big(\mathcal{R}_I(\mu), \mathcal{R}_{\widetilde{I}}(\mu)\big) \le D_{\mathcal{R}}.$$

The second condition changes the distribution supplied to recovery while holding the source instance fixed; the third changes the source instance while holding that distribution fixed.

Theorem 3.2. Let $\mathcal{X}$ be a swap-closed family of source instances and let $\mathcal{T}$ and $\mathcal{R}_I$ be a reduction and recovery maps with sensitivity-transfer parameters $(C_{\mathcal{T}}, C_{\mathcal{R}}, D_{\mathcal{R}})$. Let $\mathcal{Y}'$ be a swap-closed family of target instances containing $\mathcal{T}(\mathcal{X})$, and let A be an algorithm on $\mathcal{Y}'$. Then $\mathcal{R}_I$ transforms the algorithm A to an algorithm B for the source instance, whose output distribution is given by

$$\mathsf{Law}(B(I)) := \mathcal{R}_I\big(\mathsf{Law}(A(\mathcal{T}(I)))\big).$$

Then

$$\mathsf{SwapSens}(B, \mathcal{X}) \le C_{\mathcal{T}} C_{\mathcal{R}} \mathsf{SwapSens}(A, \mathcal{Y}') + D_{\mathcal{R}}.$$

Consequently, if $C_{\mathcal{T}} C_{\mathcal{R}} > 0$ and $\mathsf{SwapSens}(B, \mathcal{X}) \ge L$, then

$$\mathsf{SwapSens}(A, \mathcal{Y}') \ge \frac{L - D_{\mathcal{R}}}{C_{\mathcal{T}} C_{\mathcal{R}}}.$$

Proof. For $I \in \mathcal{X}$ and any one-swap neighbor $\widetilde{I}$, swap-closure gives $\widetilde{I} \in \mathcal{X}$. A shortest swap path between the target instances lies in $\mathcal{Y}'$ and gives

$$\mathsf{EMD}\big(A(\mathcal{T}(I)), A(\mathcal{T}(\widetilde{I}))\big) \le C_{\mathcal{T}} \cdot \mathsf{SwapSens}(A, \mathcal{Y}').$$

Apply condition 2 of Definition 3.1 to the two target output distributions, keeping I fixed, and then change the recovery from I to $\widetilde{I}$ while holding the second distribution fixed. The triangle inequality proves the first bound; rearranging proves the second. □

A randomized recovery procedure acting on one target output is called a *pointwise recovery*. We use R_I both for this procedure and for its associated map R_I from elements in Y_I to distributions on X_I. For a distribution μ on Y_I, write

$$(R_I)_{\#}\mu := \sum_{y \in Y_I} \mu(y) R_I(y)$$

for the distribution induced by R_I, also called the pushforward of μ through R_I. Equivalently, first draw a target output from μ and then run R_I on that output with fresh internal randomness. When the input is a random target output $\boldsymbol{Y}$, we write $R_I(\boldsymbol{Y})$ for the resulting random source output. Thus,

$$\mathsf{Law}(R_I(\boldsymbol{Y})) = (R_I)_{\#}\mathsf{Law}(\boldsymbol{Y}).$$

We reserve $\mathcal{R}_I$ for maps from target output distributions to source output distributions. A pointwise recovery induces the distribution map $\mathcal{R}_I(\mu) = (R_I)_{\#}\mu$.

Lemma 3.3 (Lifting a pointwise recovery to distributions)**.** A pointwise recovery map R_I that satisfies

$$\mathsf{EMD}(R_I(y), R_I(\widetilde{y})) \le C_{\mathcal{R}} d_{\mathrm{H}}(y, \widetilde{y}),$$
$$\mathsf{EMD}(R_I(y), R_{\widetilde{I}}(y)) \le D_{\mathcal{R}}$$

for every I and all $y, \widetilde{y}$ in the first line, and for all neighboring instances $I, \widetilde{I}$ and every y in their common input domain in the second, induces a recovery on distributions $\mu \mapsto (R_I)_{\#}\mu$ with the same two constants.

Proof. Fix input distributions μ, ν on the range of $\mathcal{T}(I)$. Couple them optimally and, conditional on a coupled input pair, use the pointwise coupling for the recovery maps. This proves

$$\mathsf{EMD}\big((R_I)_{\#}\mu, (R_I)_{\#}\nu\big) \le C_{\mathcal{R}} \cdot \mathsf{EMD}(\mu, \nu).$$

Using the same input sample for neighboring instances proves the corresponding bound on the change in recovery. □

Composing Recovery Maps The next lemma bounds the sensitivity-transfer parameters under composition. Suppose $\Phi \mapsto \mathcal{T}(\Phi)$ has a recovery $\mathcal{R}_\Phi$ on distributions with parameters $(C_\mathcal{T}, C_\mathcal{R}, D_\mathcal{R})$. Let $\mathcal{J}$ be a swap-closed family containing all $\mathcal{T}(\Phi)$, with one common assignment domain. Let $\mathcal{T}_{\text{step}}$ be a further transformation defined on every instance $I \in \mathcal{J}$. We apply $\mathcal{T}$ first and $\mathcal{T}_{\text{step}}$ second, obtaining $\mathcal{T}_{\text{step}}(\mathcal{T}(\Phi))$. For each $I \in \mathcal{J}$, let R_I be a pointwise recovery that takes an output of $\mathcal{T}_{\text{step}}(I)$ and produces a random output for I. Thus recovery proceeds in the reverse order: first apply $(R_{\mathcal{T}(\Phi)})_\#$, then $\mathcal{R}_\Phi$. We state the lemma for a pointwise recovery in the second stage because the composition, alphabet-reduction, and degree-reduction steps used later all have recoveries of this form; the accumulated recovery $\mathcal{R}_\Phi$ need not be induced by a pointwise recovery.

Suppose $\mathcal{T}_{\text{step}}$ and R_I have respective bounds t, r, d for instance changes, input perturbations, and changes in the recovery rule, as follows. For all neighboring instances $I, \widetilde{I} \in \mathcal{J}$, the instances $\mathcal{T}_{\text{step}}(I)$ and $\mathcal{T}_{\text{step}}(\widetilde{I})$ have a common range,

$$\mathsf{SwapDist}(\mathcal{T}_{\text{step}}(I), \mathcal{T}_{\text{step}}(\widetilde{I})) \le t, \qquad \mathsf{EMD}(R_I(y), R_{\widetilde{I}}(y)) \le d,$$

where the second inequality holds for every y in that domain, and for every I and all $y, \widetilde{y}$ in the range of $\mathcal{T}_{\text{step}}(I)$,

$$\mathsf{EMD}(R_I(y), R_I(\widetilde{y})) \le r\, d_{\mathrm{H}}(y, \widetilde{y}).$$

For a distribution μ on assignments to $\mathcal{T}_{\text{step}}(\mathcal{T}(\Phi))$, define the recovered distribution on assignments to Φ by

$$\mathcal{R}^+_\Phi(\mu) := \mathcal{R}_\Phi\big((R_{\mathcal{T}(\Phi)})_\# \mu\big).$$

Lemma 3.4 (Composing recoveries)**.** Under the hypotheses above, the composed transformation $\mathcal{T}_{\text{step}} \circ \mathcal{T}$ with recovery $\mathcal{R}^+_\Phi$ has sensitivity-transfer parameters satisfying

$$\begin{aligned} C^+_\mathcal{T} &\le C_\mathcal{T} t, \\ C^+_\mathcal{R} &\le C_\mathcal{R} r, \\ D^+_\mathcal{R} &\le D_\mathcal{R} + C_\mathcal{T} C_\mathcal{R} d. \end{aligned}$$

Proof. By Lemma 3.3, the r-Lipschitz bound for R_I lifts to distributions. Composing it with the $C_\mathcal{R}$-Lipschitz bound for $\mathcal{R}_\Phi$ gives $C^+_\mathcal{R} \le C_\mathcal{R} r$. For neighboring instances $\Phi, \widetilde{\Phi}$, applying $\mathcal{T}_{\text{step}}$ to a path of at most $C_\mathcal{T}$ swaps from $\mathcal{T}(\Phi)$ to $\mathcal{T}(\widetilde{\Phi})$ gives a path of at most $C_\mathcal{T} t$ swaps between their images. Thus $C^+_\mathcal{T} \le C_\mathcal{T} t$.

Along this path from $\mathcal{T}(\Phi)$ to $\mathcal{T}(\widetilde{\Phi})$, at each step $I, \widetilde{I}$, apply the bound $\mathsf{EMD}(R_I(y), R_{\widetilde{I}}(y)) \le d$ and average over $y \sim \mu$. The triangle inequality gives

$$\mathsf{EMD}\big((R_{\mathcal{T}(\Phi)})_\# \mu, (R_{\mathcal{T}(\widetilde{\Phi})})_\# \mu\big) \le C_\mathcal{T} d.$$

Apply the $C_\mathcal{R}$-Lipschitz bound for $\mathcal{R}_\Phi$ to these two distributions, then replace $\mathcal{R}_\Phi$ by $\mathcal{R}_{\widetilde{\Phi}}$ at cost at most $D_\mathcal{R}$. The triangle inequality gives $D^+_\mathcal{R} \le C_\mathcal{R} C_\mathcal{T} d + D_\mathcal{R}$. □

Stable Selection Our recovery procedures select among candidate outputs by taking the first candidate whose score meets a uniformly random threshold. This rule is inspired by related procedures from replicable learning [25]. We first bound the probability that perturbing the scores changes the selected index, then deduce a Hamming-distance bound when the candidates are assignments.

Let $J \subseteq [0,1]$ be an interval of length $w > 0$. For ordered scores $q_1, \dots, q_R \in [0,1]$ and a threshold $\theta \in J$, select the smallest index i with $q_i \geq \theta$, or index 0 if no score meets the threshold. Index 0 represents a default output.

Lemma 3.5 (Random-threshold selection)**.** For two score lists $(q_i)_{i=1}^R$ and $(\widetilde{q}_i)_{i=1}^R$ in the setting above, let K and $\widetilde{K}$ be the selected indices when both lists use the same uniform threshold $\boldsymbol{\theta} \in J$. Let A be the set of indices i for which the interval between q_i and $\widetilde{q}_i$ intersects J:

$$A := \{i \in [R] : [\min\{q_i, \widetilde{q}_i\}, \max\{q_i, \widetilde{q}_i\}] \cap J \neq \varnothing\}.$$

Then

$$\Pr\Big[K \neq \widetilde{K}\Big] \leq \frac{1}{w} \sum_{i \in A} |q_i - \widetilde{q}_i|.$$

Moreover, for a finite alphabet Σ and a positive integer N, let $z_0, z_1, \dots, z_R, \widetilde{z}_1, \dots, \widetilde{z}_R \in \Sigma^N$ be candidate assignments with a common default $\widetilde{z}_0 := z_0$. Then

$$\mathbf{E}\, d_{\mathrm{H}}(z_K, \widetilde{z}_{\widetilde{K}}) \leq \sum_{i=1}^{R} d_{\mathrm{H}}(z_i, \widetilde{z}_i) + \frac{N}{w} \sum_{i \in A} |q_i - \widetilde{q}_i|.$$

Proof. For each i, the tests $q_i \geq \boldsymbol{\theta}$ and $\widetilde{q}_i \geq \boldsymbol{\theta}$ can have different outcomes only when $\boldsymbol{\theta}$ lies between q_i and $\widetilde{q}_i$, an event of probability at most $|q_i - \widetilde{q}_i|/w$ and probability zero for $i \notin A$. If all tests have the same outcomes, then $K = \widetilde{K}$. A union bound proves the probability estimate. When $K = \widetilde{K}$, the Hamming distance between the selected assignments is at most $\sum_{i=1}^R d_{\mathrm{H}}(z_i, \widetilde{z}_i)$; when $K \neq \widetilde{K}$, it is at most N. Taking expectations and using the probability estimate proves the second inequality. □

4 Right-Degree Reduction

We use the standard expander-based right-degree reduction described by Dinur and Harsha [14, Theorem 5.4], which replaces each right vertex by a cloud of copies of the prescribed degree. Their proof also gives the randomized recovery used here: retain the left labels and recover each original right label by uniformly sampling a copy in its cloud. We formulate this reduction for left-predicate Label Cover.

Our contribution is to bound the number of target constraints affected by a source edit and the expected Hamming distances between recovered labelings, separately on the left and right sides. These bounds allow us to transfer sensitivity lower bounds through degree reduction (Lemma 4.3).

Definition 4.1 (Expanders)**.** An undirected d-regular multigraph $G = (V, E)$ on n vertices is an $[n, d, \lambda]$-expander if the normalized adjacency operator has spectral norm at most λ on $\mathbf{1}^\perp$. Parallel edges are counted with multiplicity, and a loop contributes two to both its diagonal adjacency entry and the degree of its endpoint.

Proposition 4.2 (Explicit expanders of every size)**.** For every $n \geq 1$ and every even $d \geq 6$, there is a $\mathrm{poly}(n, d)$-time deterministic algorithm that constructs an $[n, d, \lambda_d]$-expander, where

$$\lambda_d := \frac{2\sqrt{d/2 - 1}}{d/2} < 1 \qquad \text{and} \qquad \lambda_d \leq \frac{2\sqrt{2}}{\sqrt{d}}.$$

Proof. Put $q := d/2$. Cohen [9] gives a deterministic polynomial-time construction of a q-regular bipartite Ramanujan multigraph on $2n$ vertices, with classes $[n]_L$ and $[n]_R$. Let B be its biadjacency matrix. The Ramanujan bound gives $\|Bx\|_2, \|B^{\mathsf{T}}x\|_2 \le 2\sqrt{q-1}\|x\|_2$ for every $x \perp \mathbf{1}$.

Identify i_L and i_R for every $i \in [n]$. The resulting undirected multigraph has adjacency matrix $A := B + B^{\mathsf{T}}$ and degree $2q = d$; edges between i_L and i_R become loops under the convention in Definition 4.1. For every $x \perp \mathbf{1}$,

$$\|Ax\|_2 \le \|Bx\|_2 + \|B^{\mathsf{T}}x\|_2 \le 4\sqrt{q-1}\,\|x\|_2.$$

Dividing by $d = 2q$ gives the claimed value of λ_d. □

Let $d \ge 6$ be the desired right degree. Degree reduction replaces each right vertex v of degree D_v by a cloud of D_v new vertices, each of degree d. A d-regular expander on the cloud determines which copies of the original edge incidences are attached to each new vertex. The recovery map samples one cloud vertex to recover the original label at v. Expansion limits the advantage of assigning different labels to different cloud vertices: the value of any target labeling is at most the expected value of its recovered labeling plus the expander parameter λ. Figure 2 illustrates the construction, which we now define formally.

Fix an even integer $d \ge 6$ and a parameter $\lambda \in [0, 1)$. Let $I = (U, V, E, \Sigma_U, \Sigma_V, \mathcal{P}, F)$ be a left-predicate Label Cover instance with no isolated right vertices, and write $G = (U, V, E)$ for its bipartite multigraph. For each right vertex $v \in V$, let D_v denote its degree and set $\delta_V := \min_{v \in V} D_v$. A *family of* (d, λ)*-expanders* for I is a choice, for each $v \in V$, of a $[D_v, d, \lambda]$-expander $H_v = ([D_v], E'')$. Once such a family is fixed, we define the transformation $\mathcal{T}_{\mathrm{DR}}(I; d)$ as follows. We fix an arbitrary ordering of the edge incidences at each right vertex $v \in V$. The maps $e_G(v, \cdot) : [D_v] \to E$ and $u_G(v, \cdot) : [D_v] \to U$ return, on input $i \in [D_v]$, the i-th incident edge and its left endpoint, respectively. Let $\Gamma_{H_v} : [D_v] \times [d] \to [D_v]$ map (i, k) to the k-th neighbor of i in H_v, where parallel edges and the two incidences of each loop appear with multiplicity in the neighbor list.

Algorithm 1: Right-Degree Reduction $\mathcal{T}_{\mathrm{DR}}$

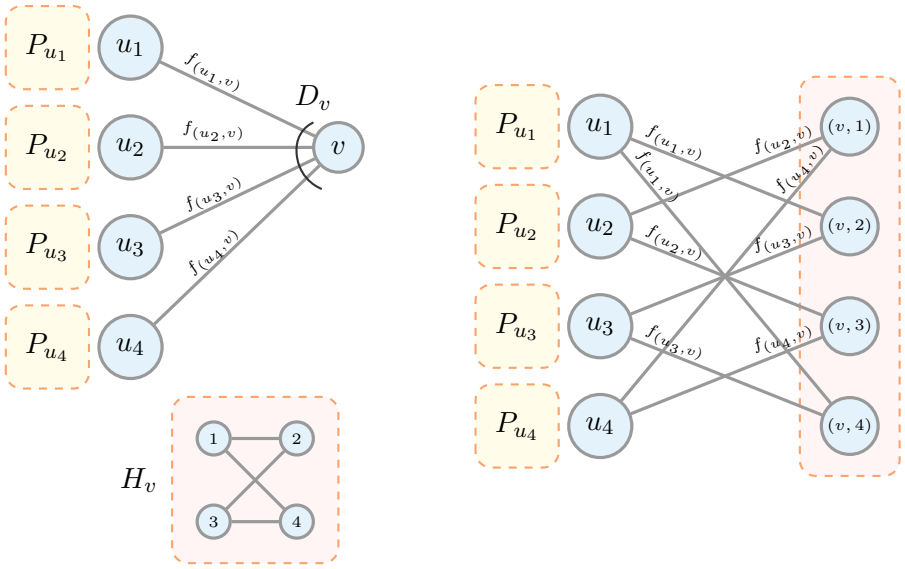
Input: A left-predicate Label Cover instance $I = (U, V, E, \Sigma_U, \Sigma_V, \mathcal{P}, F)$, an integer $d \ge 6$, a family of (d, λ)-expanders, and an ordering of the edge incidences at each right vertex

Output: An instance $I' = (U, V', E', \Sigma_U, \Sigma_V, \mathcal{P}, F')$ with right degree d.

```
for v ∈ V do
    Let D_v be the degree of v;
    Let H_v be the chosen [D_v, d, λ]-expander for v;
    Add to V' a vertex (v, i) for each i ∈ [D_v];
    for i ∈ [D_v] do
        for k ∈ [d] do
            Let j := Γ_{H_v}(i, k);
            Add a new edge incidence e' from u_G(v, j) to (v, i);
            Set f_{e'} := f_{e_G(v,j)};
return I'
```

By construction, $|V'| = |E|$. Since each H_v is d-regular, each original edge is copied d times, always with the same left endpoint. Thus the degree of each $u \in U$ in the output graph is $d \cdot \deg_G(u)$, counting parallel edges with multiplicity.

For an input Label Cover instance I, the recovery map $R_{\mathrm{DR},I}$ converts an assignment $\pi' = (\pi'_U, \pi'_V)$ for $I' = \mathcal{T}_{\mathrm{DR}}(I; d)$ into a random assignment $\boldsymbol{\pi}$ for $I = (U, V, E, \Sigma_U, \Sigma_V, \mathcal{P}, F)$. It copies the left labels, i.e.,



(a) The neighborhood of v before degree reduction and auxiliary graph H_v.

(b) The portion of the Label Cover instance corresponding to v after degree reduction.

Figure 2: Schematic of degree reduction at a right vertex v, using a 2-regular auxiliary graph H_v to illustrate the edge routing. The reduction uses an expander of the prescribed degree.

sets $\boldsymbol{\pi}_U(u) := \pi'_U(u)$ for each $u \in U$. For each right vertex $v \in V$, it samples an index $i_v \in [D_v]$ uniformly and independently, and sets $\boldsymbol{\pi}_V(v) := \pi'_V(v, i_v)$. When I is clear, we abbreviate this map by R_{DR}.

Lemma 4.3 (Sensitivity transfer through right-degree reduction)**.** Let $d \geq 6$ be even and let $\lambda \in [0, 1)$. Let $\mathcal{I}$ be a swap-closed family of left-predicate Label Cover instances on a common underlying graph and alphabets, with minimum right degree $\delta_V \geq 1$. For each right vertex v, let D_v be its common degree across the family, and fix a $[D_v, d, \lambda]$-expander and an ordering of the source edge incidences at v. Put

$$\mathcal{I}' := \mathsf{SwapClo}(\mathcal{T}_{\mathrm{DR}}(\mathcal{I}; d)).$$

The family $\mathcal{I}'$ has right degree d, and $\mathcal{T}_{\mathrm{DR}}$ maps satisfiable instances to satisfiable instances.

For every $I \in \mathcal{I}$ and every labeling π' of $\mathcal{T}_{\mathrm{DR}}(I; d)$, recovery satisfies

$$\mathsf{val}_{\mathcal{T}_{\mathrm{DR}}(I;d)}(\pi') \leq \mathbf{E}\big[\mathsf{val}_I\big(R_{\mathrm{DR},I}(\pi')\big)\big] + \lambda. \tag{2}$$

Fix $\varepsilon > 0$ and let A' be an algorithm on $\mathcal{I}'$ whose output has expected value at least $\varepsilon + \lambda$ on every satisfiable member. Define

$$A(I) := R_{\mathrm{DR},I}\big(A'(\mathcal{T}_{\mathrm{DR}}(I; d))\big).$$

Then A has expected value at least ε on every satisfiable member of $\mathcal{I}$. Moreover, suppose that for all neighboring instances $J, \widetilde{J} \in \mathcal{I}'$, there is one coupling $(\boldsymbol{\pi}', \widetilde{\boldsymbol{\pi}}')$ of the two output distributions such that

$$\mathbf{E}[d_{\mathrm{H}}(\boldsymbol{\pi}'_U, \widetilde{\boldsymbol{\pi}}'_U)] \leq x_U, \qquad \mathbf{E}[d_{\mathrm{H}}(\boldsymbol{\pi}'_V, \widetilde{\boldsymbol{\pi}}'_V)] \leq x_V.$$

For all neighboring instances $I, \widetilde{I} \in \mathcal{I}$, there is then a coupling $(\boldsymbol{\pi}, \widetilde{\boldsymbol{\pi}})$ of the output distributions of $A(I)$ and $A(\widetilde{I})$ satisfying

$$\mathbf{E}[d_{\mathrm{H}}(\boldsymbol{\pi}_U, \widetilde{\boldsymbol{\pi}}_U)] \leq d x_U, \qquad \mathbf{E}[d_{\mathrm{H}}(\boldsymbol{\pi}_V, \widetilde{\boldsymbol{\pi}}_V)] \leq \frac{d}{\delta_V} x_V.$$

Proof. If (π_U, π_V) satisfies I, then the assignment $\pi'_U := \pi_U$ and $\pi'_V(v,i) := \pi_V(v)$ satisfies $\mathcal{T}_{\mathrm{DR}}(I;d)$.

We next prove (2). Fix an assignment π' to $\mathcal{T}_{\mathrm{DR}}(I;d)$. For each $v \in V$, write $e_G(v,j) = (u_{v,j}, v)$ for $j \in [D_v]$, and, for $\sigma \in \Sigma_V$, set

$$\begin{aligned} X_{v,\sigma} &:= \{j \in [D_v] : P_{u_{v,j}}(\pi'_U(u_{v,j})) = 1 \text{ and } f_{e_G(v,j)}(\pi'_U(u_{v,j})) = \sigma\}, \\ Y_{v,\sigma} &:= \{i \in [D_v] : \pi'_V(v,i) = \sigma\}, \\ \alpha_{v,\sigma} &:= |X_{v,\sigma}|/D_v, \qquad\qquad \beta_{v,\sigma} := |Y_{v,\sigma}|/D_v. \end{aligned}$$

Let M_v be the normalized adjacency matrix of H_v. The fraction of transformed edges in the cloud of v that are satisfied is

$$s_v := \frac{1}{D_v} \sum_{\sigma \in \Sigma_V} \mathbf{1}_{Y_{v,\sigma}}^{\mathsf{T}} M_v \mathbf{1}_{X_{v,\sigma}},$$

whereas the expected fraction of the original incidences at v satisfied after recovery is

$$r_v := \sum_{\sigma \in \Sigma_V} \alpha_{v,\sigma} \beta_{v,\sigma}.$$

To compare s_v and r_v, center the two indicator vectors at their means. Both centered vectors lie in $\mathbf{1}^{\perp}$, so the expander norm bound gives, for each $\sigma \in \Sigma_V$,

$$\begin{aligned} &\frac{1}{D_v} \mathbf{1}_{Y_{v,\sigma}}^{\mathsf{T}} M_v \mathbf{1}_{X_{v,\sigma}} - \alpha_{v,\sigma}\beta_{v,\sigma} \\ &\quad = \frac{1}{D_v} (\mathbf{1}_{Y_{v,\sigma}} - \beta_{v,\sigma}\mathbf{1})^{\mathsf{T}} M_v (\mathbf{1}_{X_{v,\sigma}} - \alpha_{v,\sigma}\mathbf{1}) \\ &\quad \le \lambda \sqrt{\alpha_{v,\sigma}\beta_{v,\sigma}}. \end{aligned}$$

Summing over σ and applying Cauchy–Schwarz gives

$$s_v - r_v \le \lambda \sum_{\sigma} \sqrt{\alpha_{v,\sigma}\beta_{v,\sigma}} \le \lambda \sqrt{\sum_{\sigma} \alpha_{v,\sigma}} \sqrt{\sum_{\sigma} \beta_{v,\sigma}} \le \lambda,$$

since $\sum_\sigma \alpha_{v,\sigma} \le 1$ and $\sum_\sigma \beta_{v,\sigma} = 1$. Averaging with weights $D_v/|E|$ gives (2). This calculation uses the adjacency matrix with multiplicity, so it also covers loops and parallel edges under the convention in Definition 4.1. Averaging over the output distribution of $A'(\mathcal{T}_{\mathrm{DR}}(I;d))$ gives $\mathbf{E}[\mathsf{val}_I(A(I))] \ge \varepsilon$ for every satisfiable $I \in \mathcal{I}$.

Fix neighboring instances $I, \widetilde{I} \in \mathcal{I}$. A projection swap changes exactly the d copied projections, while a left-predicate swap changes only that predicate. Hence the transformed instances are joined in $\mathcal{I}'$ by a swap path

$$\mathcal{T}_{\mathrm{DR}}(I;d) = J_0, J_1, \ldots, J_m = \mathcal{T}_{\mathrm{DR}}(\widetilde{I};d) \qquad \text{with } m \le d.$$

For each consecutive pair, choose the coupling in the hypothesis. Repeatedly glue these couplings. The coordinatewise triangle inequalities then give an endpoint coupling $(\boldsymbol{\pi}', \widetilde{\boldsymbol{\pi}}')$ satisfying

$$\mathbf{E}[d_{\mathrm{H}}(\boldsymbol{\pi}'_U, \widetilde{\boldsymbol{\pi}}'_U)] \le m x_U, \qquad \mathbf{E}[d_{\mathrm{H}}(\boldsymbol{\pi}'_V, \widetilde{\boldsymbol{\pi}}'_V)] \le m x_V.$$

Conditioned on this endpoint pair, couple the recoveries by using the same uniform index $i_v \in [D_v]$ at every v. The left labels are copied, and

$$\Pr\big[\boldsymbol{\pi}_V(v) \ne \widetilde{\boldsymbol{\pi}}_V(v) \mid \boldsymbol{\pi}', \widetilde{\boldsymbol{\pi}}'\big] \le \frac{1}{\delta_V} \big|\{i \in [D_v] : \boldsymbol{\pi}'_V(v,i) \ne \widetilde{\boldsymbol{\pi}}'_V(v,i)\}\big|.$$

Summing over v and averaging proves the two bounds in the statement. The same coupling also shows that the pointwise recovery is 1-Lipschitz under total Hamming distance. For a fixed target labeling π', the distribution of $R_{\mathrm{DR},I}(\pi')$ is independent of the projections and predicates of I. □

5 Alphabet and Degree Reduction

This section gives a sensitivity-preserving right alphabet reduction technique for Label Cover.

After recalling the required coding-theoretic facts, we define the transformation $\mathcal{T}_{\mathrm{AR}}$ and its recovery maps $R_{\mathrm{AR},I}$. We then establish bounds on size, degrees, alphabets, value recovery, and the sensitivity-transfer parameters $C_{\mathcal{T}}, C_{\mathcal{R}}, D_{\mathcal{R}}$ from Definition 3.1, for use in Section 11. Alphabet reduction leaves each left predicate P_u unchanged. We omit $\mathcal{P}$ from the instance notation.

5.1 Alphabet Reduction

A mapping $\mathcal{C} : \Sigma \to \hat{\Sigma}^k$ is called a *code* with minimum relative distance $1 - \delta$ if for every $a \neq b \in \Sigma$ the strings $\mathcal{C}(a)$ and $\mathcal{C}(b)$ differ in at least $(1 - \delta)k$ coordinates.

Proposition 5.1 (Code with polynomial-size encoding and membership circuits)**.** For every $0 < \delta < 1$ and alphabet Σ, there exists a code $\mathcal{C} : \Sigma \to \hat{\Sigma}^k$ with relative distance $1 - \delta$ where $|\hat{\Sigma}| = O(\delta^{-2})$ and $k = O(\delta^{-2} \log |\Sigma|)$. The code may be chosen explicitly so that encoding, testing whether a word is a codeword, and decoding valid codewords have circuits of size $\mathrm{poly}(k, \log |\Sigma|, \delta^{-1})$.

Proof. Remark 5.4 of [15] obtains the stated alphabet, length, and distance by concatenating a rate-optimal outer code with a Reed–Solomon inner code. To obtain the circuit bounds asserted here, choose both component codes to be linear and put their generator matrices in systematic form, so that each component message appears unchanged in designated coordinates of its codeword. Identify Σ with the first $|\Sigma|$ strings in the resulting message space.

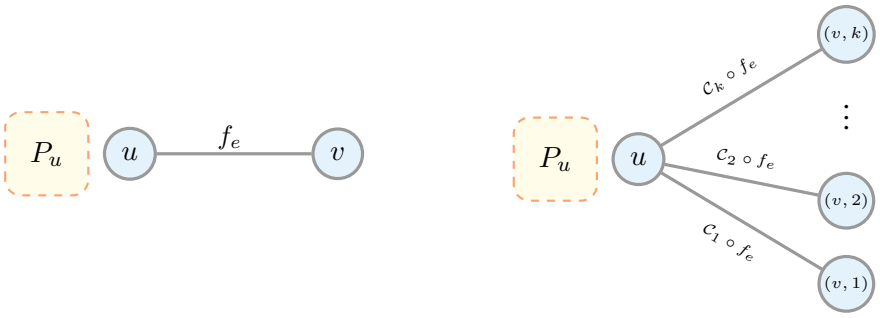
To encode, multiply by the component generator matrices. To test whether a word is a codeword, first check the linear equations defining the two component codes, given by their parity-check matrices, and then check that the systematic message represents an element of Σ. For a valid codeword, decoding reads the systematic message and inverts this identification. Standard Boolean circuits for finite-field arithmetic, matrix–vector multiplication, and comparison have size polynomial in the block length, message length, and field description length. Their composition has size $\mathrm{poly}(k, \log |\Sigma|, \delta^{-1})$. □

For each alphabet Σ and distance parameter $0 < \delta < 1$, fix an explicit code from Proposition 5.1, together with its encoding and decoding circuits.

Suppose $\mathcal{C} : \Sigma_V \to \hat{\Sigma}_V^k$ is a code with (relative) distance $1 - \eta^3$ for some $0 < \eta < 1$ satisfying the guarantees of Proposition 5.1 for $\delta = \eta^3$. We now describe our alphabet reduction transformation, which we denote by $\mathcal{T}_{\mathrm{AR}}$. The transformation maps Label Cover instances $I = (U, V, E, \Sigma_U, \Sigma_V, F)$ to Label Cover instances $I' = (U, V \times [k], E', \Sigma_U, \hat{\Sigma}_V, F')$. For every edge incidence $e = (u, v) \in E$ and every $i \in [k]$, the edge set E' contains a distinct incidence (e, i) from u to (v, i), with projection

$$f'_{(e,i)}(a) := \mathcal{C}(f_e(a))_i.$$

Writing $\mathcal{C}_i(b) := \mathcal{C}(b)_i$ for the ith coordinate map, this projection is $\mathcal{C}_i \circ f_e$.



(a) Edge incidence $e = (u, v)$ before $\mathcal{T}_{AR}$ (b) The incidences (e, i) after $\mathcal{T}_{AR}$

Figure 3: Alphabet reduction $\mathcal{T}_{AR}$ on one edge incidence.

For an instance $I = (U, V, E, \Sigma_U, \Sigma_V, F)$ we write $\bar{d}_U := \frac{|E|}{|U|}$, $\bar{d}_V := \frac{|E|}{|V|}$, $\Delta_V := \max_{v \in V} \deg(v)$, $\Delta_U := \max_{u \in U} \deg(u)$, and $\delta_V := \min_{v \in V} \deg(v)$ for the average left and right degrees, the maximum right and left degrees, and the minimum right degree, respectively.

For recovery, assume that the source instance has no isolated right vertices, so that $\delta_V \geq 1$. The labels on the k reduced copies of a source right vertex need not form a codeword. We recover original right labels from the projections of the left labels; the code distance will relate agreement with the reduced right labels to the value after this recovery. In particular, recovery itself reads only the left labels. For a source instance I, write R_I for $R_{\mathrm{AR},I}$ within this subsection. Given an assignment π' for $I' = \mathcal{T}_{\mathrm{AR}}(I)$, the recovery procedure R_I constructs an assignment $\boldsymbol{\pi}$ for I as follows. For each $v \in V$ and $b \in \Sigma_V$, define $p_v(b) := \Pr_{e=(u,v) \text{ incident to } v}[f_e(\pi'(u)) = b]$. Here the probability is over a uniformly random edge incident to v (counting multiplicity). We include projections of all left labels in p_v, even those failing their predicates. This makes recovery independent of the predicates; the value analysis below counts only projected labels $f_e(\pi'(u))$ with $P_u(\pi'(u)) = 1$. We set

$$\boldsymbol{\pi}(u) = \pi'(u) \qquad \text{for every } u \in U.$$

For each $v \in V$, we then sample $\boldsymbol{\pi}(v)$ independently from the distribution p_v.

The recovery map R_I depends on the source instance I through the projections f_e used to define p_v. To bound $C_{\mathcal{R}}$, we vary the input assignment π' with I held fixed; to bound $D_{\mathcal{R}}$, we change I by one swap with π' held fixed.

Lemma 5.2 (Sensitivity transfer through alphabet reduction)**.** Let $\mathcal{I}$ be a swap-closed family of Label Cover instances $I = (U, V, E, \Sigma_U, \Sigma_V, F)$ with average left degree $\bar{d}_U$, average right degree $\bar{d}_V$, maximum left degree Δ_U, and minimum right degree $\delta_V \geq 1$. Fix $\eta \in (0, 1/4)$ and choose a code $\mathcal{C} : \Sigma_V \to \hat{\Sigma}_V^k$ from Proposition 5.1 with relative distance $1 - \eta^3$. Using this code, let $\mathcal{I}' = \mathcal{T}_{\mathrm{AR}}(\mathcal{I})$. Its right alphabet satisfies $|\hat{\Sigma}_V| = O(\eta^{-6})$, and the block length satisfies $k = O(\eta^{-6} \log |\Sigma_V|)$. Then the following hold.

1. If $I \in \mathcal{I}$ is satisfiable, then $\mathcal{T}_{\mathrm{AR}}(I)$ is satisfiable.
2. For every $I \in \mathcal{I}$ and every possibly random labeling $\boldsymbol{\pi}'$ of $I' = \mathcal{T}_{\mathrm{AR}}(I)$,
$$\mathbf{E}[\mathsf{val}_I(R_I(\boldsymbol{\pi}'))] \geq \left(\mathbf{E}[\mathsf{val}_{I'}(\boldsymbol{\pi}')]\right)^2 - \eta^3.$$
3. The transformation $\mathcal{T}_{\mathrm{AR}}$ and the induced recovery maps on distributions $\mathcal{R}_I(\mu) := (R_I)_{\#}\mu$ have sensitivity-transfer parameters
$$C_{\mathcal{T}} = k, \qquad C_{\mathcal{R}} = 1 + \frac{\Delta_U}{\delta_V}, \qquad D_{\mathcal{R}} = \frac{1}{\delta_V}.$$

Proof. The alphabet and code-length bounds follow from Proposition 5.1 with $\delta = \eta^3$. A satisfying labeling π extends to the reduced instance by $\pi'(u) = \pi(u)$ and $\pi'(v, i) = \mathcal{C}(\pi(v))_i$.

For Item 2, fix a labeling π' of $I' = \mathcal{T}_{\mathrm{AR}}(I)$. For each $v \in V$, define the subdistribution $x_v : \Sigma_V \to [0, 1]$ by

$$x_v(b) := \Pr_{e=(u,v) \text{ incident to } v} [P_u(\pi'(u)) = 1,\ f_e(\pi'(u)) = b].$$

Thus $x_v(b) \le p_v(b)$ and $\sum_b x_v(b) \le 1$. For a uniform code coordinate $i \in [k]$, put

$$Z_v(i) := \sum_b x_v(b) \mathbf{1}[\pi'(v, i) = \mathcal{C}(b)_i], \qquad \alpha_v := \mathop{\mathbf{E}}_i Z_v(i), \qquad \nu_v := \sum_b x_v(b) p_v(b).$$

The reduced value and expected recovered value are the averages of α_v and ν_v, respectively, with weights $\deg(v)/|E|$. Distinct codewords agree on at most an η^3 fraction of coordinates, so

$$\alpha_v^2 \le \mathop{\mathbf{E}}_i Z_v(i)^2 \le \sum_b x_v(b)^2 + \eta^3 \sum_{b \ne b'} x_v(b) x_v(b') \le \nu_v + \eta^3.$$

Averaging over v and the input labeling, Jensen's inequality proves Item 2.

A projection swap changes at most k reduced projections, and a predicate swap changes one reduced predicate, so $C_{\mathcal{T}} = k$ is valid. For $C_{\mathcal{R}}$ and $D_{\mathcal{R}}$, implement recovery by sampling a uniform incident edge at each v and returning its projected left label. Couple two recoveries using the same sampled incidences. For instances $I, \widetilde{I}$ with projections $f_e, \widetilde{f}_e$ and labelings $\pi', \widetilde{\pi}'$ of their reduced instances, this gives

$$\mathsf{EMD}(R_I(\pi'), R_{\widetilde{I}}(\widetilde{\pi}')) \le d_{\mathrm{H}}(\pi'_U, \widetilde{\pi}'_U) + \sum_{e=(u,v) \in E} \frac{\mathbf{1}[f_e(\pi'_U(u)) \ne \widetilde{f}_e(\widetilde{\pi}'_U(u))]}{\deg(v)}.$$

For $I = \widetilde{I}$, the sum is at most $(\Delta_U/\delta_V) d_{\mathrm{H}}(\pi'_U, \widetilde{\pi}'_U)$. Together with the first term, this gives the pointwise Lipschitz constant $1 + \Delta_U/\delta_V$. For a fixed input labeling, one projection swap contributes at most $1/\delta_V$, whereas a predicate swap leaves the sampling rule unchanged. By Lemma 3.3, these two pointwise bounds give the claimed $C_{\mathcal{R}}$ and $D_{\mathcal{R}}$ for the induced maps $\mathcal{R}_I$. □

5.2 Combined Alphabet-and-Degree Reduction

We now combine alphabet reduction with right-degree reduction. Every family to which this combined map is applied has a common graph and alphabets and no isolated right vertices. The expanders of every size from Proposition 4.2 allow us to apply degree reduction regardless of the minimum right degree. For the recovery thresholds recorded below, we henceforth assume $0 < \varepsilon < 1/16$, so that $3\sqrt{\varepsilon}$ and $4\sqrt{\varepsilon}$ both lie in $(0, 1)$. Let $I = (U, V, E, \Sigma_U, \Sigma_V, \mathcal{P}, F)$ be a left-predicate Label Cover instance with $|U| = n$, $|V| = m$, average left degree $\bar{d}_U$, average right degree $\bar{d}_V$, maximum right degree Δ_V, maximum left degree Δ_U, and minimum right degree δ_V. Table 2 tracks how these parameters change when we first apply the alphabet-reduction map induced by a code $\mathcal{C} : \Sigma_V \to \hat{\Sigma}_V^k$ of relative distance $1 - \eta^3$, and then run the degree-reduction step based on $[D_v, d, \lambda]$-expanders tailored to the right degree D_v of each vertex $v \in V$. The last row records required expected values and is read from right to left: attaining a threshold after a reduction allows recovery to attain at least the preceding threshold before that reduction.

We choose $\mathcal{C}$ from Proposition 5.1 with $\delta = \eta^3$ and $\eta = \varepsilon/4$. The resulting right alphabet $\hat{\Sigma}_V$ has size $O(\varepsilon^{-6})$, and the block length is $k = O(\varepsilon^{-6} \log |\Sigma_V|)$. Since $(3\sqrt{\varepsilon})^2 - (\varepsilon/4)^3 \ge \varepsilon$, the value-recovery bound in Lemma 5.2 gives recovery threshold $3\sqrt{\varepsilon}$.

Table 2: Parameters after the alphabet-then-degree reduction with $0 < \varepsilon < 1/16$, $\eta = \varepsilon/4$, $k = O(\varepsilon^{-6}\log|\Sigma_V|)$, and $d = \Theta(\varepsilon^{-1})$ chosen so that $\lambda_d \le \sqrt{\varepsilon}$. Degrees are averages unless indicated otherwise.

Parameter	I	alphabet $\to \hat{\Sigma}_V$	degree $\to d$
# left vertices	n	n	n
# right vertices	m	$O(\varepsilon^{-6} m \log\|\Sigma_V\|)$	$O(\varepsilon^{-6} m\bar{d}_V \log\|\Sigma_V\|)$
Average left degree	$\bar{d}_U$	$O(\varepsilon^{-6}\bar{d}_U \log\|\Sigma_V\|)$	$O(\varepsilon^{-7}\bar{d}_U \log\|\Sigma_V\|)$
Average right degree	$\bar{d}_V$	$\bar{d}_V$	$\Theta(\varepsilon^{-1})$
Left alphabet	Σ_U	Σ_U	Σ_U
Right alphabet	Σ_V	$\hat{\Sigma}_V$ ($\|\hat{\Sigma}_V\| = O(\varepsilon^{-6})$)	$\hat{\Sigma}_V$
Recovery threshold	ε	$3\sqrt{\varepsilon}$	$4\sqrt{\varepsilon}$

For $0 < \varepsilon < 1/16$ and an absolute constant c_{AR} to be chosen below, set $d_\varepsilon := 2\lfloor c_{\mathrm{AR}}/(2\varepsilon)\rfloor$. Define $\mathrm{red}_\varepsilon(I)$ by first applying alphabet reduction with $\eta = \varepsilon/4$, transporting the left predicates unchanged, and then applying right-degree reduction with target degree d_ε. Write $R_{\varepsilon,I}$ for the recovery obtained by composing the two recovery maps in the reverse order.

We use the following standing convention whenever the combined reduction is applied to a family with a common graph and alphabets. Fix one code for the common right alphabet. For each right degree D attained in the alphabet-reduced graph, use at every vertex of degree D the same expander H_D of target degree d_ε, returned by Proposition 4.2. Also fix the edge-incidence orderings across the family. Thus the degree-reduction recovery depends only on its input assignment and these fixed data; the combined recovery depends on the source instance only through the alphabet-reduction recovery.

Lemma 5.3 (Alphabet-then-degree reduction)**.** There are sufficiently large absolute constants $c_{\mathrm{AR}}, C_{\mathrm{red}} > 0$ for which the following holds. Let $0 < \varepsilon < 1/16$ and $\mathcal{I}$ be a swap-closed family of left-predicate Label Cover instances $I = (U, V, E, \Sigma_U, \Sigma_V, \mathcal{P}, F)$ on a common graph and common alphabets, with no isolated right vertices, $|U| = n$, $|V| = m$, average left degree $\bar{d}_U$, average right degree $\bar{d}_V$, maximum right degree Δ_V, maximum left degree Δ_U, and minimum right degree δ_V. Then every instance $\mathrm{red}_\varepsilon(I) = (U', V', E', \Sigma'_U, \Sigma'_V, \mathcal{P}', F')$ satisfies

$$|U'| = n, \qquad |V'| = O(\varepsilon^{-6} m\bar{d}_V \log|\Sigma_V|), \qquad \bar{d}_{U'} = O(\bar{d}_U \varepsilon^{-7}\log|\Sigma_V|),$$

has right degree $d_\varepsilon = \Theta(\varepsilon^{-1})$ and right alphabet $\hat{\Sigma}_V$ of size $O(\varepsilon^{-6})$. The recovery $R_{\varepsilon,I}$ and the transformation satisfy the following guarantees.

1. If I is satisfiable, then $\mathrm{red}_\varepsilon(I)$ is satisfiable.
2. If a possibly random assignment $\boldsymbol{\pi}'$ for $\mathrm{red}_\varepsilon(I)$ satisfies $\mathbf{E}[\mathsf{val}_{\mathrm{red}_\varepsilon(I)}(\boldsymbol{\pi}')] \ge 4\sqrt{\varepsilon}$, then
$$\mathbf{E}\big[\mathsf{val}_I\big(R_{\varepsilon,I}(\boldsymbol{\pi}')\big)\big] \ge \varepsilon.$$
3. For all $I, \tilde{I} \in \mathcal{I}$, we have
$$\mathsf{SwapDist}\big(\mathrm{red}_\varepsilon(I), \mathrm{red}_\varepsilon(\tilde{I})\big) \le C_{\mathrm{red}}\varepsilon^{-7}\log|\Sigma_V| \cdot \mathsf{SwapDist}(I, \tilde{I}).$$
4. For every $I \in \mathcal{I}$ and every two assignments $\pi', \tilde{\pi}'$ on $\mathrm{red}_\varepsilon(I)$, we have
$$\mathsf{EMD}\big(R_{\varepsilon,I}(\pi'), R_{\varepsilon,I}(\tilde{\pi}')\big) \le \left(1 + \frac{\Delta_U}{\delta_V}\right) d_{\mathrm{H}}(\pi', \tilde{\pi}').$$

5. If $I, \tilde{I} \in \mathcal{I}$ differ by one swap, then for every assignment π' on the common domain of $\mathrm{red}_\varepsilon(I)$ and $\mathrm{red}_\varepsilon(\tilde{I})$, we have

$$\mathsf{EMD}\big(R_{\varepsilon,I}(\pi'), R_{\varepsilon,\tilde{I}}(\pi')\big) \le \frac{1}{\delta_V}.$$

Proof. Set $\eta = \varepsilon/4$ and choose c_{AR} large enough that $d = d_\varepsilon = \Theta(\varepsilon^{-1}) \ge 6$ and $\lambda_d \le \sqrt{\varepsilon}$ in Proposition 4.2. The alphabet-reduction code has length $k = O(\varepsilon^{-6} \log |\Sigma_V|)$ and alphabet size $O(\varepsilon^{-6})$. The two reductions give

$$|U'| = n, \qquad |V'| = k|E| = km\bar{d}_V, \qquad \bar{d}_{U'} = dk\bar{d}_U,$$

and right degree d, proving the structural bounds. Both reductions preserve satisfiability. Their value-recovery guarantees in Lemmas 5.2 and 4.3 give $4\sqrt{\varepsilon} \to 3\sqrt{\varepsilon} \to \varepsilon$ in expectation. A source swap causes at most kd target swaps, proving Item 3.

The degree-reduction recovery copies the left labels, and the alphabet-reduction recovery reads only these labels. Thus the composite recovery has the same distribution as applying the alphabet-reduction sampling rule directly to the final left labels. Applying the $C_{\mathcal{R}}$ and $D_{\mathcal{R}}$ bounds from Lemma 5.2 to point masses proves Items 4 and 5. □

For a family $\mathcal{I}$, write

$$\mathrm{red}_\varepsilon(\mathcal{I}) := \{\mathrm{red}_\varepsilon(I) : I \in \mathcal{I}\}$$

for its pointwise image before swap closure. Using the same code, expanders, and incidence orderings along a source swap path gives

$$\mathrm{red}_\varepsilon(\mathsf{SwapClo}(\mathcal{I})) \subseteq \mathsf{SwapClo}(\mathrm{red}_\varepsilon(\mathcal{I})).$$

These choices and the common input graph determine one common output graph and common alphabets.

For later composition rounds in the Dinur-Harsha framework, we also need circuits that decide whether the right labels around a left vertex u belong to $\mathcal{N}_I(u)$: whether they are the projections of one left label satisfying its predicate, as in Definition 2.4. The next lemma bounds the description lengths of these circuits after reduction. The bound depends nearly linearly on the existing description length S, so repeated reductions do not raise S to a new constant power at each round.

Lemma 5.4 (Local circuits and paths of predicate swaps after reduction)**.** There are absolute constants $C, c_{\mathrm{red}} \ge 1$ with the following property. Suppose I has no isolated vertices, local description complexity at most S, left degree at most Δ_U, and right alphabet Σ_V. For $0 < \varepsilon < 1/16$, set

$$T := \Delta_U\big(\varepsilon^{-1} \log(2|\Sigma_V|)\big)^{c_{\mathrm{red}}}.$$

Then $\mathrm{red}_\varepsilon(I)$ has local description complexity at most

$$C(S+T)\log^{c_{\mathrm{red}}}(2+S+T). \tag{3}$$

It has no isolated vertices. If a family has common graph, alphabets, and edge projections, the reduced family also has common graph, alphabets, and edge projections. In particular, one predicate swap changes exactly one transported predicate and leaves the recovery map unchanged:

$$C^{\mathrm{pred}}_{\mathcal{T},\mathrm{red}} = 1, \qquad D^{\mathrm{pred}}_{\mathcal{R},\mathrm{red}} = 0, \qquad C_{\mathcal{R},\mathrm{red}} \le 1 + \Delta_U/\delta_V. \tag{4}$$

The bounds (3) and (4) hold along a path of predicate swaps whenever every instance on the path has local description complexity at most S.

Proof. Let $k = O(\varepsilon^{-6} \log(2|\Sigma_V|))$ be the code length and $d = \Theta(\varepsilon^{-1})$ the degree parameter. At one left vertex, the reduced neighborhood has one group of dk input symbols for each original incidence. First check equality among the d routed copies of each code coordinate. For the resulting length-k word, use the circuit for codeword membership and decode its unique original symbol. Do this for every original incidence, and apply the original circuit for membership in $\mathcal{N}_I(u)$ *once* to the decoded tuple, where u is the left vertex under consideration.

By Proposition 5.1, the work per original incidence is polynomial in k, $\log |\Sigma_V|$, and ε^{-1}. Choose c_{red} sufficiently large that the number of added gates and the total input and output lengths of the added circuits are at most T. The combined circuit has $O(S + T)$ gates and terminals. Encoding its wiring gives a total description length of $O((S + T) \log^{c_{\mathrm{red}}}(2 + S + T))$, proving (3).

Each left degree is multiplied by dk, and every reduced right vertex has degree d, so no vertex becomes isolated. Fixing the code, cloud expanders, and incidence orderings fixes the graph and projections throughout the family. The left predicate is transported unchanged. The alphabet-reduction recovery samples from the empirical distributions p_v of projected labels, which do not depend on the predicates; the right-degree-reduction recovery also does not inspect them. Thus a predicate-only swap leaves the composite recovery exactly unchanged. The Lipschitz bound is the one in Lemma 5.3. Each vertexwise circuit construction uses only the corresponding original circuit, so the same reasoning applies to every intermediate instance on a path of predicate swaps. □

Part II
The IKW Route to Low-Soundness Label Cover

In this part we prove the sensitivity lower bound for Label Cover in Theorem 6.1 using the direct-product analysis of Impagliazzo, Kabanets, and Wigderson [24], along with new recovery and degree reduction procedures.

6 The IKW Direct-Product Construction

We amplify a constraint system by using assignments to tuples of k constraints as labels and imposing consistency constraints on overlaps. Increasing k lets us obtain sensitivity lower bounds for algorithms with smaller value guarantees. Here n denotes the total number of vertices in the final Label Cover instance.

The direct-product analysis of Impagliazzo, Kabanets, and Wigderson [24] shows that sufficiently frequent agreement between local assignments yields an assignment to the original variables. We construct a randomized recovery procedure based on their analysis and combine it with degree reduction in Section 7 to prove the sensitivity lower bound of Theorem 6.1.

In the theorem below, N counts variables of the original 2-CSP.

Theorem 6.1 (Formal version of Theorem 1.1)**.** There exist constants $C, a, b, \eta > 0$ and $k_0 \in \mathbb{N}$ with the following property. Let $k = k(N)$ be any integer-valued function such that $k(N) \geq k_0$ and $k(N) = o(\log N)$. For all sufficiently large N, there is a swap-closed family of left-predicate Label Cover instances whose members have a common number n of vertices satisfying

$$N^{ak} \leq n \leq N^{bk}.$$

Every algorithm defined on this family that, on each satisfiable member, outputs a possibly random assignment of expected value at least $\exp\left(-C\sqrt{k}\right)$ has sensitivity at least $n^{\eta/k}$. Moreover, the instances are biregular, and both alphabet sizes and the maximum left and right degrees are bounded by $2^{O(k)}$.

6.1 The Tuple Label Cover Construction

We first give a biregular variant of the direct-product construction of Impagliazzo, Kabanets, and Wigderson (IKW) [24, Section 6.3]. Biregularity ensures that a uniformly sampled edge has a uniformly distributed endpoint on each side. We use this uniformity to relate constraint satisfaction to averages over right vertices and to bound the effect of label changes on the randomized recovery in Section 7. Each left vertex is an ordered tuple of edges of the original constraint graph. Each incident Label Cover edge specifies which endpoint coordinates are selected and their order in the right vertex.

Let Φ be a 2-CSP on a fixed d-regular multigraph $G = ([N], E)$ with alphabet Σ; self-loops are allowed. Degrees count endpoint incidences, so $M := |E| = dN/2$. Fix an ordering of the two endpoint incidences of each edge e. Write $\mathrm{end}_c(e)$ for the vertex of incidence (e, c), where $c \in \{0, 1\}$. For a self-loop $e = (v, v)$, the two incidences are distinct, although both have vertex v.

For $k \geq 4$, put

$$k' := \lfloor \sqrt{k} \rfloor, \qquad r := k - k', \qquad p_k := (k)_{k'} = \frac{k!}{(k-k')!}.$$

We define the encoding map $\mathcal{T}^k_{\mathrm{IKW}}$ from 2-CSPs to left-predicate Label Cover instances as follows. The instance $\mathcal{T}^k_{\mathrm{IKW}}(\Phi)$ has vertex sets and alphabets

$$U := E^k, \qquad V := [N]^{k'}, \qquad \Sigma_U := \Sigma^{2k}, \qquad \Sigma_V := \Sigma^{k'}.$$

For a left vertex $S = (e_1, \ldots, e_k)$, index the coordinates of a label $y \in \Sigma_U$ by $(h, c) \in [k] \times \{0, 1\}$. The coordinate $y_{(h,c)}$ specifies the symbol assigned to $\mathrm{end}_c(e_h)$. The predicate P_S requires that all coordinates corresponding to the same vertex of G receive the same symbol and that all k constraints of Φ indexed by $e_1, \ldots, e_k$ are satisfied. A right label assigns a symbol to each of the k' coordinates of its vertex tuple.

For every $S \in U$, injection $\iota : [k'] \hookrightarrow [k]$, and $b \in \{0, 1\}^{k'}$, add an edge $\omega = (S, \iota, b)$ to $\mathcal{T}^k_{\mathrm{IKW}}(\Phi)$, with right endpoint

$$A(\omega) := \left(\mathrm{end}_{b_i}(e_{\iota(i)})\right)_{i \in [k']}.$$

The projection f_ω selects the corresponding coordinates of the left label:

$$f_\omega(y) = \left(y_{(\iota(i), b_i)}\right)_{i \in [k']}.$$

A satisfying assignment to Φ induces a satisfying labeling of $\mathcal{T}^k_{\mathrm{IKW}}(\Phi)$ by restricting that assignment to each tuple.

Let D_U and D_V denote the left and right degrees of $\mathcal{T}^k_{\mathrm{IKW}}(\Phi)$, respectively, and let E_k be its edge set. We have

$$\begin{gathered} |U| = M^k, \qquad |V| = N^{k'}, \\ D_U = p_k 2^{k'}, \qquad D_V = p_k d^{k'} M^r, \qquad |E_k| = D_U |U| = D_V |V|. \end{gathered} \tag{5}$$

For each right vertex A, there are p_k choices of ι, $d^{k'}$ choices of endpoint incidences at its coordinates, and M^r choices of edges in the remaining positions. Thus every right vertex has degree $D_V = p_k d^{k'} M^r$.

We next bound the number of predicate swaps needed to transform the output instance when one constraint of Φ changes. Let Φ and $\widetilde{\Phi}$ differ only in the relation on an edge $e_\star$. Their images have the same graph and projections, and their left predicates can differ only at tuples containing $e_\star$. Hence

$$\mathsf{SwapDist}\big(\mathcal{T}^k_{\mathrm{IKW}}(\Phi), \mathcal{T}^k_{\mathrm{IKW}}(\widetilde{\Phi})\big) \le F_k := M^k - (M-1)^k \le \frac{k|U|}{M}. \tag{6}$$

This will give the bound $C_{\mathcal{T}} \le F_k$ for the full transformation, since the subsequent right-degree reduction preserves each left predicate.

6.2 Conditional Agreement on the Original Variables

Following the ideas of [24], we show that for many right vertices A there is a corresponding assignment to Φ that extends the label at A to all of $[N]$ and is close, on average, to the neighboring left labels whose projections agree with the label at A. This agreement will imply both a value guarantee and a recovery map with bounded sensitivity.

Let $N(A) := \{\omega \in E_k : A(\omega) = A\}$ be the set of edges incident to A in $\mathcal{T}^k_{\mathrm{IKW}}(\Phi)$. For a labeling $\pi = (\pi_U, \pi_V)$, define

$$\mathrm{Cons}_A := \{\omega = (S, \iota, b) \in N(A) : f_\omega(\pi_U(S)) = \pi_V(A)\}, \qquad \mu_A := \frac{|\mathrm{Cons}_A|}{|N(A)|}. \tag{7}$$

We call an edge $\omega \in N(A)$ consistent if $\omega \in \mathrm{Cons}_A$; thus consistency means that the projection of the left label equals the right label. This condition alone does not require the left label to satisfy P_S. For an assignment $g : [N] \to \Sigma$ to Φ and an edge $\omega = (S, \iota, b)$, let $\mathrm{err}(\omega, g)$ be the fraction of the $r = k - k'$ constraint positions of S not selected by ι at which the left label $\pi_U(S)$ disagrees with g:

$$\mathrm{err}(\omega, g) := \frac{1}{r} \sum_{h \notin \mathrm{im}\, \iota} \mathbf{1}[\pi_U(S)|_h \neq g|_{e_h}],$$

where $\pi_U(S)|_h$ is the pair of symbols at position h, and $g|_{e_h}$ is the pair of symbols assigned by g to the endpoints of e_h in the fixed order.

We first analyze labelings whose left labels all satisfy their predicates. When Φ is satisfiable, every P_S is satisfied by at least one label, so replacing a left label that violates P_S by one satisfying P_S cannot decrease value. We make this replacement before applying the next lemma in Section 7.2.

Lemma 6.2 (Conditional agreement for the tuple construction)**.** There are absolute constants $c_0, c_1, c_2 > 0$ and $C_{\mathrm{IKW}}, C_{\mathrm{agr}} > 0$ with the following property. Suppose $0 < \eta < 1/100$, $0 < \varepsilon < 1/4$, and

$$k \ge C_{\mathrm{IKW}} \eta^{-2} \log^2(2/\eta), \qquad k = o(\log N), \qquad \log(1/\varepsilon) \le c_0 \eta \sqrt{k}. \tag{8}$$

For all sufficiently large N, if $P_S(\pi_U(S)) = 1$ for every $S \in U$ and $\mathsf{val}_{\mathcal{T}^k_{\mathrm{IKW}}(\Phi)}(\pi) \ge \varepsilon$, then there is a set $\mathcal{G}_\pi \subseteq V$ of uniform measure at least $c_1\varepsilon$ such that every $A = (a_1, \dots, a_{k'}) \in \mathcal{G}_\pi$ has distinct coordinates, satisfies $\mu_A \ge c_2\varepsilon$, and admits an assignment $g_A : [N] \to \Sigma$ with

$$g_A(a_i) = \pi_V(A)_i \quad (i \in [k']), \qquad \mathop{\mathbf{E}}_{\omega \in \mathrm{Cons}_A} \mathrm{err}(\omega, g_A) \le C_{\mathrm{agr}} \eta. \tag{9}$$

Proof. We apply the IKW agreement analysis to random assignments on k'- and k-subsets of the original variable set $[N]$, obtained from the right and left labels, respectively. After conditioning on the sampled vertices being distinct, we identify the projection test with the test that compares these assignments on a uniform k'-subset of a uniform k-set. The IKW analysis then yields an assignment g_A close to the local assignments consistent with $\pi_V(A)$. Finally, we use the bound on disagreement at individual vertices to bound err, which compares pairs of endpoint symbols.

Sample a uniform edge $\omega = (S, \iota, b) \in E_k$, with $S = (e_1, \dots, e_k)$ and right endpoint $A = A(\omega) = (a_1, \dots, a_{k'})$. At each position $h = \iota(i)$ selected for comparison, set $c_h = b_i$; at every remaining position $h \notin \operatorname{im} \iota$, choose $c_h \in \{0, 1\}$ independently and uniformly. Thus we choose one end of each edge e_h, obtaining vertices

$$z_h := \mathrm{end}_{c_h}(e_h) \quad (h \in [k]), \qquad z_{\iota(i)} = \mathrm{end}_{b_i}(e_{\iota(i)}) = a_i.$$

The uniform choice of ω makes S, ι, b independent and uniform. For any ι, the pairs (e_h, c_h) are therefore independent uniform incidences, giving each vertex probability $d/(2M) = 1/N$. Hence $z_1, \dots, z_k$ are independent uniform vertices, also independent of ι.

To obtain a k-element vertex set, condition on the event Distinct that all z_h are distinct. We will later remove this conditioning using the bounds

$$\Pr[\neg\mathsf{Distinct}] \le \binom{k}{2}/N, \qquad \Pr[\neg\mathsf{Distinct} \mid A] = O(k^2/N) \tag{10}$$

for every A with distinct coordinates. Here conditioning on A means $A(\omega) = A$; the second bound counts collisions among the remaining vertices and with the vertices of A.

Conditional on Distinct, $W := \{z_1, \dots, z_k\}$ is a uniform k-set, and A is a uniform ordering of a uniform k'-subset of W. For each k'-subset $Q \subseteq [N]$, $C_1(Q)$ is a random assignment $Q \to \Sigma$: choose a uniform ordering $A = (a_1, \dots, a_{k'})$ of Q and assign $\pi_V(A)_i$ to a_i. We also write $\pi_V(A)$ for this assignment on $\underline{A} := \{a_1, \dots, a_{k'}\}$, retaining the order A in the analysis. For each k-subset $W \subseteq [N]$, $C_2(W)$ is a random assignment $W \to \Sigma$: independently choose an incidence uniformly among the d incidences at each vertex of W, uniformly order the resulting edge occurrences into S, and return $\pi_U(S)$ restricted to W. This restriction is well defined because P_S requires equal symbols at all occurrences of the same vertex.

Choose a uniform k-set W and a uniform k'-subset of W, and generate assignments using C_1 on that subset and C_2 on W with independent randomness. Agreement of these assignments on the subset corresponds to satisfaction of the projection constraint on ω, conditional on Distinct. Indeed, a uniform incidence conditioned on its vertex is uniform among the d incidences there, independently at each position. Conditional on W, A, the vertex order is also uniform: each order determines exactly one injection ι placing A in it. Thus $C_2(W)$ has the same distribution as the restricted left label and can be generated without using A.

Until we remove the conditioning on Distinct, probabilities over A use the uniform distribution on ordered tuples with distinct coordinates. Put $\mu^\circ_A := \Pr[\omega \in \mathrm{Cons}_A \mid A, \mathsf{Distinct}]$, and call A *good* if $\mu^\circ_A \ge \varepsilon/4$. Since every left label satisfies its predicate, an edge is satisfied if and only if it is consistent. By (10), the probability that ω is satisfied conditional on Distinct is at least $\varepsilon/2$ for large N. Thus $\mathbf{E}_A\, \mu^\circ_A \ge \varepsilon/2$, and good right vertices have measure at least $\varepsilon/4$.

We next compare two local assignments on an additional shared set H when both agree with $\pi_V(A)$ on $\underline{A}$.

For a fixed A, choose a uniform k'-set $H \subseteq [N] \setminus \underline{A}$ and, conditional on H, independent uniform $(k - 2k')$-sets $B_1, B_2 \subseteq [N] \setminus (\underline{A} \cup H)$. Each set $\underline{A} \cup H \cup B_i$ has size k; let $w_i = C_2(\underline{A} \cup H \cup B_i)$, using independent randomness for the two executions of C_2. For $\alpha, \gamma > 0$, call a good A *excellent* if

$$\Pr\big[w_1|_{\underline{A}} = w_2|_{\underline{A}} = \pi_V(A),\ d_{\mathrm{H}}(w_1|_H, w_2|_H) > \alpha k'\big] \le \gamma.$$

Here α is the threshold for the disagreement fraction on H, and γ (to be set) bounds the probability of this joint event over H, B_1, B_2 and the randomness of C_2.

We use two IKW bounds, viewing C_1, C_2 as the two oracles in [24, Section 6.3] and including their randomness as in [24, Section 3.5]. First, [24, Lemma 3.6] gives

$$\Pr_A[A \text{ is good but not excellent}] \le \frac{\exp(-\Omega(\alpha k'))}{\gamma}. \tag{11}$$

To prove this bound for ordered tuples A, first choose a uniform $2k'$-set $T = \underline{A} \cup H$, then B_1, B_2 outside T and the randomness of C_2, then split T uniformly into $\underline{A}, H$ and order $\underline{A}$. The assignments w_1, w_2 are already fixed. If they disagree on more than $\alpha k'$ vertices of T, the chance that $\underline{A}$ avoids all disagreements is at most $\exp(-\Omega(\alpha k'))$. The joint event defining excellence requires this avoidance; Markov's inequality gives the factor $1/\gamma$.

For the second bound, let $B \subseteq [N] \setminus \underline{A}$ be a uniform r-set, and let $w = C_2(\underline{A} \cup B)$. Define $g_A(x)$, for $x \notin \underline{A}$, as a most likely value of $w(x)$ conditional on $x \in B$ and $w|_{\underline{A}} = \pi_V(A)$. Break ties arbitrarily, use a fixed default symbol $\sigma_0 \in \Sigma$ when this event has probability zero, and set $g_A(a_i) = \pi_V(A)_i$. Thus g_A is the coordinatewise plurality of consistent local assignments, including the randomness of C_2. By [24, Lemma 3.8], extended to arbitrary alphabets by [24, Lemma 5.1], there are absolute constants $C_*, c_* > 0$ such that every excellent A satisfies

$$\Pr_{B,w}\left[\frac{d_{\mathrm{H}}(w|_B, g_A|_B)}{r} > C_*\alpha \,\middle|\, w|_{\underline{A}} = \pi_V(A)\right] \le C_*\frac{\gamma}{\varepsilon^2}, \tag{12}$$

provided $\alpha \ge C_*(k'/k)\log(4/\varepsilon)$ and $\gamma \le c_*\varepsilon^3$. Here $\mu_A^\circ \ge \varepsilon/4$, and the remaining sets have size $r = k - k'$; replacing r by k in the parameter bound only changes absolute constants since $r = \Theta(k)$.

Set $\alpha = \eta$ and $\gamma = c\min\{\eta, \varepsilon\}\varepsilon^2$ for a sufficiently small absolute $c > 0$. Then $\gamma \le c_*\varepsilon^3$, and for each excellent A the exceptional fraction of consistent local assignments in (12) is at most η. The lower bound on α follows from $k'/k \le 1/\sqrt{k}$ and (8), with c_0 sufficiently small. For the fraction of good but nonexcellent right vertices in (11) to be at most $\varepsilon/20$, it suffices that

$$\alpha k' \ge C\log\frac{20}{\varepsilon\gamma} = O\big(\log(1/\eta) + \log(1/\varepsilon)\big)$$

for a sufficiently large absolute C. Since $\alpha k' = \Theta(\eta\sqrt{k})$, this follows by taking C_{IKW} large and c_0 small in (8). Let $\mathcal{G}_\pi \subseteq V$ be the set of excellent right vertices with distinct coordinates. Its measure among distinct tuples is at least $\varepsilon/4 - \varepsilon/20 = \varepsilon/5$. Distinct tuples occupy a $1 - O(k'^2/N)$ fraction of V, so $\mathcal{G}_\pi$ has uniform measure $\Omega(\varepsilon)$ in V.

Fix $A \in \mathcal{G}_\pi$. For uniform B of size r outside $\underline{A}$ and $w = C_2(\underline{A} \cup B)$, the tail bound (12) gives

$$\mathop{\mathbf{E}}_{B,w}\left[\frac{d_{\mathrm{H}}(w|_B, g_A|_B)}{r} \,\middle|\, w|_{\underline{A}} = \pi_V(A)\right] \le C_*\eta + \eta = O(\eta). \tag{13}$$

Return to the experiment sampling ω and the extra endpoint choices. Let $Z_A \in [0,1]$ be the fraction of positions $h \notin \operatorname{im}\iota$ where the symbol at z_h in $\pi_U(S)$ differs from $g_A(z_h)$. Under Distinct, these vertices form $B = W \setminus \underline{A}$, and the restriction $w = \pi_U(S)|_W$ has the distribution generated by $C_2(W)$. Thus $Z_A = d_{\mathrm{H}}(w|_B, g_A|_B)/r$, so (13) bounds its expectation conditional on consistency and Distinct.

To restore uniform extra endpoint choices conditional on consistency, we now remove the conditioning on Distinct. By (10), $|\mu_A - \mu_A^\circ| = O(k^2/N)$, so $\mu_A \ge \varepsilon/8$ for large N. Conditioning on consistency

increases the collision probability by at most $1/\mu_A$; since $Z_A \le 1$, we obtain

$$\mathbf{E}[Z_A \mid \omega \in \mathrm{Cons}_A, A] \le O(\eta) + \frac{\Pr[\neg\mathsf{Distinct} \mid A]}{\mu_A} = O(\eta) + O\left(\frac{k^2}{N\varepsilon}\right) = O(\eta).$$

Indeed, (8) gives $k^2/(N\varepsilon\eta) \le O(k^{5/2}) \exp\Big(O(\sqrt{k})\Big)/N = o(1)$ because $k = o(\log N)$. The extra endpoint choices are independent of consistency. Whenever $\pi_U(S)|_h \ne g_A|_{e_h}$, a uniform endpoint detects a disagreement with probability at least $1/2$. Therefore $\mathbf{E}_{\omega\in\mathrm{Cons}_A} \mathrm{err}(\omega, g_A) \le 2\,\mathbf{E}[Z_A \mid \omega \in \mathrm{Cons}_A, A] = O(\eta)$. Together with $g_A(a_i) = \pi_V(A)_i$ and $\mu_A \ge \varepsilon/8$, this proves the lemma. □

7 Randomized Decoding and Sensitivity Analysis

Using the conditional agreement in Lemma 6.2, we construct a randomized recovery procedure with two guarantees. First, when the original 2-CSP is satisfiable, an algorithm attaining the required value on the constructed Label Cover instance yields an algorithm attaining high value on that 2-CSP. Second, we bound the recovery parameters $C_{\mathcal{R}}$ and $D_{\mathcal{R}}$ from Definition 3.1. For fixed Φ, the recovery map from Label Cover labeling distributions to assignment distributions is $C_{\mathcal{R}}$-Lipschitz with respect to EMD. With the supplied labeling distribution held fixed, $D_{\mathcal{R}}$ bounds the EMD between the recovered output distributions when one constraint relation of Φ changes. Recovery may use several independent samples from the supplied labeling distribution.

We apply right-degree reduction to $\mathcal{T}^k_{\mathrm{IKW}}(\Phi)$, replacing each right vertex by copies. We recover each original right label by choosing one copy uniformly. A change to one copy can affect recovery only when that copy is chosen. Combining this observation with bounds on how often the decoder reads each left label gives the sensitivity-transfer bound.

7.1 The Decoder and Its Value Guarantee

Recall that each edge $\omega = (S, \iota, b) \in E_k$ has a left endpoint $S = (e_1, \dots, e_k) \in U = E^k$. To recover the value at a variable x of Φ, the decoder chooses an incident edge e of G and samples ω together with a position $h \notin \mathrm{im}\,\iota$ such that $e_h = e$. Different positions h count separately even when they contain the same edge. For $A \in V$ and $e \in E$, the sampling set is

$$N(A; e) := \{(\omega, h) \in N(A) \times [k] : \omega = (S, \iota, b),\ h \notin \mathrm{im}\,\iota,\ e_h = e\}.$$

Use the same default symbol σ_0 as above. For parameters ε, η as in (8) and a labeling $\pi = (\pi_U, \pi_V)$, the decoder chooses a uniform $A = (a_1, \dots, a_{k'}) \in V$. For each $x \in [N]$, it independently does the following.

1. If $x = a_i$ for some i, return $\pi_V(A)_i$, using the least such index when necessary.
2. Otherwise choose a uniform endpoint incidence at x, with underlying edge e, and sample

   $$L := \lceil C_{\mathrm{dec}} \varepsilon^{-1} \log(2/\eta) \rceil$$

   independent uniform pairs $(\omega_\ell, h_\ell) \in N(A; e)$, with S_ℓ the left endpoint of ω_ℓ.
3. Choose an index ℓ_* uniformly among those with $\omega_{\ell_*} \in \mathrm{Cons}_A$, and return the symbol of $\pi_U(S_{\ell_*})$ in edge position h_{ℓ_*} at the endpoint chosen in Step 2. If there is no such index, return σ_0.

Here C_{dec} is a sufficiently large absolute constant. Denote the resulting assignment by $\boldsymbol{h}_A : [N] \to \Sigma$.

Each uniform sample from $N(A; e)$ in Step 2 can be generated as follows. Choose ι and one of the d endpoint incidences at each coordinate a_i of A independently and uniformly. The incidence at a_i specifies $(e_{\iota(i)}, b_i)$. Choose h uniformly outside $\operatorname{im} \iota$, set $e_h = e$, and fill the other unselected positions with independent uniform edges of E.

For coupling purposes, sample the unused randomness also when x is a coordinate of A.

Lemma 7.1 (Coupling bound for the randomized decoder)**.** Let dist denote normalized Hamming distance. For two labelings $\pi = (\pi_U, \pi_V)$ and $\widetilde{\pi} = (\widetilde{\pi}_U, \widetilde{\pi}_V)$ on the common graph of $\mathcal{T}^k_{\mathrm{IKW}}(\Phi)$ with its fixed projections, use the same right vertex A, endpoint incidences, and sampled pairs (ω_ℓ, h_ℓ) in both decoders. For each x, use a shared uniform random ordering of $[L]$ to select the first successful index in each execution. The corresponding decoded assignments $\boldsymbol{h}_A$ and $\widetilde{\boldsymbol{h}}_A$ satisfy

$$\mathbf{E} \operatorname{dist}(\boldsymbol{h}_A, \widetilde{\boldsymbol{h}}_A) \le \operatorname{dist}(\pi_V, \widetilde{\pi}_V) + L \operatorname{dist}(\pi_U, \widetilde{\pi}_U).$$

The bound also holds in expectation when $(\pi, \widetilde{\pi})$ is drawn from any coupling of two distributions of labelings.

Proof. The two symbols recovered at x agree unless the labels at A differ or some sampled left label differs. Averaging over a uniform vertex $x \in [N]$, a uniform endpoint incidence makes e uniform on E. Placing this uniform edge in a uniform unselected position and choosing all other unselected edges uniformly recovers the original distribution on $N(A)$, together with a uniform unselected position. Averaging over A therefore makes every sampled ω_ℓ a uniform edge of E_k. By biregularity, its left tuple is uniform on U. The union bound proves the assertion. □

To prove the decoder's value guarantee, we must control the probability that a sample in Step 2 is consistent with the label at A. Sampling from $N(A; e)$ fixes one unselected edge occurrence to e, so this probability may differ from μ_A. The following lemma shows that it remains close to μ_A for most edges e. We defer its proof to the end of this subsection.

Lemma 7.2 (Conditioning on one coordinate)**.** Let $\mathcal{X}$ be a finite set of size m, let $r \ge 1$, and let $f : \mathcal{X}^r \to [0, 1]$ have mean $\mu > 0$ under the uniform product distribution. Let $B \sim \mathrm{Unif}(\mathcal{X}^r)$ and define $q : \mathcal{X} \to [0, 1]$ by

$$q(x) := \frac{1}{r} \sum_{h=1}^{r} \mathbf{E}[f(B) \mid B_h = x].$$

For every constant $\theta \in (0, 1)$,

$$\frac{1}{m} \big| \{x \in \mathcal{X} : |q(x) - \mu| > \theta\mu\} \big| \le C_\theta \frac{\log(1/\mu)}{r}$$

for a constant C_θ depending only on θ.

Lemma 7.3 (Value of a decoded assignment)**.** There is an absolute constant $C_{\mathrm{val}} > 0$ such that, under the hypotheses of Lemma 6.2, every $A \in \mathcal{G}_\pi$ satisfies

$$\Pr\big[\mathsf{val}_\Phi(\boldsymbol{h}_A) \ge 1 - C_{\mathrm{val}}\eta\big] \ge 0.99.$$

Proof. Fix $A \in \mathcal{G}_\pi$ and the assignment g_A from Lemma 6.2. We first show that g_A has value at least $1 - O(\eta)$. We then show that, with probability at least 0.99, the decoded assignment $\boldsymbol{h}_A$ differs from g_A on at most an $O(\eta)$-fraction of the variables. Since Φ is regular, this implies that $\boldsymbol{h}_A$ also has value at least $1 - O(\eta)$.

Define $q_A : E \to [0, 1]$ by

$$q_A(e) := \Pr_{(\omega,h)\in N(A;e)}[\omega \in \mathrm{Cons}_A].$$

For $B \in E^r$, let $f_A(B)$ be the probability of consistency after averaging over ι and the endpoint incidences at A, with B placed in increasing order of the unselected positions. Its mean is μ_A, and $q_A(e)$ is the average of its conditional means when one coordinate of B is fixed to e. Lemma 7.2 gives an exceptional set $E_{\mathrm{irr}} \subseteq E$ with

$$b_A := |E_{\mathrm{irr}}|/M \le C\frac{\log(1/\mu_A)}{r} = O(\eta), \qquad \frac{\mu_A}{2} \le q_A(e) \le \frac{3\mu_A}{2} \quad (e \notin E_{\mathrm{irr}}). \tag{14}$$

The estimate $b_A = O(\eta)$ uses $\mu_A \ge c_2\varepsilon$ and (8).

Every right vertex has degree D_V from (5). Writing $n_e := |N(A; e)|$, the construction gives

$$|N(A)| = D_V, \qquad n_e = D_V r/M. \tag{15}$$

Let $\mathcal{D}_A$ be the distribution on pairs $(\omega, h) \in \mathrm{Cons}_A \times [k]$ with $h \notin \operatorname{im}\iota$, where $\omega = (S, \iota, b)$, obtained by first choosing ω uniformly in Cons_A and then an unselected position h uniformly. Its mass at each such pair (ω, h) is $1/(\mu_A D_V r)$, and its marginal at $e_h = e$ is $q_A(e)/(M\mu_A)$. If g_A violates e, every left label satisfying the predicate of a tuple containing e disagrees with g_A there, because the predicate requires that e be satisfied. Thus (9) and (14) imply

$$1 - \mathsf{val}_\Phi(g_A) \le b_A + 2C_{\mathrm{agr}}\eta = O(\eta). \tag{16}$$

We next bound $\mathbf{E}\,\mathrm{dist}(\boldsymbol{h}_A, g_A)$. Since $\mu_A \ge c_2\varepsilon$, for a nonexceptional edge e, the probability that none of the L sampled pairs is consistent is at most

$$(1 - q_A(e))^L \le \exp(-L\mu_A/2) \le \eta.$$

The samples are independent and uniform, and Step 3 selects uniformly among successful indices. Thus, conditional on success, every consistent pair in $N(A; e)$ is equally likely to be selected. For the sampling rule in Steps 2–3, average over a uniform vertex of G and a uniform endpoint incidence there, including samples unused when the vertex lies in A. The probability of successfully selecting a particular pair $(\omega, h) \in N(A; e)$ with $\omega \in \mathrm{Cons}_A$ is

$$\frac{1 - (1 - q_A(e))^L}{Mq_A(e)n_e} \le \frac{1}{Mq_A(e)n_e}.$$

Summing these probabilities over all e and consistent pairs gives the success probability, which may be less than one. By (15), the upper bound divided by the mass under $\mathcal{D}_A$ is

$$\frac{1/(Mq_A(e)n_e)}{1/(\mu_A D_V r)} = \frac{\mu_A}{q_A(e)} \le 2 \qquad (e \notin E_{\mathrm{irr}}).$$

Consequently (9) bounds the probability of returning a symbol different from $g_A(x)$ through a nonexceptional edge by $2C_{\mathrm{agr}}\eta$. For x equal to a coordinate of A, the decoder returns $g_A(x)$ by definition, so

$$\mathbf{E}\,\mathrm{dist}(\boldsymbol{h}_A, g_A) \le b_A + 2C_{\mathrm{agr}}\eta + \eta = O(\eta).$$

Markov's inequality bounds this distance by a sufficiently large constant times η with probability at least 0.99. Changing the labels at a ρ-fraction of the vertices of a regular 2-CSP changes the assignment's value by at most 2ρ. Together with (16), this proves the lemma. □

Proof of Lemma 7.2. Let P be the uniform distribution on $\mathcal{X}^r$ and define a distribution Q on the same space by $Q(z) := P(z)f(z)/\mu$ for $z \in \mathcal{X}^r$. For probability distributions on a common finite set, use natural logarithms to define $D(Q\|P) := \sum_z Q(z)\ln(Q(z)/P(z))$, with zero terms omitted. Since $f \le 1$, we have $D(Q\|P) \le \ln(1/\mu)$. For $h \in [r]$, let Q_h be the h-th coordinate marginal of Q, a distribution on $\mathcal{X}$, and let $\overline{Q} = r^{-1}\sum_{h=1}^{r} Q_h$ be their average on $\mathcal{X}$. Entropy subadditivity and convexity of relative entropy give

$$D(\overline{Q}\|\,\mathrm{Unif}(\mathcal{X})) \le \frac{1}{r}\sum_{h} D(Q_h\|\,\mathrm{Unif}(\mathcal{X})) \le \frac{\ln(1/\mu)}{r}.$$

Moreover, $m\overline{Q}(x) = q(x)/\mu$. Since the latter quantities average to one, the relative entropy on the left equals

$$\frac{1}{m}\sum_{x}\left(\frac{q(x)}{\mu}\ln\frac{q(x)}{\mu} - \frac{q(x)}{\mu} + 1\right).$$

The function $z\ln z - z + 1$, extended continuously at zero, has a positive minimum outside $[1-\theta, 1+\theta]$. This proves the bound. □

7.2 Recovery after Right-Degree Reduction

We apply the right-degree reduction of Section 4 to $\mathcal{T}^k_{\mathrm{IKW}}(\Phi)$, replacing each right vertex by a cloud of copies. To recover the label of an original right vertex, we choose one copy uniformly. If D_V is the original right degree and V the original right vertex set, choosing a uniform vertex of V and then a uniform copy of that vertex selects any particular copy with probability $1/(D_V|V|) = 1/|E_k|$. Thus one changed right label contributes at most $N/|E_k|$ to the expected Hamming distance between the N-variable assignments in one run of the decoder.

Proposition 7.4 (Sensitivity-transfer parameters for the IKW route)**.** There is an absolute constant $K \ge 100$ such that the following holds. Let η, ε, k, N satisfy (8), with $0 < \eta < 1/K$ and $0 < \varepsilon < 1/4$. Consider 2-CSPs Φ on the common d-regular multigraph $G = ([N], E)$, with the common alphabet Σ and endpoint orderings from the tuple construction; only their constraint relations vary. Choose an even right degree $d_\star \ge 6$ with expander parameter $\lambda_\star \le \varepsilon$ as in Definition 4.1. Use the same expander and incidence ordering on each cloud as Φ varies, and define the full transformation by

$$\mathcal{T}(\Phi) := \mathcal{T}_{\mathrm{DR}}(\mathcal{T}^k_{\mathrm{IKW}}(\Phi); d_\star).$$

There is a recovery map $\mathcal{R}_\Phi$ mapping distributions of labelings of $\mathcal{T}(\Phi)$ to distributions of assignments to Φ. If Φ is satisfiable and an input distribution μ has expected value at least 4ε, then $\mathcal{R}_\Phi(\mu)$ has expected value at least $1 - K\eta$ on Φ. The sensitivity-transfer parameters (Definition 3.1) satisfy

$$C_{\mathcal{T}}C_{\mathcal{R}},\ D_{\mathcal{R}} \le O\left(\frac{k\log^2(2/\eta)}{\eta\varepsilon^3}\right). \tag{17}$$

As the constraint relations of Φ vary, all images have the same graph, alphabets, and projections. The images of 2-CSPs differing in one relation are connected by paths of predicate swaps.

Proof. To sample from $\mathcal{R}_\Phi(\mu)$, we generate independent candidate assignments to Φ and select one using a random value threshold. Each trial generates one candidate as follows. Draw a labeling of $\mathcal{T}(\Phi)$ from μ and apply R_{DR} from Section 4 to obtain a labeling of $\mathcal{T}^k_{\mathrm{IKW}}(\Phi)$. The value guarantee in Lemma 7.3 requires every left label to satisfy its predicate. We therefore replace each left label that violates its predicate by the first label satisfying that predicate in an ordering independent of Φ, if one exists, and otherwise leave it unchanged. Run the randomized decoder of Section 7.1 on the resulting labeling to obtain the candidate assignment to Φ.

The replacement of left labels cannot decrease value, is 1-Lipschitz in each left coordinate, and depends on Φ only through the corresponding predicate.

If Φ is satisfiable, every left predicate is satisfied by at least one label. Thus the replacement ensures that all left labels satisfy their predicates. The value-recovery bound in Lemma 4.3 shows that the labeling supplied to the decoder has expected value at least 3ε. Its value is at least ε with probability at least ε. Conditional on any such labeling, Lemmas 6.2 and 7.3 imply that the trial returns an assignment to Φ of value at least $1 - C_{\mathrm{val}}\eta$ with probability at least $a\varepsilon$, for an absolute $a > 0$. Hence a trial succeeds with probability at least $a\varepsilon^2$.

Run

$$T := \left\lceil \frac{1}{a\varepsilon^2} \log \frac{1}{\eta} \right\rceil$$

independent trials. We select among their candidate assignments by a random value threshold: a small change in a candidate's value then changes its eligibility only with small probability. Lemma 3.5 will bound the sensitivity of this selection rule. Choose t independently of the trials and uniformly in $[C_{\mathrm{val}}, 2C_{\mathrm{val}}]$ and return the first candidate of value at least $1 - t\eta$, or the constant assignment $x \mapsto \sigma_0$ if no candidate passes. With probability at least $1 - \eta$, some trial succeeds, in which case a candidate passes and its value is at least $1 - 2C_{\mathrm{val}}\eta$. Choosing $K \geq \max\{100, 4C_{\mathrm{val}} + 2\}$ therefore gives expected value at least

$$(1 - \eta)(1 - 2C_{\mathrm{val}}\eta) \geq 1 - K\eta.$$

We next bound the expected Hamming distance between the candidate assignments in one coupled trial. The right vertex set V' of $\mathcal{T}(\Phi)$ consists of D_V copies of each vertex in V, so $|V'| = |E_k|$. For fixed Φ and two labelings of $\mathcal{T}(\Phi)$, let $H_U, H_{V'}$ be their Hamming distances on U and V', respectively. For each vertex in V, choose the same random copy in both recoveries and use the coupling from Lemma 7.1. Let $\boldsymbol{g}, \widetilde{\boldsymbol{g}} : [N] \to \Sigma$ be the two candidate assignments obtained in one coupled trial. They satisfy

$$\begin{aligned} \mathbf{E}\, d_{\mathrm{H}}(\boldsymbol{g}, \widetilde{\boldsymbol{g}}) &\leq N \left(\frac{LH_U}{|U|} + \frac{H_{V'}}{D_V|V|} \right) \\ &\leq \frac{NL}{|U|}(H_U + H_{V'}), \end{aligned} \tag{18}$$

where $L = \lceil C_{\mathrm{dec}}\varepsilon^{-1} \log(2/\eta) \rceil$, $D_V|V| = |E_k|$, and $|E_k| \geq |U|$. If instead one relation of Φ changes while the labeling of $\mathcal{T}(\Phi)$ is held fixed, R_{DR} is unchanged, and the replacement of left labels can differ only at tuples containing the changed edge. There are $F_k = M^k - (M-1)^k$ such tuples by (6). The same coupling bounds the expected Hamming distance between the two candidate assignments by $NLF_k/|U|$.

For any neighboring 2-CSPs $\Phi, \widetilde{\Phi}$ and assignments $g, \widetilde{g}$, regularity gives

$$|\mathsf{val}_\Phi(g) - \mathsf{val}_{\widetilde{\Phi}}(\widetilde{g})| \leq \frac{2}{N} d_{\mathrm{H}}(g, \widetilde{g}) + \frac{1}{M}.$$

A candidate $g \in \Sigma^N$ recovered from Φ has score $\mathsf{val}_\Phi(g) \in [0,1]$. Apply the Hamming-distance bound in Lemma 3.5 to the T coupled trials, conditional on their candidate assignments, with threshold interval $[1 - 2C_{\mathrm{val}}\eta, 1 - C_{\mathrm{val}}\eta]$ of width $C_{\mathrm{val}}\eta$. For fixed Φ and labeling distributions μ, ν on $\mathcal{T}(\Phi)$, draw each trial's pair of labelings independently from a coupling attaining $\mathsf{EMD}(\mu, \nu)$, and omit the $1/M$ term. When one relation of Φ changes, use the same labeling drawn from μ in both recoveries and include that term. Averaging the conditional Hamming-distance bound over the coupled trials gives

$$C_{\mathcal{R}} \le O\left(\frac{TNL}{\eta|U|}\right),$$
$$D_{\mathcal{R}} \le O\left(\frac{TNLF_k}{\eta|U|} + \frac{TN}{\eta M}\right).$$

These bounds hold on all 2-CSP instances Φ; only the value guarantee above used satisfiability.

Right-degree reduction retains each left predicate, so the path of predicate swaps from (6) still has length at most F_k. Thus $C_{\mathcal{T}} \le F_k \le k|U|/M$. Multiplying the preceding bound for $C_{\mathcal{R}}$ by $C_{\mathcal{T}}$, and using $N/M = 2/d \le 2$, proves (17). □

7.3 Proof of the Label Cover Consequence

Both the multiplicative factor $C_{\mathcal{T}}C_{\mathcal{R}}$ and the additive term $D_{\mathcal{R}}$ in the sensitivity-transfer bound are $N^{o(1)}$, where N is the number of variables of Φ. We apply this bound to the 2-CSP family of Lemma 2.3 and express the resulting sensitivity in terms of the final number of vertices.

Proof of Theorem 6.1. Use the family $\mathcal{F}_N$ from Lemma 2.3, with degree d_0, alphabet Σ_{src}, and sensitivity $\mathsf{SwapSens}_{1-\eta_{\mathrm{src}}}(\mathcal{F}_N) \ge N^{c_{\mathrm{src}}}$. Let K be the constant in Proposition 7.4, and fix

$$\eta := \frac{\eta_{\mathrm{src}}}{2K}, \qquad C := \frac{c_0\eta}{2}, \qquad \varepsilon_k := \frac{1}{4}\exp\left(-C\sqrt{k}\right),$$

where c_0 is from Lemma 6.2. Choose k_0 large enough that the lower bound on k in (8) and the inequality $\log(1/\varepsilon_k) = C\sqrt{k} + \log 4 \le c_0\eta\sqrt{k}$ hold for every $k \ge k_0$. The remaining condition $k = o(\log N)$ is a hypothesis of the theorem.

By Proposition 4.2, choose an even $d_\star = \exp\left(O(\sqrt{k})\right)$ large enough that $\lambda_\star \le \varepsilon_k$. Define the swap-closed family

$$\mathcal{I}_{N,k} := \mathsf{SwapClo}\big(\{\mathcal{T}_{\mathrm{DR}}(\mathcal{T}^k_{\mathrm{IKW}}(\Phi); d_\star) : \Phi \in \mathcal{F}_N\}\big).$$

Let $\mathcal{A}$ be an algorithm on $\mathcal{I}_{N,k}$ with expected value at least $4\varepsilon_k = \exp\left(-C\sqrt{k}\right)$ on every satisfiable member. Applying recovery to its outputs gives expected value at least $1 - K\eta > 1 - \eta_{\mathrm{src}}$ on every satisfiable $\Phi \in \mathcal{F}_N$. For this constant η and $k = o(\log N)$, (17) gives

$$C_{\mathcal{T}}C_{\mathcal{R}},\ D_{\mathcal{R}} \le O(k\varepsilon_k^{-3}) = N^{o(1)}.$$

The sensitivity-transfer theorem Theorem 3.2 therefore yields

$$\mathsf{SwapSens}(\mathcal{A}, \mathcal{I}_{N,k}) \ge \frac{N^{c_{\mathrm{src}}} - D_{\mathcal{R}}}{C_{\mathcal{T}}C_{\mathcal{R}}} \ge N^{c_{\mathrm{src}}/2}$$

for all sufficiently large N.

It remains to check the degrees, alphabet sizes, and number of vertices. By (5), the original left degree is $D_U = p_k 2^{k'} \le (2k)^{k'}$. Degree reduction makes the right degree $d_\star$ and the left degree $D_U d_\star$. The graph stays biregular, both alphabets are unchanged, and

$$D_U d_\star = \exp\Big(O(\sqrt{k}\log k)\Big) \le 2^{O(k)}, \qquad d_\star = \exp\Big(O(\sqrt{k})\Big),$$
$$|\Sigma_U| = |\Sigma_{\mathrm{src}}|^{2k} = 2^{O(k)}, \qquad |\Sigma_V| = |\Sigma_{\mathrm{src}}|^{k'} = 2^{O(k)}.$$

Finally,

$$n = |U| + |V'| = M^k(1 + D_U) = N^{\Theta(k)},$$

since $M = d_0 N/2$ and $k = o(\log N)$. Choose absolute $a, b > 0$ with $N^{ak} \le n \le N^{bk}$. The sensitivity bound is then at least $n^{c_{\mathrm{src}}/(2bk)}$. Taking the sensitivity exponent in Theorem 6.1 to be $c_{\mathrm{src}}/(2b)$ proves all its assertions. □

Part III
The Dinur–Harsha Route to Low-Soundness Label Cover

In this part we prove the sensitivity lower bound for Label Cover in Theorem 11.1 using the composition framework of Dinur and Harsha [15].

We start from the same 2-CSP family in Lemma 2.3, for which satisfying almost all constraints requires polynomial sensitivity. Our goal is to retain a polynomial sensitivity lower bound when the required fraction of satisfied constraints is only $1/g$, for g potentially a function of n, while obtaining the degree and alphabet bounds in the theorem.

In Sections 8 and 9, we encode assignments to these 2-CSPs by polynomials and construct tests that check parts of this encoding. The construction lets us convert an algorithm with a small value guarantee on satisfiable instances into one that satisfies almost all constraints of the original 2-CSP. To transfer the sensitivity lower bound, we also control how the recovered output distribution changes when either the supplied output distribution or one constraint of the original 2-CSP changes.

The degrees and alphabet sizes of this initial construction still depend on the number of variables in the original 2-CSP. To reduce them, in Section 10 we view predicates as large circuits deciding which tuples of neighboring right labels can be realized by a left label satisfying its predicate. These are then replaced by smaller circuits that query a small number of proof symbols. The graph specifying which labels are compared remains fixed when the 2-CSP constraints change, allowing us to bound the effect of each such change.

Finally, Section 11 repeats this construction together with alphabet and degree reduction. We control both the size of the checking circuits and the accumulated losses in the sensitivity-transfer bound of Section 3. This yields degree and alphabet bounds depending only on g, while preserving a sensitivity lower bound that is polynomial in the final instance size, with an exponent independent of g.

8 Constraint Groups and Low-Degree Testing

We prepare two ingredients for converting an algorithm with a small value guarantee on satisfiable instances into one that satisfies almost all constraints of the 2-CSP family in Lemma 2.3.

First, we group the constraints of these 2-CSPs so that, for any assignment violating a constant fraction of all constraints, a uniformly chosen group is unlikely to contain only satisfied constraints. Second, we state a polynomial-testing theorem: agreement between low-degree polynomials on lines and proposed values at points can be explained, apart from a small error, by a short list of global polynomials. In Section 9, these ingredients let us identify a small collection of assignments with high value on the original 2-CSP from local polynomial labels.

We use the following guarantee for the Arora–Sudan low-degree test, stated in terms of a list of global polynomials. A polynomial on an affine line is represented in any fixed affine parameterization; equality of two such restrictions does not depend on this choice.

Proposition 8.1 (Arora–Sudan low-degree test)**.** There are absolute constants $C_0, C_1 > 0$ such that, for a finite field $\mathbb{F}$, whenever $q = |\mathbb{F}| \geq C_0((D+1)/\kappa)^{C_1}$ and $0 < \kappa \leq 1/2$, every $f : \mathbb{F}^m \to \mathbb{F}$ has a list $\mathcal{L}_f \subseteq \{P \in \mathbb{F}[x_1, \dots, x_m] : \deg P \leq D\}$ of at most $4/\kappa$ polynomials with the following property. For every assignment $\ell \mapsto B_\ell$ of polynomial functions $B_\ell : \ell \to \mathbb{F}$ of degree at most D to affine lines $\ell \subseteq \mathbb{F}^m$,

$$\Pr_{\ell,\, z \in \ell}\big[B_\ell(z) = f(z) \ \wedge \ B_\ell \notin \{P|_\ell : P \in \mathcal{L}_f\}\big] \leq 2\kappa, \tag{19}$$

where the line and the point on it are uniform. The list depends only on f, not on the assignment of line polynomials.

Proof. This is the remark following Theorem 17 in the August 1, 2001 author version of [2], obtained there by using Lemma 14 in place of Lemma 13. The list may be taken to contain all polynomials of degree at most D having agreement at least $\kappa/2$ with f; the same field-size condition, after enlarging the absolute constants, bounds its size by $4/\kappa$. Lemma 14 applies to every assignment $\ell \mapsto B_\ell$ with this fixed list. □

For later use, let $\beta_f(\ell)$ be the maximum agreement with $f|_\ell$ of a line polynomial of degree at most D outside the listed restrictions, with maximum zero if this set is empty. Choosing a maximizer separately on each line in (19) gives

$$\mathop{\mathbf{E}}_{\ell} \beta_f(\ell) \leq 2\kappa. \tag{20}$$

This observation allows different representations of the same line to carry different answers: every answer outside the fixed list is bounded by the same maximum.

We next choose groups of constraint indices so that an assignment violating a constant fraction of constraints is unlikely to satisfy every constraint in a uniformly chosen group. The bound on how often each index occurs will also control the number of tests affected by changing one constraint relation.

Lemma 8.2 (Constraint groups from expander walks)**.** Let E be a nonempty finite set of constraint indices, and fix $0 < \eta < 1/2$ and $0 < p \leq 1/4$. There is a fixed list of ordered groups $S_1, \dots, S_M \in E^r$ of constraint indices, with repetitions allowed, such that

$$r = 1 + \left\lceil \frac{2}{\eta} \log \frac{8}{p} \right\rceil, \qquad M = |E| p^{-O_\eta(1)},$$

every $e \in E$ belongs to at most $rM/|E|$ groups, and, for every $W \subseteq E$ with $|W| \geq \eta|E|$,

$$\Pr_{j \in [M]}[S_j \text{ contains no member of } W] \leq p/8. \tag{21}$$

The list depends only on E, η, and p, not on the constraint relations.

Proof. Fix a constant even degree $d_{\exp}$ large enough that Proposition 4.2 gives an expander on E with normalized nontrivial spectral norm at most $1/2$. Enumerate all length-$(r-1)$ walks, retaining the choice of edge incidence at each step, so $M = |E|d_{\exp}^{r-1}$. Every walk has the same weight in the stationary walk distribution. Regularity implies that exactly $M/|E|$ indexed walks visit a prescribed vertex at each position, proving the occurrence bound.

Let $T \in \mathbb{R}^{E\times E}$ be the normalized adjacency matrix and $A = E \setminus W$. For a vector $x \in \mathbb{R}^E$ supported on A, decomposition into the constant vector and its orthogonal complement gives

$$x^{\mathsf{T}}Tx \le \tfrac{1}{2}\|x\|_2^2 + \frac{1}{2|E|}\Big(\sum_{e\in E} x_e\Big)^2 \le (1-\eta/2)\|x\|_2^2.$$

The principal submatrix $T_A \in \mathbb{R}^{A\times A}$ is symmetric and entrywise nonnegative. For $y \in \mathbb{R}^A$, $|y^{\mathsf{T}}T_A y| \le |y|^{\mathsf{T}}T_A|y|$; applying the preceding bound to the zero extension of $|y|$ to E gives $\|T_A\| \le 1-\eta/2$. Write $\mathbf{1}_A \in \mathbb{R}^A$ for the all-ones vector. Hence the probability that the stationary walk stays in A is at most $|E|^{-1}\mathbf{1}_A^{\mathsf{T}}T_A^{r-1}\mathbf{1}_A \le (1-\eta/2)^{r-1} \le p/8$. □

9 Initialization by Polynomial Restrictions

We construct the initial Label Cover family from which the later composition steps start, using the base 2-CSP family from Lemma 2.3. On satisfiable 2-CSPs, an algorithm with a small value guarantee for the constructed Label Cover instances can be converted into one that satisfies almost all constraints of the original 2-CSP. We bound the changes in both the transformed instance and the recovered output distribution, so that the 2-CSP sensitivity lower bound transfers.

We encode assignments to the 2-CSP variables by polynomial interpolation. Each left label is a polynomial intended to describe part of this encoding; each right label gives a value at one point. The tests from the preceding section use the Arora–Sudan guarantee to relate agreement between these labels to a short list of candidate global polynomials. For sensitivity transfer, we combine the constraint groups with a recovery over a common polynomial space chosen independently of the constraint relations and supplied labels. We score every polynomial by the fraction of left labels that pass their tests and agree with its restrictions. We return the first polynomial in a common order whose score meets a random threshold. For labelings of sufficiently large value, the short-list guarantee ensures that some polynomial has a sufficiently large score. Changes to labels or relations affect the scores, and the random threshold lets us bound the resulting change in recovery.

Fix the common graph $G = (X, E)$ of the family $\mathcal{F}_N$ in Lemma 2.3, with alphabet Σ_{src}. As in that lemma, $N = |X|$ is the number of variables, so $|E| = d_0 N/2$ is the number of constraints. Each edge has a fixed ordering of its endpoints, and parallel edges retain distinct indices. For relations $b = (b_e)_{e\in E}$, with $b_e \subseteq \Sigma_{\mathrm{src}} \times \Sigma_{\mathrm{src}}$, let $\Phi_b \in \mathcal{F}_N$ have relation b_e on edge e. An assignment $a : X \to \Sigma_{\mathrm{src}}$ satisfies an edge $e = (x, y)$ exactly when $(a(x), a(y)) \in b_e$; the endpoints may coincide. Only the relations b_e vary across this family. We choose all query locations from the fixed graph G, so changing one relation will affect local tests while leaving the constructed graph unchanged.

Theorem 9.1 (Initialization by polynomial restrictions)**.** For all sufficiently large N and every $(\log|E|)^{-1} \le \gamma \le 1/2$, the family $\mathcal{F}_N$ on $G = (X, E)$ with alphabet Σ_{src} admits a transformation $b \mapsto \widehat{J}_b$ and a recovery map $\mathcal{R}_{b,\gamma}$ from distributions on full labelings of $\widehat{J}_b$ to distributions on Σ_{src}^X, with the following properties. The instances have common graph, alphabets, and projections; all vertices have positive degree;

and satisfiability is preserved. For every distribution μ on full labelings of $\widehat{J}_b$, if

$$\mathop{\mathbf{E}}_{Y\sim\mu} \mathsf{val}_{\widehat{J}_b}(Y) \geq \gamma,$$

then the recovered distribution has expected value greater than $1-\eta_{\mathrm{src}}$ on Φ_b. The number of vertices on each side is $N^{O(1)}$, and both degrees, the right alphabet size, the logarithm of the left alphabet size, and the local description complexity (Definition 2.4) are $\mathrm{poly}(\log N, \gamma^{-1})$. The sensitivity-transfer parameters satisfy

$$C_{\mathcal{T}} C_{\mathcal{R}},\ D_{\mathcal{R}} \leq O\left(\gamma^{-5}\log\frac{2}{\gamma}\right). \tag{22}$$

If b and b' differ in one relation, their images are connected by paths of predicate swaps of length at most $C_{\mathcal{T}}$ with the same local-description bound. The constants depend only on the constants in Lemma 2.3.

We prove the theorem in the remainder of this section. Recovery is allowed to enumerate a finite set of polynomials; no polynomial-time recovery guarantee is needed for the sensitivity lower bound. The graph and local circuits of $\widehat{J}_b$ have polynomial-size descriptions.

9.1 The Polynomial Restriction Tests

Each group of source constraints determines a polynomial map with two parameters. At designated parameter values, the map visits the variables in the group, so a left polynomial label can be checked against the source constraints. Away from these designated values, randomizing the map produces uniform affine lines, on which the Arora–Sudan test compares the left label with right labels at individual points. This connects the source constraints to the global polynomials supplied by the list guarantee. We define the maps and labels before collecting the field-size conditions.

Fix a constant $0 < \eta < \eta_{\mathrm{src}}/2$, the allowed fraction of violated source constraints after recovery. We use τ as the value threshold before right-degree reduction, κ as the error parameter in Proposition 8.1, and $L_{\max}$ as its list-size bound. We choose a score threshold p small enough that value at least τ forces some polynomial on this list to agree with accepted left labels on at least a p fraction of vertices. Set

$$\tau = \gamma/4, \qquad \kappa = \tau/16, \qquad L_{\max} = \lceil 64/\tau \rceil, \qquad p = \frac{\tau}{2L_{\max}} = \Theta(\gamma^2). \tag{23}$$

Apply Lemma 8.2 to E with parameters η, p to obtain constraint groups $S_1, \ldots, S_M \in E^r$. The avoidance bound (21) will ensure that any polynomial agreeing with accepted left labels on more than a $p/8$ fraction decodes to source value greater than $1-\eta$. Put $k = 2r$, the number of variable occurrences in each group. We place the N source variables in a grid of size $h^m \geq N$ and encode their values by a polynomial of total degree at most $D = m(h-1)$. Choose

$$h = \lceil \log N \rceil, \qquad m = \left\lceil \frac{\log N}{\log h} \right\rceil, \qquad D = m(h-1).$$

Work over $\mathbb{F} = \mathrm{GF}(q)$, where q is a power of two with $q > D$, $q > k$, and $q \geq \max\{h, |\Sigma_{\mathrm{src}}|\}$; further lower bounds on q are specified below. Fix $H \subseteq \mathbb{F}$ of size h, and an injection $\iota : X \hookrightarrow H^m$. Fix an injection $\jmath : \Sigma_{\mathrm{src}} \hookrightarrow \mathbb{F}$ and a map $\beta : \mathbb{F} \to \Sigma_{\mathrm{src}}$ with $\beta(\jmath(\sigma)) = \sigma$ for every $\sigma \in \Sigma_{\mathrm{src}}$; outside the image of $\jmath$, use a fixed default symbol. Tensor-product interpolation, with the unused grid positions assigned zero, extends every encoded assignment $\jmath \circ a$ to a polynomial of total degree at most D. Let

$\mathcal{P}_D = \{P \in \mathbb{F}[x_1, \ldots, x_m] : \deg P \le D\}$ be the finite space of such polynomials. For $P \in \mathcal{P}_D$, define the assignment $a_P : X \to \Sigma_{\mathrm{src}}$ by

$$a_P(x) = \beta(P(\iota(x))). \tag{24}$$

Since $D < q$, different formal polynomials in $\mathcal{P}_D$ define different functions.

Choose distinct $t_1, \ldots, t_k \in \mathbb{F}$, and write $T_0 = \{t_1, \ldots, t_k\} \subseteq \mathbb{F}$. Define $Z \in \mathbb{F}[t]$ by $Z(t) = \prod_{s=1}^{k}(t - t_s)$. For each group S_j, list the embedded variable locations of its r constraints, retaining all repetitions, as $z_{j,1}, \ldots, z_{j,k} \in H^m \subseteq \mathbb{F}^m$. Interpolate a vector-valued polynomial $C_j : \mathbb{F} \to \mathbb{F}^m$ of degree less than k with $C_j(t_s) = z_{j,s}$. For $a \in \mathbb{F}^m$ and $v \in \mathbb{F}^m \setminus \{0\}$, define $\Gamma_{j,a,v} : \mathbb{F}^2 \to \mathbb{F}^m$ by

$$\Gamma_{j,a,v}(t, u) = C_j(t) + Z(t)a + uv. \tag{25}$$

Its designated values are $\Gamma_{j,a,v}(t_s, 0) = z_{j,s}$. For every fixed j and $t \notin T_0$, varying u traces a uniform affine line when a, v are uniform: $C_j(t) + Z(t)a$ is a uniform point, and v is an independent uniform nonzero direction.

The initial instance J_b has vertex sets $U(J_b) = [M] \times \mathbb{F}^m \times (\mathbb{F}^m \setminus \{0\})$ and $V(J_b) = \mathbb{F}^m$. Write $\omega = (j, a, v) \in U(J_b)$ for a left vertex and $z \in V(J_b)$ for a right vertex. A left label $B_\omega \in \mathbb{F}[t, u]$ is a bivariate polynomial with $\deg_t B_\omega \le kD$ and $\deg_u B_\omega \le D$, represented by its coefficient vector. A right label belongs to $\mathbb{F}$. For each $(t, u) \in (\mathbb{F} \setminus T_0) \times \mathbb{F}$, add an edge incidence from ω to $\Gamma_\omega(t, u)$ projecting B_ω to $B_\omega(t, u)$. Distinct parameter pairs give distinct incidences even when their endpoints coincide. The predicate $Q_{b,\omega}$ checks all r constraints in S_j: for its ℓth edge e, it requires

$$\big(\beta(B_\omega(t_{2\ell-1}, 0)), \beta(B_\omega(t_{2\ell}, 0))\big) \in b_e.$$

The graph, alphabets, and projections are independent of b.

Writing $Q = q^m$ and denoting the left and right degrees of J_b by d_L, d_R, we have

$$A := |U(J_b)| = MQ(Q - 1), \quad |V(J_b)| = Q, \quad d_L = q(q - k), \quad d_R = M(Q - 1)d_L. \tag{26}$$

Indeed, for fixed z, j, v, t, u with $t \notin T_0$, the equation $\Gamma_{j,a,v}(t, u) = z$ has exactly one solution a. Thus the graph is biregular and has no isolated vertices. If a_* satisfies Φ_b, let P interpolate $\jmath \circ a_*$. The labels $B_\omega = P \circ \Gamma_\omega$ and $f = P$ satisfy J_b: the composition has bidegree at most (kD, D), and every designated symbol equals $\jmath(a_*(x))$, so applying β recovers $a_*(x)$. This proves completeness.

We now choose the field size. The line test requires the first bound below; the remaining bounds ensure that evaluations determine the polynomials and that distinct candidates rarely agree. Take the least power of two q satisfying

$$\begin{gathered} q \ge C_0((D+1)/\kappa)^{C_1}, \qquad q \ge \max\{h, |\Sigma_{\mathrm{src}}|\}, \qquad q > D, \qquad q - k > kD, \\ \frac{L_{\max} kD}{q - k} \le \frac{\tau}{8}, \qquad \frac{D}{q} \le \frac{p^2}{1000}. \end{gathered} \tag{27}$$

Here C_0, C_1 are the constants in Proposition 8.1. The bound $L_{\max} kD/(q - k) \le \tau/8$ controls agreement after fixing t: if a bivariate left label differs from all restrictions of the at most $L_{\max}$ candidate global polynomials, it can agree identically in u with one of these restrictions for at most a $\tau/8$ fraction of the allowed t-values. The bound $D/q \le p^2/1000$ limits the fraction of points on which two distinct global polynomials agree. The bounds $q > D$ and $q - k > kD$ ensure that evaluations determine the global and bivariate polynomials, respectively. These requirements give $q = \mathrm{poly}(D, k, \gamma^{-1})$.

The inverse-polylogarithmic lower bound on γ gives

$$q = \log^{O(1)} N, \qquad q^m = \exp\left(O(\log\log N) \cdot O\left(\frac{\log N}{\log\log N}\right)\right) = N^{O(1)}. \tag{28}$$

Lemma 9.2 (From line agreement to bivariate restrictions)**.** For every right labeling $f : \mathbb{F}^m \to \mathbb{F}$ there are $P_1, \dots, P_L \in \mathcal{P}_D$, with $L \le L_{\max}$, such that, for every assignment of left polynomials,

$$\Pr_{\omega,t,u}\big[B_\omega(t,u) = f(\Gamma_\omega(t,u)), \quad B_\omega \notin \{P_i \circ \Gamma_\omega : i \in [L]\}\big] \le \tau/4. \tag{29}$$

Here ω is uniform on the left side, t is uniform outside T_0, and u is uniform in $\mathbb{F}$. The condition on B_ω refers to equality of bivariate polynomials.

Proof. Use the list from Proposition 8.1 with parameter κ. For fixed $j, t \notin T_0$, varying u traces a uniform affine line, and $B_\omega(t,\cdot)$ has degree at most D. Even if different left vertices supply different polynomial answers on the same geometric line, their agreement when outside the list is bounded by $\beta_f(\ell)$ after changing to a canonical line parameterization. Thus (20), averaged over j, t, bounds the probability of agreement with f when the line polynomial is outside the list by $2\kappa = \tau/8$.

Fix ω for which B_ω differs from every $P_i \circ \Gamma_\omega$. For each i, the nonzero bivariate polynomial $B_\omega - P_i \circ \Gamma_\omega$ has at least one nonzero coefficient when expanded in powers of u. That coefficient has degree at most kD in t. Consequently, for this i, the two polynomials become identical as polynomials in u for at most kD values of t. A union bound gives at most $L_{\max} kD/(q-k) \le \tau/8$ for the fraction of t-values at which the restricted polynomial is in the list. Adding the two bounds proves the claim. □

9.2 Scores for Candidate Polynomials and Lipschitz Bounds for Recovery

For each global polynomial $P \in \mathcal{P}_D$, we score the fraction of left vertices ω for which B_ω satisfies the left predicate $Q_{b,\omega}$ and equals $P \circ \Gamma_\omega$. The predicate checks all r constraints of Φ_b in S_j, where $\omega = (j, a, v)$. These scores will identify assignments with high value on Φ_b and bound the change in recovery when either the labels or the relations b_e change. For a left labeling $B = (B_\omega)_{\omega \in U(J_b)}$, define $s_{b,B} : \mathcal{P}_D \to [0,1]$ by

$$s_{b,B}(P) = \frac{1}{A} \sum_{\omega} \mathbf{1}\big[Q_{b,\omega}(B_\omega) = 1, \quad B_\omega = P \circ \Gamma_\omega\big]. \tag{30}$$

The recovery will use these scores on all of $\mathcal{P}_D$, rather than restricting its candidates to the list determined by the right labeling f. That list supplies a high-scoring candidate when the labeling has large value. For sensitivity, we separately bound the number of high-scoring polynomials in the entire space $\mathcal{P}_D$, so that the perturbation bound does not depend on the size of this space. Neither the scores nor the recovery rule use f.

Lemma 9.3 (Score guarantees)**.** For every initial instance J_b and every labeling (B, f) of J_b, the following hold.

1. If $\mathsf{val}_{J_b}(B, f) \ge \tau$, some $P \in \mathcal{P}_D$ has $s_{b,B}(P) \ge p$.
2. Every $P \in \mathcal{P}_D$ with $s_{b,B}(P) > p/8$ satisfies $\mathsf{val}_{\Phi_b}(a_P) > 1 - \eta$.
3. The number of $P \in \mathcal{P}_D$ with $s_{b,B}(P) \ge p/4$ is at most $B_{\text{list}} := \lceil 20/p \rceil$.

Proof. For the first assertion, let $P_1, \dots, P_L$ be the list supplied by Lemma 9.2 for the right labeling f. At least a $3\tau/4$ fraction of all edges are satisfied and have a left endpoint ω with $B_\omega = P_i \circ \Gamma_\omega$ for some $i \in [L]$. Since each left vertex has the same degree,

$$\sum_{i=1}^{L} s_{b,B}(P_i) \ge 3\tau/4, \qquad \max_i s_{b,B}(P_i) \ge \frac{3\tau}{4L_{\max}} \ge p.$$

If a left vertex $\omega = (j, a, v)$ contributes to $s_{b,B}(P)$, the assignment a_P satisfies every constraint in S_j. Every group S_j indexes the same number $Q(Q-1)$ of left vertices. If a_P violated at least $\eta|E|$ constraints, (21) would therefore imply $s_{b,B}(P) \le p/8$. This proves the first two assertions.

For distinct $P, P' \in \mathcal{P}_D$, fix any $t_* \notin T_0$. Equality of their full restrictions to $\Gamma_{j,a,v}$ implies equality at $C_j(t_*) + Z(t_*)a$. This is a uniform point of $\mathbb{F}^m$ when a is uniform, so Schwartz–Zippel gives

$$\Pr_{\omega}[P \circ \Gamma_\omega = P' \circ \Gamma_\omega] \le D/q \le p^2/1000. \tag{31}$$

Let $A_P \subseteq U(J_b)$ be the set of left vertices counted in (30) and put $\alpha = p/4$. Suppose there is a set $\mathcal{C} \subseteq \mathcal{P}_D$ of $K = \lceil 4/\alpha \rceil$ candidates, each with score at least α. The first two inclusion–exclusion terms give

$$1 \ge \frac{1}{A}\left|\bigcup_{P \in \mathcal{C}} A_P\right| \ge K\alpha - \binom{K}{2}\frac{p^2}{1000} > 1.$$

The last inequality uses $K\alpha \ge 4$ and $Kp < 17$. This contradiction bounds the number of candidates by $K \le 20/p$, as required. □

Fix a total order on $\mathcal{P}_D$ independently of b. A recovery trial on a left labeling B samples θ uniformly from $[p/2, p]$ and outputs $R_b(B;\theta) \in \mathcal{P}_D \cup \{\perp\}$: the first polynomial P with $s_{b,B}(P) \ge \theta$, or $\perp$ if none exists. By Lemma 9.3, for every B, each nonfailure output P satisfies $\mathsf{val}_{\Phi_b}(a_P) > 1 - \eta$. If some right labeling together with B gives value at least τ, the trial never fails.

Lemma 9.4 (Perturbation bounds for recovery from one labeling)**.** Using the same uniform threshold θ in both runs, for fixed b and arbitrary left labelings B, B', and for b, b' differing in one relation with B fixed, respectively, we have

$$N \Pr\big[R_b(B;\theta) \ne R_b(B';\theta)\big] \le C^{\mathrm{one}}_{\mathcal{R}} d_{\mathrm{H}}(B, B'), \qquad C^{\mathrm{one}}_{\mathcal{R}} := \frac{8NB_{\mathrm{list}}}{pA}, \tag{32}$$

$$N \Pr[R_b(B;\theta) \ne R_{b'}(B;\theta)] \le D^{\mathrm{one}}_{\mathcal{R}}, \qquad D^{\mathrm{one}}_{\mathcal{R}} := \frac{8NB_{\mathrm{list}}r}{p|E|}. \tag{33}$$

Changing one relation b_e changes at most $C_{\mathcal{T}} := rA/|E|$ left predicates of J_b, and

$$C_{\mathcal{T}} C^{\mathrm{one}}_{\mathcal{R}},\ D^{\mathrm{one}}_{\mathcal{R}} \le O(r/p^2). \tag{34}$$

Proof. Let $s, s' : \mathcal{P}_D \to [0,1]$ be the two score functions being compared, with $\|s - s'\|_\infty \le \nu$. By Lemma 9.3, at most $2B_{\mathrm{list}}$ polynomials have score at least $p/4$ under either function. Apply the probability bound in Lemma 3.5 to the two score lists in the same polynomial order, with index 0 representing $\perp$ and threshold interval $[p/2, p]$. Only those polynomials can have score intervals meeting $[p/2, p]$. Different selected indices give different trial outputs, so multiplying the probability bound by N gives

$$N \Pr[\text{trial outputs differ}] \le 4NB_{\mathrm{list}}\nu/p.$$

Changing $d_{\mathrm{H}}(B, B')$ left labels changes each score by at most $d_{\mathrm{H}}(B, B')/A$. An edge $e \in E$ occurs in at most $rM/|E|$ groups, so changing b_e changes at most $rA/|E|$ left predicates and each score by at most $r/|E|$. These observations imply the stated (slightly looser) constants. Finally $N/|E| = 2/d_0$ and $B_{\mathrm{list}} = O(1/p)$ give (34). □

9.3 Degree Reduction and Recovery from Independent Samples

We reduce the right degree while preserving recovery from the left labels. We then repeat recovery on independent input samples and return the assignment from the first successful trial.

Choose an even $d_\gamma = \Theta(\gamma^{-2})$ for which Proposition 4.2 gives error $\lambda_\gamma \le \gamma/4$, and set

$$\widehat{J}_b = \mathcal{T}_{\mathrm{DR}}(J_b; d_\gamma).$$

All expanders and incidence orderings are fixed across b. Left labels and left predicates are unchanged. We apply R_b directly to the left part of a labeling of $\widehat{J}_b$, ignoring every right label.

For a fixed labeling Y of $\widehat{J}_b$, (2) gives

$$\mathsf{val}_{\widehat{J}_b}(Y) \le \mathbf{E}\,\mathsf{val}_{J_b}(R_{\mathrm{DR},J_b}(Y)) + \lambda_\gamma.$$

The degree-reduction recovery preserves all left labels. Thus if $\mathsf{val}_{\widehat{J}_b}(Y) \ge \gamma/2$, some right labeling of J_b gives value at least $\gamma/4 = \tau$ with those same left labels. Applying R_b to these left labels therefore returns a polynomial rather than $\perp$ almost surely. We therefore need neither to execute R_{DR,J_b} nor to use its generic Lipschitz bound. Also, since degree reduction leaves left predicates unchanged, changing b_e still changes at most $rA/|E|$ left predicates of $\widehat{J}_b$.

Fix a default assignment $a_0 : X \to \Sigma_{\mathrm{src}}$ independently of b. For any distribution μ over full labelings of $\widehat{J}_b$, take

$$R_{\mathrm{amp}} = \left\lceil \frac{2}{\gamma} \log \frac{4}{\eta_{\mathrm{src}}} \right\rceil$$

independent samples $Y_1, \dots, Y_{R_{\mathrm{amp}}}$ from μ. Write B_j for the left part of Y_j, and independently sample uniform thresholds $\theta_j \in [p/2, p]$ to obtain $P_j = R_b(B_j; \theta_j)$. Return a_{P_j} for the first j with $P_j \ne \perp$, or a_0 if all trials fail. This defines $\mathcal{R}_{b,\gamma}(\mu)$ on all input distributions. The failure symbol is distinct from every polynomial, including those that decode to a_0.

If $\mathbf{E}_{Y\sim\mu}\,\mathsf{val}_{\widehat{J}_b}(Y) \ge \gamma$, then

$$\Pr_{Y\sim\mu}[\mathsf{val}_{\widehat{J}_b}(Y) \ge \gamma/2] \ge \frac{\gamma}{2-\gamma} \ge \gamma/2.$$

Each trial thus returns a polynomial with probability at least $\gamma/2$, and independence bounds the probability of all trials failing by $\exp(-\gamma R_{\mathrm{amp}}/2) \le \eta_{\mathrm{src}}/4$. Every polynomial returned by a trial decodes to value greater than $1-\eta$, so the output has expected value at least

$$(1 - \eta_{\mathrm{src}}/4)(1-\eta) > 1 - \eta_{\mathrm{src}}.$$

For two input distributions with b fixed, take R_{amp} independent copies of an optimal input coupling and share the thresholds within each pair. If all paired trial outputs agree in $\mathcal{P}_D \cup \{\perp\}$, the final assignments agree. Their Hamming distance is at most N otherwise. Restriction to left labels contracts Hamming distance, so a union bound and Lemma 9.4 give the first bound below. When one relation changes, use the same input samples and shared thresholds; the same argument gives the second bound:

$$C_{\mathcal{R}} \le R_{\mathrm{amp}} C^{\mathrm{one}}_{\mathcal{R}}, \qquad D_{\mathcal{R}} \le R_{\mathrm{amp}} D^{\mathrm{one}}_{\mathcal{R}}. \tag{35}$$

Together with (34), $R_{\mathrm{amp}} = O(\gamma^{-1})$, $p = \Theta(\gamma^2)$, and $r = O(\log(2/\gamma))$, this proves (22).

Structural parameters and local description. For $\widehat{J}_b$, we have

$$\begin{aligned} |U(\widehat{J}_b)| = A, \qquad |V(\widehat{J}_b)| = Ad_L, \\ \deg_U = d_\gamma d_L, \qquad \deg_V = d_\gamma, \qquad \Sigma_V = \mathbb{F}, \qquad |\Sigma_U| = q^{(kD+1)(D+1)}. \end{aligned} \tag{36}$$

By (28) and (26), the number of vertices on each side is $N^{O(1)}$. Both degrees, the right alphabet size, and the logarithm of the left alphabet size are $\mathrm{poly}(\log N, \gamma^{-1})$.

For the local description complexity of Definition 2.4, we must test whether a tuple of incident right labels at ω is the projection tuple of a left label satisfying $Q_{b,\omega}$. On the query grid $(\mathbb{F} \setminus T_0) \times \mathbb{F}$ of J_b, the evaluations uniquely determine a polynomial of bidegree at most (kD, D), since $q - k > kD$ and $q > D$. Interpolate from a fixed $(kD+1) \times (D+1)$ subgrid, check agreement on all remaining grid positions, and check the r constraints of Φ_b in S_j on the recovered polynomial, where $\omega = (j, a, v)$. All interpolation matrices and points are hardwired. The map β compares a field element with the constantly many encoded symbols, and each relation b_e is a constant-size table. Together with finite-field arithmetic, these tests give a circuit of size $\mathrm{poly}(q, k, D, \log q)$, independent of the large right degree. After degree reduction, first check that the right labels at the endpoints of the d_γ copies of each incidence of J_b agree, then apply this circuit to the common label from each group of copies. The resulting size is $\mathrm{poly}(q, d_\gamma, k, D, \log q) = \mathrm{poly}(\log N, \gamma^{-1})$.

Finally, for b, b' differing in one relation, a path of predicate swaps between $\widehat{J}_b$ and $\widehat{J}_{b'}$ has length at most $rA/|E|$. Each local predicate on the path is chosen from one endpoint and retains the common projections, so the same circuit bound holds along the path. Completeness follows by copying honest right labels through the degree reduction. This completes the proof of Theorem 9.1.

10 Query-Oblivious Decodable PCPs and Composition

At a left vertex of a Label Cover instance I, we must check whether the neighboring right labels are the projections of a left label satisfying its predicate. We express this test as a circuit and replace it by smaller tests on an auxiliary proof. These tests use a decoder that queries a small number of proof symbols to recover a requested input symbol or reject.

For sensitivity transfer, the proof locations read by the decoder must stay fixed when the circuit being checked changes. We obtain this property, called query-obliviousness, by encoding the circuit description and confining its effect to one local comparison. We also bound the size of each decoding circuit. Using one right vertex per proof symbol then produces a Label Cover instance with smaller local checking circuits and a graph that is fixed across the family of input instances I. These properties are needed to repeat the construction in Section 11. Figure 1 shows the new vertices and the recovery dependencies: new right labels form a separate proof for each old left vertex, while a new left label supplies answers to tests that decode the same old right label.

10.1 Decoder Guarantees Used in Composition

We first state the decoder guarantees, then give the composition and recovery, and finally prove the decoder lemmas. For a finite alphabet Σ, the input to the decoder is a circuit $C : \Sigma^k \to \{0,1\}$, with the encoding convention in Section 2.

Definition 10.1 (PCP decoder)**.** A nonadaptive PCP decoder has a proof alphabet σ, proof length m, and a finite seed set $\mathscr{R}$. Given $C : \Sigma^k \to \{0,1\}$, an index $j \in [k]$, and a uniform seed $\rho \in \mathscr{R}$, it produces

a query tuple $I_{C,j,\rho} \in [m]^q$ and a local circuit $g_{C,j,\rho} : \sigma^q \to \Sigma \cup \{\perp\}$. On a proof $p \in \sigma^m$, its output is $g_{C,j,\rho}(p|_{I_{C,j,\rho}})$. For given size and alphabet parameters, the local circuit complexity s is the maximum encoding length of $g_{C,j,\rho}$ over all admissible C, j, ρ. A decoder is *query-oblivious* if $I_{C,j,\rho}$ depends on C only through its input arity and the common size and alphabet parameters. We use uniform finite seed sets to avoid rounding their cardinalities to powers of two. A uniform seed can be sampled using $\log|\mathscr{R}| + O(1)$ random bits in expectation.

Definition 10.2 (Robust decodable PCP)**.** For $w \in \sigma^q$ and $B \subseteq \sigma^q$, define

$$\operatorname{agr}(w, B) := \max_{z \in B} \frac{|\{h \in [q] : w_h = z_h\}|}{q},$$

with agreement zero when $B = \varnothing$. Query occurrences, including repetitions of a proof position, are counted with multiplicity. For a local circuit $g : \sigma^q \to \Sigma \cup \{\perp\}$, a set $Y \subseteq \Sigma^k$, and $j \in [k]$, put

$$\operatorname{Bad}_Y(g, j) := \big\{w \in \sigma^q : g(w) \notin \{\perp\} \cup \{y_j : y \in Y\}\big\}.$$

Thus $\operatorname{Bad}_Y(g, j)$ consists of local answer words that do not reject and decode to a symbol different from y_j for every $y \in Y$. The quantity $\operatorname{agr}(w, \operatorname{Bad}_Y(g, j))$ is the largest fraction of coordinates that w shares with any such word; it can be positive even when $g(w) = \perp$.

The decoder has perfect completeness if, for every y with $C(y) = 1$, there is a proof that decodes y_j for every j, ρ. It has robust soundness error δ and list bound L if, for every proof p, there is a possibly empty set $Y \subseteq C^{-1}(1)$ with $|Y| \le L$ such that, for uniform independent $j \in [k]$ and $\rho \in \mathscr{R}$,

$$\mathop{\mathbf{E}}_{j,\rho} \operatorname{agr}\big(p|_{I_{C,j,\rho}}, \operatorname{Bad}_Y(g_{C,j,\rho}, j)\big) \le \delta. \tag{37}$$

In the composition of Section 10.2, a left label supplies local answer words that can match some of the queried proof symbols without matching the whole query word. The agreement bound controls the contribution of these partial matches with words in $\operatorname{Bad}_Y(g, j)$. Robust soundness also implies ordinary list-decoding soundness for the same list Y:

$$\Pr_{j,\rho}\big[g_{C,j,\rho}(p|_{I_{C,j,\rho}}) \notin \{\perp\} \cup \{y_j : y \in Y\}\big] \le \delta. \tag{38}$$

The following decoder supplies all the guarantees used in composition. We derive it from the algebraic dPCP of [15, Theorem 6.5 and Section 6.1.2]: Section 10.3 bounds the local circuit size, and Section 10.4 makes the query locations independent of which circuit is being checked.

Lemma 10.3 (Query-oblivious robust decoder)**.** There are absolute constants $\alpha_*, \gamma_*, A_* > 0$ such that, for all sufficiently large M, all $M^{-\alpha_*} \le \delta \le 1/2$, and $|\Sigma| \le M^{\gamma_*}$, circuits $C : \Sigma^k \to \{0, 1\}$ of complexity at most M have a query-oblivious robust dPCP with perfect completeness and

$$L \le 4/\delta, \qquad q, s \le M^{1/32}, \qquad m,\ R := |\mathscr{R}|,\ |\sigma| \le M^{A_*}. \tag{39}$$

The parameters $L, q, s, m, \mathscr{R}, \sigma$ depend only on M, δ, Σ. The only dependence of the local circuit on C is a comparison with one symbol of a fixed encoding of its description.

The proof is given in Section 10.4, after we use the decoder to construct the composed Label Cover instance and its recovery. The query bound q controls the length of a new left label, and the local circuit bound s controls the cost of checking it. These are the bounds needed for the next composition step; m and R control the number of new vertices. Robust soundness with parameters L, δ will give recovered value at least $(\text{composed value} - \delta)/L$.

10.2 Composition with One Proof Symbol per Right Vertex

We use the Dinur–Harsha composition rule [15, Section 4], which compares tests decoding the same old right label. We give the construction in Label Cover form and bound changes to its instances and recovered labelings. For each left vertex $u \in U$, with $k_u = \deg_I(u)$, let $C_u : \Sigma_V^{k_u} \to \{0,1\}$ recognize $\mathcal{N}_I(u)$, the set of projection tuples of left labels satisfying P_u, as defined in Definition 2.4. Using one right vertex per proof symbol will make the local description complexity depend on the number of queries rather than the proof length.

Fix such a family with common graph and alphabets, positive degrees, and local circuits of complexity at most M. Let d_U and d_V bound its left and right degrees, respectively. Let $\mathcal{D}$ be the decoder of Lemma 10.3, or any query-oblivious robust decoder. For an incidence $e = (u, v)$ of I, let $j(e)$ be its index at u. Write $I_{u,j,\rho} \in [m]^q$ for the query tuple, which can depend on k_u but not on C_u. Define $I \circledast \mathcal{D}$ as follows.

1. Its left vertex set is $U' = V \times \mathscr{R}$. Start with right vertices $(u, z) \in U \times [m]$ and delete positions z that are never queried at u, using the same deletion throughout the family. Let $V' \subseteq U \times [m]$ be the remaining right vertex set. Deleted proof symbols are filled by a fixed default in recovery.
2. Put $\Sigma'_V = \sigma$ and $\Sigma'_U = \sigma^{d_V q}$. A left label at (v, ρ) has one row $w_e \in \sigma^q$ per incidence $e = (u, v)$ at v, with unused rows ignored. Its predicate accepts exactly when all $g_{C_u,j(e),\rho}(w_e)$ agree on one non-$\perp$ symbol.
3. For each such incidence and $h \in [q]$, put one edge from (v, ρ) to $(u, (I_{u,j(e),\rho})_h)$, projecting to $(w_e)_h$. These are distinct edge incidences even if some endpoints coincide.

Every right label is one proof *symbol*; no vertex label contains an entire proof. The graph and projections are independent of the checking circuits C_u of I.

Theorem 10.4 (Composition with a common output graph)**.** Assume the input family has degree bounds d_U, d_V, positive degrees, and common graph and alphabets, and use a query-oblivious robust decoder with parameters $(q, m, R, \sigma, s, L, \delta)$. The preceding transformation has perfect completeness and an instance-specific recovery R_I such that, for every possibly random composed labeling $\boldsymbol{Y}$, the recovered assignment $\boldsymbol{\pi} := R_I(\boldsymbol{Y})$ in $\Sigma_U^U \times \Sigma_V^V$ satisfies

$$\mathbf{E}[\mathsf{val}_I(\boldsymbol{\pi})] \geq \frac{\mathbf{E}[\mathsf{val}_{I \circledast \mathcal{D}}(\boldsymbol{Y})] - \delta}{L}. \tag{40}$$

If I and $\widetilde{I}$ differ in one predicate or projection, their composed instances differ only in at most $d_U R$ left predicates. The transformation and recovery have sensitivity-transfer parameters satisfying

$$C_{\mathcal{T},\mathrm{Comp}} = d_U R, \qquad C_{\mathcal{R},\mathrm{Comp}} = 1, \qquad D_{\mathcal{R},\mathrm{Comp}} \leq d_U + 1. \tag{41}$$

The structural bounds are

$$\begin{gathered} |U'| = |V|R, \quad |V'| \leq |U|m, \quad |E'| = |E|Rq, \qquad d_{U'} \leq d_V q, \quad d_{V'} \leq d_U R q, \\ |\Sigma'_V| = |\sigma|, \qquad \log|\Sigma'_U| = d_V q \log|\sigma|. \end{gathered} \tag{42}$$

Both sides have positive degree after the common deletion above. With the decoder of Lemma 10.3 and $d_V \leq M$, the local description complexity, left degree, and logarithm of the left alphabet size are, for sufficiently large M, at most

$$C d_V M^{1/16} \tag{43}$$

for an absolute constant C. These local bounds do not involve the whole proof length m.

Proof. Fix total orders on the finite candidate sets and label sets. For each u and proof $p_u \in \sigma^m$, choose the first list $Y_{I,u}(p_u) \subseteq \mathcal{N}_I(u)$ that satisfies robust soundness. Define a representative map $\mathrm{rep}_{I,u} : \mathcal{N}_I(u) \to \Sigma_U$ such that, for each $y \in \mathcal{N}_I(u)$, the label $\mathrm{rep}_{I,u}(y)$ is the first label in the fixed order that satisfies P_u and has projection tuple y. The list and representative map depend only on p_u, P_u, and the projections incident to u, so they are retained when these local data are unchanged.

Given a composed labeling Y, let $p_u \in \sigma^m$ consist of its right labels at (u, z), filling deleted positions with the fixed default. Recover the left label of I at u by choosing a uniform member of $Y_{I,u}(p_u)$ and returning its representative, or a fixed default if the list is empty. Independently, choose a uniform seed $\rho \in \mathscr{R}$. At each v, apply the local decoders to the rows of the composed left label at (v, ρ). If their outputs agree on a non-$\perp$ symbol, recover that symbol at v; otherwise use a fixed default. This defines R_I from labelings of $I \circledast \mathcal{D}$ to distributions on $\Sigma_U^U \times \Sigma_V^V$, the assignments to I.

Completeness. A satisfying labeling of I supplies a tuple in $\mathcal{N}_I(u)$ at each u. Choose an honest proof for that tuple and give its symbols to the right vertices (u, z). At (v, ρ), fill every row with the corresponding queried symbols. All decoders return the label of I at v, every left predicate accepts, and every projection is satisfied.

Soundness. Fix the composed labeling Y and sample $\boldsymbol{\pi} \sim R_I(Y)$. A uniform composed incidence is obtained by choosing a uniform incidence $e = (u, v)$ of I, a uniform seed ρ, and a uniform query occurrence h. Let $z_{v,\rho}$ be the common non-$\perp$ output when the left predicate at (v, ρ) accepts. Call the row w_e at (v, ρ) bad if the predicate at (v, ρ) accepts but $z_{v,\rho} \notin \{y_{j(e)} : y \in Y_{I,u}(p_u)\}$. A bad row w_e lies in the set $\mathrm{Bad}_{Y_{I,u}(p_u)}$ from Definition 10.2. Consequently the fraction of matching queried coordinates in that row is at most

$$\mathrm{agr}(p_u|_{I_{u,j(e),\rho}}, \mathrm{Bad}_{Y_{I,u}(p_u)}(g_{C_u,j(e),\rho}, j(e))).$$

Conditionally on u, the incidence index $j(e)$ is uniform in $[k_u]$. Robust soundness, averaged with weights $k_u/|E|$, therefore shows that bad rows contribute at most δ to the composed value. If the predicate at (v, ρ) accepts and w_e is not bad, some candidate $y \in Y_{I,u}(p_u)$ satisfies $y_{j(e)} = z_{v,\rho}$. The projection of the recovered left label along e equals $z_{v,\rho}$ with probability at least $1/L$. Since a row's matching fraction is at most one, averaging yields

$$\mathbf{E}[\mathsf{val}_I(\boldsymbol{\pi}) \mid Y] \geq L^{-1}(\mathsf{val}_{I \circledast \mathcal{D}}(Y) - \delta).$$

Averaging over Y proves the stated bound for expected value. This argument uses the robust soundness bound in (37), which controls agreement with nonrejecting local words decoding outside the list. Ordinary list-decoding soundness alone would not bound the partially matching bad rows.

Instance and recovery perturbations. Changing the predicate at u or a projection incident to u in I changes only C_u. It can therefore change predicates only at (v, ρ) with v incident to u, at most $d_U R$ vertices. Query-obliviousness keeps every edge and projection fixed. Changing one composed right symbol changes the recovered left label only at the corresponding vertex u of I. Changing one composed left label at (v, ρ_0) can change the recovered right label of I only at v, and only if the sampled seed is ρ_0. These couplings give costs at most one and $1/R$, respectively, and hence $C_{\mathcal{R},\mathrm{Comp}} = 1$ by a coordinatewise hybrid.

If the predicate or an incident projection at u in I changes while the composed labeling is fixed, use the same sampled seed and the same candidate choices at every other left vertex of I. Only the recovered left label at u and the recovered right labels at its at most d_U neighbors can change. This gives $D_{\mathcal{R},\mathrm{Comp}} \leq d_U + 1$, independently of m.

Table 3: Parameters of the composition in Theorem 10.4.

Quantity	Bound after composition
Left / right vertices	$\lvert V\rvert R$ / at most $\lvert U\rvert m$
Left / right degree	at most $d_V q$ / at most $d_U Rq$
Left / right alphabet	$\sigma^{d_V q}$ / σ
Expected recovered value	at least (expected composed value $-\ \delta)/L$
Sensitivity-transfer parameters	$C_{\mathcal{T}} = d_U R$, $C_{\mathcal{R}} = 1$, $D_{\mathcal{R}} \le d_U + 1$
Local description complexity (our decoder)	$O(d_V M^{1/16})$

Local circuits and size. The bounds in (42) follow directly from the construction. At a composed left vertex (v, ρ), a tuple of incident right labels belongs to $\mathcal{N}_{I \circledast \mathcal{D}}(v, \rho)$ exactly when its rows $w_e \in \sigma^q$, indexed by incidences $e = (u, v)$ at v, give the same non-$\perp$ value for every $g_{C_u, j(e), \rho}(w_e)$. A circuit runs at most d_V local circuits of size s, reads $d_V q$ proof symbols, and checks equality of the decoded outputs. Its encoding length is bounded by

$$O\big(d_V(s + q\log(2|\sigma|) + \log(2|\Sigma_V|))\log(2 + d_V + q + s + \log|\sigma|)\big).$$

Parallel incidences are separate input coordinates; the same test is valid for arbitrary tuples, whether or not they arise from a global labeling. Under (39), $d_V \le M$, and the input alphabet bound, the circuit encoding length, left degree, and $\log|\Sigma'_U|$ are at most $C d_V M^{1/16}$ for sufficiently large M. □

10.3 Local Circuits for the Algebraic Decoder

We now construct the decoder used above. The algebraic dPCP of [15, Theorem 6.5] supplies robust soundness with query bound $n^{1/8}$ for circuits of complexity at most n. We bound the encoding length of its local decoding circuits by counting their operations. We then choose the field size so that both the query bound and the local circuit bound are at most n^ν, for any constant $\nu > 0$. The latter bound is needed in (43).

Lemma 10.5 (Query and local circuit bounds for the algebraic decoder)**.** For every constant $\nu > 0$, there are constants $\alpha_\nu, \gamma_\nu, A_\nu > 0$ such that, for all sufficiently large n, all $n^{-\alpha_\nu} \le \delta \le 1/2$, and every alphabet $|\Sigma| \le n^{\gamma_\nu}$, circuits over Σ of complexity at most n have a robust dPCP with perfect completeness and

$$L \le 2/\delta, \qquad q \le n^\nu, \qquad s \le n^\nu, \qquad m,\ |\mathscr{R}|,\ |\sigma| \le n^{A_\nu}.$$

Here s is the local circuit complexity from Definition 10.1. The proof length, number of queries, seed set, and proof alphabet can be chosen uniformly for the size and alphabet parameters; the input arity may vary. Query-obliviousness is not asserted.

Proof. Use the algebraic decoder in [15, Theorem 6.5 and Section 6.1.2]. Write $\mathbb{F}$ for its field, $Q = |\mathbb{F}|$ for the field size, a for the ambient dimension, and d for the encoding degree. The proof is a table of values of a function $f : \mathbb{F}^a \to \mathbb{F}$, intended to be a low-degree encoding of a satisfying input together with the auxiliary proof used by the algebraic verifier. Input coordinate j is represented by a designated point $x_j \in \mathbb{F}^a$.

For each requested coordinate j and seed, the decoder reads values at the points of a polynomial map $\Psi : \mathbb{F}^b \to \mathbb{F}^a$, where b is an absolute constant. The image contains x_j, the affine plane used for the low-degree test, and the curve containing the queries of the underlying algebraic verifier. This parameterized image is

the *manifold* in the cited construction: it is formed by taking scalar linear combinations of points on the plane and curve, and then combining these with x_j. The query coordinates are indexed by its prescribed parameter grid; distinct parameters may query the same proof position. The parameterized formulation is described in [30, full version, Section 10.3, Construction 2]: edges are indexed by parameter tuples, including when two tuples have the same image. We keep these query occurrences separately, so the agreement in Definition 10.2 uses the same sampling of parameters. The robust guarantee is the agreement inequality immediately after Eq. (6.1) in [15, Section 6.1.2]. That inequality precedes the modification making query frequencies equal. We omit this modification: our composition averages over incidences at each old left vertex and does not require equal query frequencies at proof positions.

To bound the local circuit complexity, fix the input circuit and seed. The local computation has three parts.

1. Test membership in the local polynomial space prescribed by the construction. It contains $P \circ \Psi$ for encoding polynomials $P \in \mathbb{F}[x_1, \ldots, x_a]$ of degree at most d. The individual degrees of its members in the b parameters are at most $\Delta = \mathrm{poly}(a, d, \log n, \delta^{-1})$. The query grid uses coordinate sets obtained from $\mathbb{F}$ by excluding at most $\mathrm{poly}(a, d, \log n, \delta^{-1})$ prescribed values. Choose Q large enough that each coordinate set has more than Δ elements. Tensor-product interpolation then uniquely determines the local polynomial from these evaluations. This polynomial lower bound on Q is included in the field-size condition below. Fix a basis and an evaluation matrix for the permitted polynomial space. Row reduction, done when constructing the local test, produces a parity-check matrix and a linear interpolation map. Matrix–vector multiplication tests membership and recovers coefficients using $Q^{O(1)}$ field operations. Write $T \in \mathbb{F}[t_1, \ldots, t_b]$ for the recovered polynomial in the b parameters. For repeated proof positions, also check that their supplied values agree. These linear equations preserve honest answers and only remove accepting local words, so they cannot increase agreement with the set of bad local words.
2. The construction prescribes parameter slices whose images under Ψ are the plane and curve used by the verifier in [15, Section 6.1.1]. Apply these parameter substitutions to T, giving a bivariate plane polynomial $T_{\mathrm{pl}} \in \mathbb{F}[s_1, s_2]$. Let d_{pl} be the total-degree bound prescribed for plane answers by the base low-degree test; this bound is polynomial in $a, d, \log n, \delta^{-1}$ and may differ from the encoding degree d. Check that every monomial of total degree greater than d_{pl} has coefficient zero. The evaluation test in the first step already enforces agreement with all supplied point values. For the curve test, let $z_1, \ldots, z_t \in \mathbb{F}^a$ be the query points of the underlying nonrecursive algebraic verifier for this seed. Its construction puts these points on a curve $\Gamma : \mathbb{F} \to \mathbb{F}^a$, with $\Gamma(\lambda_i) = z_i$ at prescribed parameters $\lambda_i \in \mathbb{F}$; see [2, Appendix B.2]. Write $T_\Gamma : \mathbb{F} \to \mathbb{F}$ for the univariate polynomial obtained from T by the parameter substitution for Γ. Evaluate $T_\Gamma(\lambda_i)$ and apply that verifier's acceptance test to the resulting t answers. In particular, any values encoding sum-check messages are obtained from these evaluations of the supplied local polynomial, rather than from an additional transcript. Here t and the degree of Γ are polynomial in $a, d, \log n$. A degree-d encoding restricted to Γ has degree at most $d \deg \Gamma$. The allowed degree of T_Γ is bounded by a fixed polynomial in $a, d, \log n, \delta^{-1}$, so the evaluations also have polynomial cost. The nonrecursive acceptance test consists of polynomially many field equalities, polynomial evaluations, and sums over an encoding grid contained in $\mathbb{F}$. Its cost is $\mathrm{poly}(a, d, Q, \log n, \delta^{-1})$ field operations. All circuit- and seed-dependent data are computed while constructing the test and are constants of the local circuit. Thus the test acts on the recovered local answers; it does not search for a global satisfying extension.
3. The construction also specifies a parameter tuple $\xi_j \in \mathbb{F}^b$ with $\Psi(\xi_j) = x_j$. Evaluate $T(\xi_j)$ and test that the answer is in the input alphabet. Evaluation is a fixed linear map on the coefficients of

T. We identify the alphabet with an initial segment of the field's fixed binary representation, so its membership test has complexity polynomial in $\log Q$.

The parameter grid has $Q^{O(1)}$ points. Under the field-size conditions below, $d, \delta^{-1} \le Q$. With these inequalities and $a = O(\log n)$, there is therefore an absolute constant c, independent of the field-size exponent chosen below, for which both the number of queries and the binary encoding length of the local circuit are bounded by

$$Q^c \log^c n.$$

Field operations have Boolean circuits of size polynomial in $\log Q$; encoding their wiring only increases the absolute constant c. This estimate concerns operations on the supplied local answers. It is stronger than, and is not inferred from, polynomial-time construction of the whole verifier.

We next choose Q so that the number of queries and local circuit complexity are at most n^ν while allowing $\delta \ge n^{-\alpha_\nu}$ for some $\alpha_\nu > 0$. The low-degree and algebraic-verifier conditions in [2, Theorem 17 and Appendix B.2] require the field size to exceed a fixed polynomial in $a, d, \log n, \delta^{-1}$. The sampling and root-counting errors in [15, Section 6.1.2, Eq. (6.1)] impose conditions of the same form, as do the interpolation requirements above. Hence, for one sufficiently large absolute K, all these conditions are met if

$$Q \ge \left(a(d+1)\log(2n)\delta^{-1}\right)^K. \tag{44}$$

For an encoding grid $H \subseteq \mathbb{F}$ with $h = |H|$, the grid H^a has size $h^a = n^{O(1)}$ and the encoding has total degree $O(ah)$. Thus, for any constant $\beta > 0$, choosing the ambient dimension a as a sufficiently large constant permits $d \le n^\beta$. Choose a constant $\theta > 0$ with $c\theta < \nu/2$ and a power of two $Q \in [n^\theta, 2n^\theta]$. Take $\beta, \alpha_\nu > 0$ with $K(\beta + \alpha_\nu) < \theta/2$, and then choose a as above. For $\delta \ge n^{-\alpha_\nu}$, (44) holds for all sufficiently large n. The robust soundness and list bound $L \le 2/\delta$ are those of the cited decoder; decreasing θ changes the permitted α_ν and polynomial proof-length exponent, while the number b of query parameters remains an absolute constant. The local complexity bound $Q^c \log^c n$ is at most n^ν. The construction supports any input alphabet contained in the field. Decrease γ_ν so that $n^{\gamma_\nu} \le Q$. The proof length and seed-set size are polynomial in n. Padding the basic arithmetization gives the same proof length, number of queries, seed set, and proof alphabet for all input circuits with the prescribed size and alphabet parameters. This proves the lemma. □

10.4 Making Query Locations Independent of the Circuit

We prove Lemma 10.3 by applying Lemma 10.5 to a universal circuit. The universal circuit checks an encoded circuit description supplied as part of its input. The decoder then recovers one symbol of this encoding and compares it with the corresponding symbol for the intended circuit. We show that this modification preserves robust soundness while keeping the queries fixed.

Proof of Lemma 10.3. Pad the description of each circuit C to length M and denote the resulting M-bit string by p_C; invalid circuit descriptions are rejected. Set $\zeta = \delta^2/16$, and fix the code from Proposition 5.1

$$E : \{0,1\}^M \longrightarrow \Gamma^K, \qquad K = O(M\delta^{-4}), \quad |\Gamma| = O(\delta^{-4}),$$

whose distinct codewords have relative agreement at most ζ. Encoding, testing whether a word is a codeword, and decoding valid codewords have polynomial-size circuits.

Since the circuit encoding explicitly lists all input terminals, every circuit of complexity at most M has input arity $k \le M$. For each such arity, construct a *fixed* universal circuit $U_k : (\Sigma \times \Gamma)^{kK} \to \{0,1\}$.

Write its inputs as $w_{j,t} = (a_{j,t}, z_{j,t})$, $j \in [k]$, $t \in [K]$. The circuit requires that $a_{j,t}$ be independent of t, that $z_{j,t}$ be independent of j, and that the resulting string $(z_t)_{t\in[K]}$ be a codeword $E(p)$ of a valid circuit description. After decoding p, it evaluates the circuit described by p on $(a_j)_{j\in[k]}$ and requires acceptance. A universal circuit is obtained, for example, by evaluating the gates specified by p in order and selecting each gate's inputs with fixed multiplexers. Its size is polynomial in the description length M. For a circuit $C : \Sigma^k \to \{0,1\}$ with padded description $p_C \in \{0,1\}^M$ and an input $y \in \Sigma^k$, define

$$w(C, y) := \big((y_j, E(p_C)_t)\big)_{j\in[k],\, t\in[K]} \in (\Sigma \times \Gamma)^{kK},$$

where the coordinates are indexed by $(j, t) \in [k] \times [K]$. Then, with C ranging over these circuits,

$$U_k^{-1}(1) = \Big\{ w(C, y) : y \in \Sigma^k,\ C(y) = 1 \Big\}. \tag{45}$$

The universal circuit is independent of the particular C to be decoded.

Under the coarse assumptions $\delta \geq M^{-1}$ and $|\Sigma| \leq M$, all these universal circuits have complexity at most M^{c_U} for one absolute c_U. Fix an integer $C_U > c_U + 1$ and pad their descriptions to length $n_U = \lceil M^{C_U} \rceil$. We will apply Lemma 10.5 with $\nu = 1/(64C_U)$ and soundness error $\delta/2$ to these circuits; let $\mathcal{D}_U$ denote the resulting decoder for U_k. Write $\mathscr{R}_U$ for its seed set and α_ν, γ_ν for its constants. Choose positive $\alpha_*, \gamma_* < 1$ so small that

$$\alpha_* < C_U \alpha_\nu / 2, \qquad \gamma_* + 4\alpha_* < C_U \gamma_\nu / 2.$$

Then the soundness error $\delta/2$ of $\mathcal{D}_U$ is at least $n_U^{-\alpha_\nu}$, and its input alphabet $\Sigma \times \Gamma$ has size at most $n_U^{\gamma_\nu}$, for all sufficiently large M. Thus the lemma applies. The number of queries and local circuit complexity of $\mathcal{D}_U$ are $O(M^{1/64})$; its other parameters are polynomial in M.

We now construct the decoder for C by checking one symbol of $E(p_C)$. On target coordinate j, independently choose $t \in [K]$ uniformly and use $\mathcal{D}_U$ to decode coordinate (j, t). If its answer is (a, z), return a exactly when $z = E(p_C)_t$, and return $\perp$ otherwise. Use seed set $[K] \times \mathscr{R}_U$. The queries use U_k, not C. The additional local computation is one comparison of $O(\log M)$ bits, so $q, s \leq M^{1/32}$ for sufficiently large M. The parameters are common because the universal circuit descriptions are padded to the same length. Completeness follows by encoding the table in (45) for the given pair (C, y).

It remains to check *robust* soundness. Fix an arbitrary proof $p \in \sigma^m$. By robust soundness of $\mathcal{D}_U$, there is a list of satisfying assignments $w(C_a, y^{(a)})$ to U_k, indexed by $a \in [L]$, where $L \leq 4/\delta$. The inputs $y^{(a)}$ whose associated circuit description equals p_C form the set

$$Y_C := \{ y^{(a)} : a \in [L],\ p_{C_a} = p_C \} \subseteq C^{-1}(1) \subseteq \Sigma^k.$$

Define the set of positions where $E(p_{C_a})$ agrees with $E(p_C)$ for some different description $p_{C_a} \neq p_C$ by

$$T_{\mathrm{coll}} := \{ t \in [K] : \exists a \in [L] \text{ with } p_{C_a} \neq p_C \text{ and } E(p_{C_a})_t = E(p_C)_t \}.$$

The code distance and a union bound give

$$\Pr_t[t \in T_{\mathrm{coll}}] \leq L\zeta \leq \delta/4. \tag{46}$$

For $t \notin T_{\mathrm{coll}}$, if a local answer word makes the decoder for C output outside $\{\perp\} \cup \{y_j : y \in Y_C\}$, then $\mathcal{D}_U$ returns a pair different from the (j, t)-entry of every listed table $w(C_a, y^{(a)})$. Thus the set Bad_{Y_C} of nonrejecting local words decoding outside the retained list is a subset of the corresponding set for $\mathcal{D}_U$, for each j, t and seed of $\mathcal{D}_U$. On T_{coll}, agreement is at most one. Let $I \in [m]^q$ denote the query tuple for target

coordinate j and seed (t, ρ), where $\rho \in \mathscr{R}_U$. Averaging over independent uniform $j \in [k]$, $t \in [K]$, and $\rho \in \mathscr{R}_U$ proves

$$\mathbf{E}\,\mathrm{agr}(p|_I, \mathrm{Bad}_{Y_C}) \le \delta/2 + \Pr_t[t \in T_{\mathrm{coll}}] \le 3\delta/4 \le \delta.$$

This also covers an empty retained list. The list size remains at most $4/\delta$, completing the proof. □

11 Iterated Composition

We now combine the initial family from Section 9 with the construction of Section 10. We repeatedly replace large local checking circuits by smaller ones, together with alphabet and degree reduction, until the degree and alphabet bounds depend only on the required value $1/g$.

The main issue is to control the entire sequence of reductions. We bound the checking-circuit sizes so that each successive step remains applicable, and we bound the total loss in sensitivity so that a polynomial lower bound survives. These estimates prove Theorem 11.1 with a sensitivity exponent independent of g. In the formal statement below, $n = |U|$ counts left vertices and $1/g$ is the required expected fraction of satisfied edges on satisfiable inputs.

Theorem 11.1 (Formal version of Theorem 1.2)**.** There exist absolute constants $B, \delta > 0$ and $n_{\min} \in \mathbb{N}$ such that, for every integer $n \ge n_{\min}$ and every real g with $1 < g \le \log^B n$, there is a swap-closed family of left-predicate Label Cover instances with $|U| = n$. Every algorithm defined on this family that, on each satisfiable member, outputs a possibly random assignment of expected value at least $1/g$ has sensitivity $\Omega(n^\delta)$. Moreover, the instances can be chosen so that $|V| \le n \log^{O(1)} n$, the left and right degrees and the right alphabet size are $g^{O(1)}$, and $|\Sigma_U| \le \exp\big(g^{O(1)}\big)$.

11.1 Transfer Parameters and Paths of Predicate Swaps

We track the three sensitivity-transfer parameters together with bounds on the local description complexity along paths of predicate swaps, as defined in Section 2. These paths let us apply the next decoder using one common complexity bound at every intermediate instance. For the existence statements we may also choose the circuit with the lexicographically first minimum-length encoding for each set $\mathcal{N}_I(u)$ of projection tuples of left labels satisfying P_u, so unchanged local data always have unchanged chosen circuits. No bound on the time needed to make this canonical choice is required for our sensitivity lower bounds.

We retain a map $\mathcal{T}_i$ on the 2-CSP family $\mathcal{F}_{N_{\mathrm{base}}}$ from Lemma 2.3 and recovery maps $\mathcal{R}_{i,b}$. Here N_{base} counts the variables of Φ_b. As in Section 9, $b = (b_e)_{e \in E}$ records its constraint relations on the fixed edge set E. We abbreviate $\mathcal{T}_i(\Phi_b)$ by $\mathcal{T}_i(b)$ and write $\mathcal{I}_i = \{\mathcal{T}_i(\Phi) : \Phi \in \mathcal{F}_{N_{\mathrm{base}}}\}$ for the image family, with common vertex sets U_i, V_i. The map $\mathcal{R}_{i,b}$ takes distributions on labelings of $\mathcal{T}_i(b)$ to distributions on Σ_{src}^X, where X is the common variable set of the original 2-CSPs. Write

$$C_{\mathrm{mult},i} := C_{\mathcal{T}}^{(i)} C_{\mathcal{R}}^{(i)}, \qquad D_i := D_{\mathcal{R}}^{(i)}.$$

Here $C_{\mathrm{mult},i}$ is the multiplicative transfer loss and D_i is the additive recovery bound accumulated through stage i. Besides the sensitivity-transfer properties in Definition 3.1, we keep a path of predicate swaps of length at most $C_{\mathcal{T}}^{(i)}$ between the images of two 2-CSPs differing in one relation, and a common bound on the local description complexity of every instance on these paths. If one further transformation replaces each predicate swap along these paths by a path of at most t predicate swaps, has recovery Lipschitz bound r, and

changes the recovered distribution by EMD at most d under one such source swap with the supplied output held fixed, the composition rule is

$$\begin{aligned} C_{\mathcal{T}}^{(i+1)} &\le C_{\mathcal{T}}^{(i)} t, \qquad C_{\mathcal{R}}^{(i+1)} \le C_{\mathcal{R}}^{(i)} r, \\ C_{\mathrm{mult},i+1} &\le C_{\mathrm{mult},i} tr, \qquad D_{i+1} \le D_i + C_{\mathrm{mult},i} d. \end{aligned} \tag{47}$$

This is Lemma 3.4 applied along the retained paths. Explicitly, concatenating the transformed paths gives $C_{\mathcal{T}}^{(i+1)} \le C_{\mathcal{T}}^{(i)} t$. For the recovery bounds, couple recoveries along the same path while the final distribution is fixed, and then apply $\mathcal{R}_{i,b}$ to the recovered distributions. The next decoder is applied only to the image family and the retained paths, whose local circuits have the stated bound. In the final sensitivity argument, the target algorithm is defined on the larger family $\mathsf{SwapClo}(\mathcal{I}_i)$, the closure under both predicate and projection swaps. We do not apply the decoder to this entire closure, so its other instances need no local circuit bound.

11.2 Stage-Zero Initialization by Polynomial Restrictions

We choose the number N_{base} of 2-CSP variables so that every instance in $\mathcal{I}_0$ has at most n_0 vertices on each side. The degrees, right alphabet size, logarithm of the left alphabet size, and local description complexity will then be polylogarithmic in n_0, as will the multiplicative transfer loss $C_{\mathrm{mult},0}$ and the additive bound D_0. For stage i, write S_i for its local description complexity, $d_{U,i}, d_{V,i}$ for its maximum left and right degrees, and $\Sigma_{U,i}, \Sigma_{V,i}$ for its two alphabets.

Proposition 11.2 (Initialization by polynomial restrictions)**.** There are absolute constants $b_0, B_*, K_0 > 0$ such that, for all sufficiently large n_0, the following holds. Put

$$\varepsilon_0 = (\log n_0)^{-1}, \quad N_{\mathrm{base}} = \lfloor n_0^{1/B_*} \rfloor, \quad \gamma_0 = 9\sqrt{\varepsilon_0}.$$

There is a family $\mathcal{I}_0$ with a common graph, alphabets, and projections, and an instance map from $\mathcal{F}_{N_{\mathrm{base}}}$, given by

$$\Phi_b \longmapsto J_b \longmapsto \widehat{J}_b = \mathcal{T}_{\mathrm{DR}}(J_b; d_{\gamma_0}) = \mathcal{T}_0(b). \tag{48}$$

The number of vertices on each side is at most n_0, all vertices have positive degree, and

$$S_0,\ d_{U,0},\ d_{V,0},\ |\Sigma_{V,0}|, \log|\Sigma_{U,0}| \le \varepsilon_0^{-b_0}.$$

For every b, the recovery $\mathcal{R}_{0,b}$ maps any distribution on labelings of $\mathcal{T}_0(b)$ with expected value at least $9\sqrt{\varepsilon_0}$ to a distribution on assignments to Φ_b with expected value greater than $1 - \eta_{\mathrm{src}}$. Satisfiable 2-CSPs have satisfiable images. The transfer parameters obey

$$C_{\mathrm{mult},0},\ D_0 \le K_0 \varepsilon_0^{-5/2} \log(2/\varepsilon_0). \tag{49}$$

Images of 2-CSPs differing in one relation admit paths of predicate swaps of length at most $C_{\mathcal{T}}^{(0)}$, along which every instance has local description complexity at most $\varepsilon_0^{-b_0}$.

Proof. Apply Theorem 9.1 with $N = N_{\mathrm{base}}$ and $\gamma = \gamma_0$. Choose B_* larger than the absolute exponent in the theorem's $N^{O(1)}$ bound on the number of vertices on each side; then the number of vertices on each side is at most n_0, and $(\log(d_0 N_{\mathrm{base}}/2))^{-1} \le \gamma_0 \le 1/2$ for sufficiently large n_0. Its local bounds are polynomial in $\log N_{\mathrm{base}}$ and γ_0^{-1}, hence at most $\varepsilon_0^{-b_0}$ for one constant b_0. Its remaining conclusions give completeness, value recovery, and the paths of predicate swaps with this same local-description bound. Substituting $\gamma_0 = 9\sqrt{\varepsilon_0}$ into (22) proves (49). □

11.3 Decreasing the Bound on Local Description Complexity

One round first composes the current Label Cover instance with a decoder, then reduces its right alphabet and degree. Recovery runs in the reverse order. We choose the value thresholds first, then the circuit bounds that make both transformations applicable at every round.

At stage i, recovery to the original 2-CSP requires expected value $\chi_i := 9\sqrt{\varepsilon_i}$. Use the decoder of Lemma 10.3 with error $\xi_i := \varepsilon_i^{1/4}$ and list size at most $4/\xi_i$. By Theorem 10.4, expected value at least

$$\xi_i + (4/\xi_i)\chi_i = 37\varepsilon_i^{1/4}$$

in the composed instance suffices to recover value at least χ_i. We therefore take $\varepsilon_{i+1} = 40\varepsilon_i^{1/4}$. By Lemma 5.3, value at least $9\sqrt{\varepsilon_{i+1}}$ after alphabet and degree reduction recovers value at least ε_{i+1} before that reduction. Thus the successive lower bounds on expected recovered value are

$$9\sqrt{\varepsilon_{i+1}} \xrightarrow{\text{reduction recovery}} \varepsilon_{i+1} \xrightarrow{\text{composition recovery}} 9\sqrt{\varepsilon_i}.$$

For the small parameters used below, $\varepsilon_{i+1} > \varepsilon_i$: the required value increases, while an upper bound ε_i^{-c} on local circuit size decreases, for a constant exponent $c > 0$ to be chosen. The remaining choice of exponents ensures that the smaller circuit bound survives alphabet and degree reduction.

To specify the bounds that each stage must preserve, we choose the exponents before starting the iteration. Choose

$$a > \max\{b_0, 7\}.$$

The exponent a controls the bounds on the right degree and right alphabet size. Let α_*, γ_* be the constants in Lemma 10.3, and let c_{red} be the constant exponent in Lemma 5.4, increased once to be at least 7. The exponent c will control the bounds on local description complexity, left degree, and the logarithm of the left alphabet size. Choose $c > b_0 + a + 1$ so large that

$$(c+1)\alpha_* > 1/4, \qquad (c+1)\gamma_* > a, \qquad \frac{c}{4} + 4a + c_{\text{red}} + 6 < c. \tag{50}$$

Such a choice exists: the coefficient of c on the left of the last inequality is $1/4$. The exponent a and the decoder constants have already been fixed and do not change when c is chosen.

Definition 11.3 (Stage data)**.** For a parameter ε_i, a stage consists of the instance map, recoveries on distributions, and paths of predicate swaps above, with the following invariants. The graph, alphabets, and projections are common, all vertices have positive degree, and

$$\begin{aligned} d_{V,i},\ |\Sigma_{V,i}| &\le \varepsilon_i^{-a}, \\ S_i,\ d_{U,i},\ \log|\Sigma_{U,i}| &\le \varepsilon_i^{-c}. \end{aligned} \tag{51}$$

Every instance on the retained paths has local description complexity at most ε_i^{-c}. If Φ_b is satisfiable, so is $\mathcal{T}_i(b)$. For every collection of relations b, the recovery $\mathcal{R}_{i,b}$ maps any distribution on labelings of $\mathcal{T}_i(b)$ with expected value at least $\chi_i = 9\sqrt{\varepsilon_i}$ to a distribution on assignments to Φ_b with expected value at least $1 - \eta_{\text{src}}$.

When applying the decoder, we use the complexity budget $\lceil \varepsilon_i^{-(c+1)} \rceil$, leaving slack over the maintained bound ε_i^{-c}.

Proposition 11.4 (One-stage update)**.** There is an absolute $c_0 > 0$ such that the following holds. Suppose stage i has the preceding data, put

$$\xi_i = \varepsilon_i^{1/4}, \qquad \varepsilon_{i+1} = 40\varepsilon_i^{1/4}, \qquad M_i = \lceil \varepsilon_i^{-(c+1)} \rceil, \tag{52}$$

and assume $\varepsilon_{i+1} \le c_0$. Let $\mathcal{D}_i$ be the query-oblivious decoder from Lemma 10.3 with complexity bound M_i, error ξ_i, and input alphabet $\Sigma_{V,i}$. Write q_i, R_i, L_i, and σ_i for its number of queries, seed-set cardinality, bound on the list size, and proof alphabet, respectively. Applying each transformation to every instance of its input family, define

$$\mathcal{J}_{i+1} := \mathcal{I}_i \circledast \mathcal{D}_i, \qquad \mathcal{I}_{i+1} := \mathrm{red}_{\varepsilon_{i+1}}(\mathcal{J}_{i+1}).$$

This gives stage-$(i+1)$ data with the same constants a, c. For an absolute constant $b_{\mathrm{step}} \ge 1$, the combined transformation has transfer bounds (t_i, r_i, d_i) along paths of predicate swaps and size growth satisfying

$$t_i, r_i, d_i \le \varepsilon_i^{-b_{\mathrm{step}}}, \qquad \max\{|U_{i+1}|, |V_{i+1}|\} \le \max\{|U_i|, |V_i|\}\varepsilon_i^{-b_{\mathrm{step}}}. \tag{53}$$

The exponent b_{step} is independent of i, n_0, and the final target value.

Proof. We first verify the decoder conditions and the new local complexity. The first two inequalities in (50) ensure, for all sufficiently small ε_i, that

$$S_i \le M_i, \quad d_{V,i} \le M_i, \quad |\Sigma_{V,i}| \le M_i^{\gamma_*}, \quad \xi_i \ge M_i^{-\alpha_*}.$$

When the checking circuits at left vertices of a stage-i instance have different input arities, use the corresponding universal circuits from Lemma 10.3; their padded description length and all decoder parameters are common. Thus an incidence chosen uniformly at random at a left vertex u corresponds to a uniformly random input coordinate of the checking circuit at u.

For the intermediate family $\mathcal{J}_{i+1}$, write S_{i+1}^J for a common bound on local description complexity, $d_{U,i+1}^J$ and $\delta_{V,i+1}^J$ for its maximum left and minimum right degrees, and $\Sigma_{U,i+1}^J, \Sigma_{V,i+1}^J$ for its left and right alphabets. By (43), these quantities satisfy

$$S_{i+1}^J,\ d_{U,i+1}^J,\ \log|\Sigma_{U,i+1}^J| \le C\varepsilon_i^{-a} M_i^{1/16}, \qquad \log|\Sigma_{V,i+1}^J| \le A_* \log M_i. \tag{54}$$

By Lemmas 5.3 and 5.4, reduction makes the new right degree $O(\varepsilon_{i+1}^{-1})$ and right alphabet size $O(\varepsilon_{i+1}^{-6})$. Since $a > 7$, both are at most ε_{i+1}^{-a} after decreasing c_0.

The same lemmas and (54) give

$$\begin{aligned} \max\{S_{i+1}, d_{U,i+1}, \log|\Sigma_{U,i+1}|\} &\le C_1 \varepsilon_i^{-a} M_i^{1/16} \varepsilon_{i+1}^{-c_{\mathrm{red}}} \log^{C_2} M_i \\ &\le C_3 \varepsilon_{i+1}^{-(c/4+4a+c_{\mathrm{red}}+5)} \le \varepsilon_{i+1}^{-c}. \end{aligned} \tag{55}$$

For the second inequality, $M_i \le 2\varepsilon_i^{-(c+1)}$ and $\log^{C_2} M_i \le \varepsilon_i^{-1}$ for sufficiently small ε_i. Substituting $\varepsilon_i = (\varepsilon_{i+1}/40)^4$ gives an exponent of $c/4 + 4a + c_{\mathrm{red}} + 17/4$, below $c/4 + 4a + c_{\mathrm{red}} + 5$. The last inequality follows from (50) after decreasing c_0 again to absorb the multiplicative constants. The reduction leaves the left alphabet unchanged. By Theorem 10.4 and lemma 5.4, all instances in $\mathcal{I}_{i+1}$ have the same graph, alphabets, and projections, and all vertices have positive degree.

Value recovery. For each b, put $J_{i+1,b} = \mathcal{T}_i(b) \circledast \mathcal{D}_i$ and $\mathcal{T}_{i+1}(b) = \mathrm{red}_{\varepsilon_{i+1}}(J_{i+1,b})$. Write $Q_{i,b}$ for the recovery from labelings of $\mathcal{T}_{i+1}(b)$ to distributions on labelings of $J_{i+1,b}$ through alphabet-and-degree reduction, and $S_{i,b}$ for the recovery from labelings of $J_{i+1,b}$ to distributions on labelings of $\mathcal{T}_i(b)$ through composition. For a distribution μ on labelings of $\mathcal{T}_{i+1}(b)$, define

$$\mathcal{R}_{i+1,b}(\mu) := \mathcal{R}_{i,b}\big((S_{i,b})_{\#}(Q_{i,b})_{\#}\mu\big).$$

To supply a sample requested by $\mathcal{R}_{i,b}$, draw $Y \sim \mu$, then $Z \sim Q_{i,b}(Y)$, and return a sample from $S_{i,b}(Z)$. Use fresh randomness for each draw. Completeness is preserved by both transformations. If μ has expected value at least $9\sqrt{\varepsilon_{i+1}}$, then Lemma 5.3 implies that $(Q_{i,b})_{\#}\mu$ has expected value at least ε_{i+1} on $J_{i+1,b}$. The decoder has $L_i \le 4/\xi_i$, and

$$\xi_i + L_i\chi_i \le \varepsilon_i^{1/4} + 36\varepsilon_i^{1/4} \le \varepsilon_{i+1}.$$

Hence Theorem 10.4 implies that $(S_{i,b})_{\#}(Q_{i,b})_{\#}\mu$ has expected value at least χ_i on $\mathcal{T}_i(b)$. Applying $\mathcal{R}_{i,b}$ therefore gives $\mathcal{R}_{i+1,b}(\mu)$, a distribution on assignments to Φ_b with expected value at least $1 - \eta_{\mathrm{src}}$.

Paths of predicate swaps and transfer. Composition sends one predicate swap at u to at most $d_{U,i}R_i$ predicate swaps. Reduce these images with the same code and expanders. By (4), reduction transports these predicates unchanged, and its recovery rule is unchanged along this path, so its additive recovery bound is zero. The combined step therefore has

$$t_i = d_{U,i}R_i, \qquad r_i = 1 + \frac{d^J_{U,i+1}}{\delta^J_{V,i+1}}, \qquad d_i \le d_{U,i} + 1. \tag{56}$$

The minimum intermediate right degree is at least one. The proof length and number of seeds of $\mathcal{D}_i$ are polynomial in M_i, and $d^J_{U,i+1} \le d_{V,i}q_i$. Consequently, t_i, r_i, d_i are all at most $\varepsilon_i^{-b_{\mathrm{step}}}$ for an absolute constant b_{step}.

Replace the affected composed predicates one at a time. Every local circuit on the resulting path comes from one of the adjacent images and satisfies (54). The local reduction construction is vertexwise, so (55) bounds the local description complexity of the image of each instance on this path by ε_{i+1}^{-c}. Concatenation along the retained stage-i path gives the required stage-$(i+1)$ path. Applying (47) along that path proves the sensitivity-transfer bounds, including

$$C_{\mathrm{mult},i+1} \le C_{\mathrm{mult},i}t_ir_i, \qquad D_{i+1} \le D_i + C_{\mathrm{mult},i}d_i.$$

Finally, combining the bounds on the numbers of vertices and edge incidences in (42) with the code-length and degree bounds for reduction gives the size estimate in (53) after increasing b_{step}. □

11.4 The Iterated Family and the Final Size

We show that the accumulated size growth and sensitivity losses are polylogarithmic in the initial size parameter n_0. We then choose the stopping stage for target expected value $1/g$ and pad to a prescribed number of left vertices, retaining a sensitivity lower bound with a constant positive exponent.

Lemma 11.5. Let $\varepsilon_0 = (\log n_0)^{-1}$ and $\varepsilon_{i+1} = 40\varepsilon_i^{1/4}$. There are absolute constants $c_{\mathrm{size}}, c_{\mathrm{sens}}, \delta_0 > 0$ such that, for every sufficiently large n_0 and every i with $\varepsilon_i \le c_0$, there are stage-i data (Definition 11.3) with the following properties.

1. The number of vertices on each side is at most $n_0 \prod_{t=0}^{i} \varepsilon_t^{-c_{\mathrm{size}}}$.
2. The transfer parameters satisfy $C_{\mathrm{mult},i}, D_i \le \prod_{t=0}^{i} \varepsilon_t^{-c_{\mathrm{sens}}}$, and

$$\mathsf{SwapSens}_{9\sqrt{\varepsilon_i}}(\mathsf{SwapClo}(\mathcal{I}_i)) \ge \Omega\left(\frac{n_0^{\delta_0}}{\prod_{t=0}^{i} \varepsilon_t^{-c_{\mathrm{sens}}}}\right).$$

All exponents are independent of i.

Proof. Proposition 11.2 gives the base case with the bounds in (51), since $a, c > b_0$. Proposition 11.4 gives all later stages up to the stated stopping condition. Its size and transfer factors are constant powers of ε_t^{-1}. To bound D_i together with $C_{\mathrm{mult},i}$, put $H_i = 1 + C_{\mathrm{mult},i} + D_i$. The update gives, after increasing a constant exponent b',

$$H_{i+1} \le H_i \varepsilon_i^{-b'}.$$

Together with the initial bounds on the number of vertices on each side, $C_{\mathrm{mult},0}$, and D_0 in Proposition 11.2, this proves the asserted product bounds after increasing c_{size} and c_{sens}.

For any constant $z > 0$, the recursion implies $\varepsilon_t \ge \varepsilon_0^{4^{-t}}$, and hence

$$\prod_{t=0}^{i} \varepsilon_t^{-z} \le \varepsilon_0^{-z\sum_{t=0}^{i} 4^{-t}} \le \varepsilon_0^{-4z/3} = \log^{4z/3} n_0. \tag{57}$$

In particular $D_i = o(N_{\mathrm{base}}^{c_{\mathrm{src}}})$ uniformly in the allowed stages. Let $\mathcal{A}$ be an algorithm on the swap closure with the stated value guarantee, and recover its output distribution to obtain an algorithm on $\mathcal{F}_{N_{\mathrm{base}}}$. On every satisfiable Φ_b, this algorithm has expected value at least $1 - \eta_{\mathrm{src}}$. Thus Lemma 2.3 and theorem 3.2 give

$$\mathsf{SwapSens}(\mathcal{A}, \mathsf{SwapClo}(\mathcal{I}_i)) \ge \frac{N_{\mathrm{base}}^{c_{\mathrm{src}}} - D_i}{C_{\mathrm{mult},i}} \ge \frac{N_{\mathrm{base}}^{c_{\mathrm{src}}}}{2C_{\mathrm{mult},i}}.$$

Since $N_{\mathrm{base}} = \lfloor n_0^{1/B_*} \rfloor$, this proves Item 2, with $\delta_0 = c_{\mathrm{src}}/B_*$ and an adjusted absolute constant. □

Proof of Theorem 11.1. Let n be the prescribed final left-side size. Choose an absolute $K > 4c_{\mathrm{size}}/3 + 2$ and set

$$n_0 = \left\lfloor \frac{n}{\log^K n} \right\rfloor, \qquad \varepsilon_0 = (\log n_0)^{-1}.$$

Fix $0 < B < 1/2$ and $1 < g \le \log^B n$. Put

$$\tau = \min\{(81g^2)^{-1}, c_0/2\}.$$

For all sufficiently large n, uniformly over this range of g, $\varepsilon_0 \le \tau$. Let i_* be the largest index with $\varepsilon_{i_*} \le \tau$. The sequence is increasing while it is below τ, so such an index exists and is finite, and

$$(\tau/40)^4 < \varepsilon_{i_*} \le \tau, \qquad 9\sqrt{\varepsilon_{i_*}} \le 1/g. \tag{58}$$

Apply Lemma 11.5 to this index. The lower bound in (58) shows that both degrees and the right alphabet size are $g^{O(1)}$, and the left alphabet size is at most $\exp\left(g^{O(1)}\right)$. The constants are uniform even for g close to one, where these statements mean an absolute constant times the specified power. By (57), each side has at most $n_0 \log^{4c_{\mathrm{size}}/3} n_0 \le n$ vertices.

Pad with isolated left vertices to make the left side exactly n, using predicates identically one on these new vertices, and take the swap closure of the resulting family. Value and satisfiability are unchanged: both are defined by the edge constraints. The right side is unchanged, hence has size at most n, in particular at most $n \log^{O(1)} n$. An algorithm on the padded swap closure restricts to an algorithm on the unpadded family with the same value and no larger sensitivity, since original predicate and projection swaps commute with padding and restriction contracts Hamming distance. The lower bound from Lemma 11.5, together with (57), is

$$\Omega\left(\frac{n_0^{\delta_0}}{\log^{4c_{\mathrm{sens}}/3} n_0}\right) = \Omega\left(\frac{n^{\delta_0}}{\log^{O(1)} n}\right) = \Omega(n^{\delta})$$

for one constant $0 < \delta < \delta_0$. □

Part IV
Covering Problems and Locality Lower Bounds

In this part, we develop the covering applications of our two Label Cover sensitivity lower bounds. Section 12 proves the Set Cover bounds, and Section 13 transfers them to Dominating Set before deriving the locality consequences and comparing them with previous distributed upper and lower bounds.

12 Sensitivity Lower Bounds for Set Cover

This section proves two sensitivity lower bounds for Set Cover. The IKW route handles approximation ratios up to $\log \Gamma$, where Γ bounds both set size and element frequency, where the sensitivity lower bound scales weakly with Γ. The Dinur–Harsha route gives a polynomial sensitivity lower bound (independent of Γ) for approximation ratios that are a small positive power of the logarithm of the universe size. In both cases, we control set size and element frequency so that the reduction in Section 13 yields sensitivity lower bounds for Dominating Set on graphs of bounded maximum degree. Both proofs use an adaptation of the classical reduction from Label Cover to Set Cover [17, 29], so we first establish the common transformation and recovery statements.

In the *Set Cover problem*, an instance is a pair $(U, \mathcal{F} = (S_1, \ldots, S_m))$, where U is a finite universe and $\mathcal{F}$ is an indexed family of subsets of U. The goal is to find an index set $\mathcal{S} \subseteq [m]$ of minimum size such that the corresponding subfamily $\{S_i : i \in \mathcal{S}\}$ covers U, i.e., $\bigcup_{i\in\mathcal{S}} S_i = U$. We call an instance *feasible* if some subfamily covers U, equivalently if $\bigcup_{i=1}^{m} S_i = U$.

For two instances $I = (U, \mathcal{F} = (S_1, \ldots, S_m))$ and $\tilde{I} = (U, \tilde{\mathcal{F}} = (\tilde{S}_1, \ldots, \tilde{S}_m))$ over the same universe and common index set $[m]$, we write $\mathsf{SwapDist}(I, \tilde{I}) = \sum_{i=1}^{m} |S_i \triangle \tilde{S}_i|$. That is, $\mathsf{SwapDist}(I, \tilde{I})$ counts the number of *single-incidence edits* (additions or deletions of one element–set membership) across all sets; equivalently, it is the Hamming distance between the two incidence matrices. We identify a selected index set $\mathcal{S} \subseteq [m]$ with its indicator vector, and define the Hamming distance by $d_{\mathrm{H}}(\mathcal{S}, \tilde{\mathcal{S}}) := |\mathcal{S} \triangle \tilde{\mathcal{S}}|$. For a (possibly randomized) algorithm A and an instance I, define its sensitivity with respect to incidence edits, denoted again by $\mathsf{SwapSens}(A, I)$, by $\mathsf{SwapSens}(A, I) := \max_{\tilde{I}:\, \mathsf{SwapDist}(I,\tilde{I})=1} \mathsf{EMD}(A(I), A(\tilde{I}))$, where $\tilde{I}$ ranges over instances on the same universe and index set that differ from I by one incidence edit. The earth mover's distance uses d_{H} as the underlying metric on outputs. For a family $\mathcal{I}$, set $\mathsf{SwapSens}(A, \mathcal{I}) := \max_{I\in\mathcal{I}} \mathsf{SwapSens}(A, I)$. For $\alpha \geq 1$, let $\mathsf{SwapSens}_\alpha(\mathcal{I})$ be the minimum $\mathsf{SwapSens}(A, \mathcal{I})$ over all algo-

rithms A defined on all Set Cover instances and satisfying the α-approximation guarantee on every feasible $I \in \mathcal{I}$.

Parameter guide. In each lower-bound proof, n is the prescribed Set Cover universe size. We use s for the source Label Cover size, defined locally: the total number of vertices in the IKW proof and the number of left vertices in the Dinur–Harsha proof.

12.1 Reduction from Label Cover to Set Cover

Following standard reductions from label cover to set cover we use the following set system of [17, 29].

Lemma 12.1 ([17, 29])**.** Let $m, q \geq 1$ and $b \geq 2$ be integers, and let $\ell > 0$ and $0 < \theta < 1$ satisfy

$$b \exp\left(-\frac{\ell}{q(1-\theta)}\right) > mq \ln 2.$$

There is a universe B of size b with m partitions into q indexed parts

$$B = C_{x,1} \sqcup \cdots \sqcup C_{x,q}, \qquad x \in [m],$$

such that, if a collection of at most ℓ indexed parts covers B, then the collection must contain more than θq parts from one partition.

The construction and proof follow [17, Lemma 3.2], which makes the slightly weaker conclusion that any collection of ℓ indexed parts covering B must have at least two parts from one partition. We need the stronger θq bound for our recovery procedure. The proof is standard and deferred to the end of the section.

The Set Cover Instance Fix a left-predicate Label Cover instance $I = (U, V, E, \Sigma_U, \Sigma_V, \mathcal{P}, F)$ whose multigraph is biregular with positive left and right degrees d_L, d_R. Fix a set system $(B, \{C_{i,j}\})$ as in Lemma 12.1 with $m = |\Sigma_V|$, $q = d_R$, and parameters $\ell > 0$, $0 < \theta < 1$, indexing its partitions by the right-alphabet Σ_V. At each $v \in V$, order the incident edges as $e_{v,1}, \ldots, e_{v,d_R}$. The Set Cover instance $\mathcal{T}_{\mathrm{SC}}(I)$ has universe $V \times B$ and sets indexed by

$$\Lambda_I := U \times \Sigma_U.$$

We define $S_{(u,y)}$ in the following way. If $P_u(y) = 0$, i.e., if label y fails the predicate at u, we will leave $S_{(u,y)}$ empty. Otherwise, for each edge $e_{v,i} = (u, v)$ touching u, include $\{v\} \times C_{f_{e_{v,i}}(y),i} \subset V \times B$, i.e.,

$$S_{(u,y)} := \bigcup_{e_{v,i}=(u,v)} \{v\} \times C_{f_{e_{v,i}}(y),i}$$

The Recovery Map Given a target output for the above set cover instance, indexed by $\mathcal{S} \subseteq \Lambda_I$, let

$$L_u := \{y \in \Sigma_U : (u, y) \in \mathcal{S}\}.$$

The recovery $R_{\mathrm{SC},I}$ independently chooses, at each left vertex u, a uniform label from L_u. At each right vertex v, it independently chooses a uniform incident edge $e = (u, v)$, then a fresh uniform label from L_u, and outputs its projection under f_e.[3] Whenever L_u is empty, a fixed default label in Σ_U is used.

[3] We remark one could instead re-use the label for u chosen in the first step, but independence will make the recovery simpler to analyze.

Lemma 12.2 (Size, completeness, and sensitivity)**.** The construction has universe size N_0 and number of sets m_0 given by

$$N_0 = |V||B|, \qquad m_0 = |U||\Sigma_U|.$$

Set sizes are at most $d_L|B|$, and element frequencies are at most $d_R|\Sigma_U|$. If I is satisfiable, $\mathcal{T}_{\mathrm{SC}}(I)$ has a cover of size $|U|$. On a family with common graph and alphabets, the transformation and recovery have sensitivity-transfer parameters

$$C_{\mathcal{T}} = d_L|\Sigma_U||B|, \qquad C_{\mathcal{R}} = 1 + \frac{d_L}{d_R}, \qquad D_{\mathcal{R}} = 1.$$

Proof. The sizes N_0 and m_0 are immediate from the construction. A set indexed by (u, y) is a union of at most d_L sets of size at most $|B|$, so the set sizes are at most $d_L|B|$. Each element $(v, b) \in V \times B$ belongs only to sets indexed by labels at neighbors of v, so each element appears in at most $d_R|\Sigma_U|$ sets. Given a satisfying labeling of I, select at each left vertex the set corresponding to its label. At a right vertex v labeled x, these sets contain $\{v\} \times C_{x,i}$ for every $i \in [d_R]$ and therefore cover $\{v\} \times B$.

To bound $C_{\mathcal{T}}$, observe a projection swap on $e = (u, v)$ affects incidences only between sets indexed by labels at u and elements of $\{v\} \times B$. A predicate swap at u affects such incidences for at most d_L choices of v. In either case, at most $|\Sigma_U||B|$ incidences change for each v, so

$$C_{\mathcal{T}} \le d_L|\Sigma_U||B|.$$

To bound $C_{\mathcal{R}}$, observe that adding or removing a selected set index (u, y) changes only the list L_u. Couple the sampled edges identically and couple the sampled labels identically whenever their lists agree. The recovered label can then change only at u and at right vertices whose sampled edge has left endpoint u. Summing over the d_L edges incident to u, including parallel edges, the expected number of changed right labels is at most d_L/d_R. Thus

$$C_{\mathcal{R}} \le 1 + \frac{d_L}{d_R}.$$

Finally, the recovery rule does not depend on predicate swaps, and a projection swap can only change the recovered label at its right endpoint so

$$D_{\mathcal{R}} \le 1.$$

□

We now argue that applying our recovery procedure to a randomized solution to the set cover instance of small enough expected size results in a high value solution to the original label cover instance.

Lemma 12.3 (Recovery Value)**.** If a distribution over set covers $\boldsymbol{\mathcal{S}}$ has expected size at most

$$\mathbf{E}\,|\boldsymbol{\mathcal{S}}| \le (1-\theta)\frac{\ell}{d_R}|U|,$$

then the expected recovered value is at least

$$\mathbf{E}\big[\mathsf{val}_I(R_{\mathrm{SC},I}(\boldsymbol{\mathcal{S}}))\big] \ge \theta^5 \frac{d_R^2}{\ell^2}.$$

Proof. Choose a uniform right vertex v independent of the cover, and write $T := \sum_{e=(u,v)} |L_u|$. Biregularity gives

$$\mathbf{E}\, T = \frac{d_L}{|V|}\, \mathbf{E}\, |\boldsymbol{\mathcal{S}}| = \frac{d_R}{|U|}\, \mathbf{E}\, |\boldsymbol{\mathcal{S}}| \le (1-\theta)\ell.$$

We will argue that, for every fixed cover and right vertex v with $T \le \ell$, the set-system property of Lemma 12.1 implies that the expected fraction of edges incident to v satisfied by the recovered label-cover assignment is at least $\theta^4 d_R^2/\ell^2$. Since $\Pr[T \le \ell] \ge \theta$ by Markov, this implies

$$\mathbf{E}\big[\mathsf{val}_I(R_{\mathrm{SC},I}(\boldsymbol{\mathcal{S}}))\big] \ge \frac{\theta^4 d_R^2}{\ell^2} \Pr[T \le \ell] \ge \frac{\theta^5 d_R^2}{\ell^2}$$

as desired.

When $T \le \ell$, the parts used by $\boldsymbol{\mathcal{S}}$ to cover $\{v\} \times B$ form a cover of B by at most ℓ indexed parts of our system from Lemma 12.1. By the set-system property, these contain more than θd_R parts from one partition, so there exists some $x \in \Sigma_V$ and $H \subseteq [d_R]$ with $|H| > \theta d_R$ such that, for each $i \in H$, some $y \in L_{u_i}$ satisfies $P_{u_i}(y) = 1$ and $f_{e_{v,i}}(y) = x$, where u_i is the left endpoint of $e_{v,i}$.

For the fixed cover and right vertex v, define

$$p_v(x) := \frac{1}{d_R} \sum_{e=(u,v)} \frac{|\{y \in L_u : P_u(y) = 1,\ f_e(y) = x\}|}{|L_u|},$$

where a summand is zero when L_u is empty. Thus $p_v(x)$ is the probability that a uniform edge $e = (u, v)$ has $L_u \neq \varnothing$ and a uniform label from L_u satisfies P_u and projects to x. By the defining property of H and Cauchy–Schwarz,

$$p_v(x) \ge \frac{1}{d_R} \sum_{i \in H} \frac{1}{|L_{u_i}|} \ge \frac{|H|^2}{d_R T} \ge \frac{\theta^2 d_R}{\ell}.$$

Now observe that by construction, the probability that recovery outputs x at v is at least $p_v(x)$. Similarly, the probability over a random edge (u, v) that the recovered label at u satisfies its predicate and projects to x is at least $p_v(x)$. These choices are independent conditioned on the cover and v, so the conditional expected fraction of satisfied edges incident to v is at least $p_v(x)^2 \ge \theta^4 \frac{d_R^2}{\ell^2}$ as desired. □

Set Cover of any Size Finally, we put these lemmas together with a basic padding procedure to produce a sensitivity lower bound for set cover of any (sufficiently large) size from a low-soundness label cover bound. In the following sections, we will instantiate this result with our two label cover bounds.

Lemma 12.4. Let $\mathcal{I}$ be a swap-closed Label Cover family with common biregular graph and alphabets, with $d_L, d_R > 0$ as above, and fix a common set system with parameters $\ell > 0$, $0 < \theta < 1$, and $m_0 \le N_0$. Suppose every algorithm on $\mathcal{I}$ that outputs a labeling of expected value at least $\theta^5 d_R^2/\ell^2$ on each satisfiable member has swap sensitivity at least $L > 1$. Let Γ, n be integers satisfying

$$\Gamma \ge \max\{d_L|B|, d_R|\Sigma_U|\}, \qquad n \ge N_0.$$

Then, for every α-approximation algorithm A defined on all Set Cover instances of universe size n, with

$$\alpha \le (1-\theta)\frac{\ell}{d_R}\left(1 + \frac{N_0/\Gamma + 1}{|U|}\right)^{-1},$$

there are two instances J_0, J_1 differing by one incidence edit such that

$$\mathsf{EMD}\big(A(J_0), A(J_1)\big) \geq \frac{L-1}{d_L|\Sigma_U||B|(1+d_L/d_R)}.$$

The instances J_0, J_1 have a common universe and index set, at most n indexed sets, and set sizes and element frequencies at most Γ.

Proof. Let A be the given Set Cover algorithm. For $J_I = \mathcal{T}_{\mathrm{SC}}(I)$, put $\rho = \lfloor n/N_0 \rfloor$, $t = n - \rho N_0$, and $r = \lceil t/\Gamma \rceil$. Form J_I^{pad} from ρ disjoint copies of J_I and t additional elements partitioned into r blocks of size at most Γ, adding one set equal to each block. The parameters N_0, m_0, ρ, t, r and all padding choices are common to the family. Since $m_0 \leq N_0$, the number of sets is $\rho m_0 + r \leq \rho N_0 + t = n$. Each added set has size at most Γ, and each additional element has frequency one. Together with Lemma 12.2, this proves the set-size and frequency bounds.

Run A on J_I^{pad}, select a random one of the ρ copies, and apply $R_{\mathrm{SC},I}$. This defines a source algorithm A_{LC} on all of $\mathcal{I}$. On satisfiable I, a cover of size $|U|$ exists before padding, and $r \leq N_0/\Gamma + 1$. The selected indices $\boldsymbol{\mathcal{S}}$ in a uniform copy form a random set cover satisfying

$$\mathbf{E}\,|\boldsymbol{\mathcal{S}}| \leq \alpha(|U| + r/\rho) \leq \alpha(|U| + N_0/\Gamma + 1) \leq \frac{(1-\theta)\ell}{d_R}|U|.$$

By Lemma 12.3, the recovered labeling has expected value at least $\theta^5 d_R^2/\ell^2$ on every satisfiable source instance.

The padded transformation changes at most $\rho C_{\mathcal{T}}$ incidences per source swap, where $C_{\mathcal{T}} = d_L|\Sigma_U||B|$. Choosing the same uniform copy in two recoveries shows that their Lipschitz constant $C_{\mathcal{R}}$ is at most $(1 + d_L/d_R)/\rho$ on all selected subsets, and the source dependence $D_{\mathcal{R}}$ contributes at most one. By the source lower bound, choose neighbors $I, \widetilde{I}$ with $\mathsf{EMD}(A_{\mathrm{LC}}(I), A_{\mathrm{LC}}(\widetilde{I})) \geq L$. Their padded images $J, \widetilde{J}$ satisfy

$$\frac{\mathsf{EMD}(A(J), A(\widetilde{J}))}{\mathsf{SwapDist}(J, \widetilde{J})} \geq \frac{L-1}{C_{\mathcal{T}}(1+d_L/d_R)}.$$

Delete incidences present only in J, then insert those present only in $\widetilde{J}$. Each intermediate incidence set is a subset of one endpoint's, so the size and frequency bounds persist. This is a shortest path; Lemma 2.2 supplies the desired neighboring pair. □

Remaining Proofs

Proof of Lemma 12.1. Consider the following random choice of the partitions $\{C_{i,j}\}$. For each point in B and each of the m partitions, independently choose a uniform part in which to place the point. We will argue that for any fixed set of ℓ index pairs $(i,j) \in [m] \times [q]$ with at most θq pairs per partition, the probability that the corresponding ℓ random indexed parts $\{C_{i,j}\}$ cover B is small. Union bounding over all possible such index sets then gives the existence of the desired system.

Formally, fix any collection of at most ℓ indexed parts (formally, index pairs as above), and let t_x be the number in the x-th partition. Suppose $t_x \leq \theta q$ for every x. The probability over the choice of the $\{C_{i,j}\}$ that a fixed point is uncovered by this collection is

$$p = \prod_x (1 - t_x/q) \geq \exp\left(-\frac{\sum_x t_x}{q(1-\theta)}\right) \geq \exp\left(-\frac{\ell}{q(1-\theta)}\right),$$

where we used $\ln(1-z) \geq -z/(1-\theta)$ for $0 \leq z \leq \theta$. The choices for different points are independent, so the probability that the collection covers B is at most $(1-p)^b \leq e^{-bp}$. There are at most 2^{mq} collections and by assumption $e^{-bp}2^{mq} < 1$, so there exists a set system where no such collection covers B as desired. □

12.2 Sensitivity Bound via IKW

We now use the IKW Label Cover family to prove a sensitivity lower bound for Set Cover algorithms with approximation ratio up to $\log \Gamma$, where Γ bounds both set size and element frequency. Choosing the product parameter $k = \Theta((\log\log\Gamma)^2)$ transfers the Label Cover lower bound to $n^{\Omega(1/(\log\log\Gamma)^2)}$ sensitivity, where n is the prescribed universe size, in the parameter range stated below.

Theorem 12.5 (Sensitivity lower bound for Set Cover via IKW)**.** For every constant $\varepsilon \in (0,1)$, there exist a constant $c_0 = c_0(\varepsilon) > 0$ and an integer $\Gamma_0 = \Gamma_0(\varepsilon) \geq 2$ such that the following holds. For all sufficiently large n and every integer $\Gamma \geq \Gamma_0$ satisfying

$$\log\Gamma \cdot (\log\log\Gamma)^2 \leq c_0 \log n,$$

any algorithm A_{SC} defined on all Set Cover instances of universe size n that has approximation ratio $(1-\varepsilon)\ln\Gamma$ in expectation on every feasible instance must have sensitivity with respect to incidence edits

$$2^{\Omega_\varepsilon\left(\frac{\log n}{(\log\log\Gamma)^2}\right)}.$$

Moreover, the lower bound is already witnessed by neighboring instances J_0, J_1 on the same universe and with a common index set such that

$$|\mathcal{F}(J_0)| = |\mathcal{F}(J_1)| = O(n),$$

every set in J_0 and J_1 has size at most Γ, every element has frequency at most Γ, and

$$\mathsf{EMD}(A_{\mathrm{SC}}(J_0), A_{\mathrm{SC}}(J_1)) = 2^{\Omega_\varepsilon\left(\frac{\log n}{(\log\log\Gamma)^2}\right)}.$$

Proof of Theorem 12.5. Use the constants $C_{\mathrm{IKW}}, a, b, \eta, k_0$ from Theorem 6.1. Put $\theta = \varepsilon/10$. Choosing $C_k = C_k(\varepsilon)$ sufficiently large let

$$k = \lceil C_k(\log\log\Gamma)^2 \rceil$$

such that the value requirement from Lemma 12.3 follows from $\exp\left(-C_{\mathrm{IKW}}\sqrt{k}\right) \leq \theta^5/(\ln\Gamma)^2$.

Fix $c_{\deg} > 0$ so that $D_\Gamma := 2^{c_{\deg}k}$ bounds the source degrees and the sum of its alphabet sizes. Choose N (the source 2-CSP size in Theorem 6.1) to be

$$N = \left\lfloor \left(\frac{n}{100\Gamma}\right)^{1/(bk)} \right\rfloor.$$

The stated parameter range gives for all $\Gamma \geq \Gamma_0$,

$$\log N = \Theta(\log n/k), \qquad k^2 = o(\log n), \qquad k = o(\log N).$$

so Theorem 6.1 gives a common biregular source label-cover family with s vertices and positive degrees d_L, d_R satisfying

$$N^{ak} \leq s \leq \frac{n}{100\Gamma}.$$

Every algorithm on this family that outputs a labeling of expected value at least $\exp\!\left(-C_{\mathrm{IKW}}\sqrt{k}\right)$ on each satisfiable member has sensitivity at least $L = s^{\eta/k} = n^{\Omega(1/k)}$. Set

$$|B| = \left\lfloor \frac{\theta\Gamma}{D_\Gamma^4} \right\rfloor, \qquad \ell = d_R(1-2\theta)\ln|B|.$$

Since $D_\Gamma = \Gamma^{o(1)}$, increasing Γ_0 ensures

$$\ln|B| \geq (1-\theta)\ln\Gamma, \qquad |B| \geq D_\Gamma^2, \qquad |B|^{\theta/(1-\theta)} > d_R|\Sigma_V|\ln 2.$$

The last inequality is the hypothesis of Lemma 12.1, so choose partitions of B with these parameters. The value requirement holds because

$$\frac{\theta^5 d_R^2}{\ell^2} = \frac{\theta^5}{(1-2\theta)^2(\ln|B|)^2} \geq \frac{\theta^5}{(\ln\Gamma)^2} \geq \exp\!\left(-C_{\mathrm{IKW}}\sqrt{k}\right).$$

We also have

$$d_L|\Sigma_U||B|(1+d_L/d_R) \leq D_\Gamma^4|B| \leq \theta\Gamma.$$

In particular, $d_L|B| \leq \Gamma$ and $d_R|\Sigma_U| \leq \Gamma$, and Lemma 12.2 gives

$$N_0 = |V||B| \leq s\Gamma \leq n/100, \qquad m_0 = |U||\Sigma_U| \leq N_0,$$

where the last inequality uses biregularity and $|B| \geq D_\Gamma^2$.

Put $\alpha = (1-\varepsilon)\ln\Gamma$. For all sufficiently large n, we have $|U| \geq s/(1+D_\Gamma) \geq 1/\theta$. Hence

$$\frac{N_0/\Gamma + 1}{|U|} = \frac{d_L|B|}{d_R\Gamma} + \frac{1}{|U|} \leq 2\theta.$$

Since $\theta = \varepsilon/10$,

$$\begin{aligned}\alpha\left(1 + \frac{N_0/\Gamma+1}{|U|}\right) &\leq (1-\varepsilon)(1+2\theta)\ln\Gamma \\ &\leq (1-\theta)^2(1-2\theta)\ln\Gamma \\ &\leq (1-\theta)\frac{\ell}{d_R}.\end{aligned}$$

All hypotheses of Lemma 12.4 now hold, so there are neighboring instances J_0, J_1 with the stated size and frequency bounds such that

$$\mathsf{EMD}\big(A_{\mathrm{SC}}(J_0), A_{\mathrm{SC}}(J_1)\big) \geq \frac{L-1}{d_L|\Sigma_U||B|(1+d_L/d_R)} \geq \frac{s^{\eta/k}-1}{\Gamma} = n^{\Omega_\varepsilon(1/k)}$$

as desired. □

12.3 Sensitivity Bound via Dinur–Harsha

We now transfer the polynomial sensitivity lower bound for Label Cover from Theorem 11.1 to Set Cover up to $\log^B n$-approximation for some constant $B \in (0,1)$.

Theorem 12.6 (Sensitivity lower bound for Set Cover via Dinur–Harsha)**.** There exist an integer $n_0 \in \mathbb{N}$ and positive absolute constants B, c, C, δ such that the following holds. For every integer $n \geq n_0$ and every real q with

$$2 \leq q \leq \log^B n,$$

any $C\sqrt{q}$-approximation algorithm A defined on all Set Cover instances of universe size n has sensitivity $\Omega(n^\delta)$. Moreover, the lower bound already holds on instances in which every set has size at most $\exp(q^c)$, every element belongs to at most $\exp(q^c)$ sets, and the number of sets satisfies $|\mathcal{F}| = O(n)$. More specifically, for every such algorithm A, there are two instances J_0, J_1 with the same universe of size n and a common index set that differ by one incidence edit, satisfy these structural bounds, and obey

$$\mathsf{EMD}\big(A(J_0), A(J_1)\big) = \Omega(n^\delta).$$

Proof of Theorem 12.6. Let $B_0, \delta_{\mathrm{LC}} > 0$ be the constants from Theorem 11.1. Fix $C \geq 1$, choose $c \geq 1$ large enough, then take $0 < B < B_0/4$ with $2Bc < 1$. Set $\delta = \delta_{\mathrm{LC}}/2$ and

$$g = \lceil 64C\sqrt{q} \rceil^2, \qquad \Lambda = \lfloor \exp(q^c) \rfloor, \qquad s = \left\lfloor \frac{n}{8\Lambda^2} \right\rfloor.$$

For $2 \leq q \leq \log^B n$, we have $\Lambda = n^{o(1)}$, $\log s \sim \log n$, and $g = O(q) \leq \log^{B_0} s$ for sufficiently large n. Theorem 11.1 therefore supplies a family with $|U| = s$ and source degree and alphabet bounds

$$D, |\Sigma_V| = q^{O(1)}, \qquad |\Sigma_U| \leq \exp\Big(q^{O(1)}\Big), \qquad m := |E| \leq sD.$$

Discard the isolated left vertices introduced by the final padding in Section 11, so that $|U| \leq s$. The remaining graph is biregular with positive degrees d_L, d_R: this holds at initialization and is preserved by each composition followed by alphabet and degree reduction. Any algorithm on the remaining family extends to the padded family by assigning fixed labels to the discarded vertices, with unchanged value and no greater sensitivity. Set $\theta = 1/2$ and $\ell = d_R\sqrt{g}/8$, so that $\theta^5 d_R^2/\ell^2 = 2/g$. Every algorithm on this family that outputs a labeling of expected value at least $2/g$ on each satisfiable member has sensitivity at least $L = \Omega(s^{\delta_{\mathrm{LC}}})$.

Choose a universe B_{sys} of size $\exp\big(q^{O(1)}\big)$ satisfying

$$|B_{\mathrm{sys}}|e^{-\sqrt{g}/4} > d_R|\Sigma_V| \ln 2, \qquad |B_{\mathrm{sys}}| \geq \frac{d_R}{d_L}|\Sigma_U|.$$

By Lemma 12.1, choose $|\Sigma_V|$ partitions of B_{sys} into d_R parts with parameters ℓ, θ. The second inequality gives $m_0 \leq N_0$ and $d_R|\Sigma_U| \leq d_L|B_{\mathrm{sys}}|$. The absolute constant c can be chosen so that, for all $q \geq 2$,

$$D, \quad D^2|B_{\mathrm{sys}}|, \quad d_L|\Sigma_U||B_{\mathrm{sys}}|(1 + d_L/d_R) \leq \Lambda.$$

Consequently the constructed universe has size

$$N_0 = |V||B_{\mathrm{sys}}| \leq m|B_{\mathrm{sys}}| \leq m\Lambda \leq s\Lambda^2 \leq n/8.$$

Apply Lemma 12.4 with set-size and frequency bound $\Gamma = \Lambda$, target universe size n, and approximation factor $\alpha = C\sqrt{q}$. Its approximation condition holds because

$$\begin{aligned} \alpha\left(1 + \frac{N_0/\Lambda + 1}{|U|}\right) &= C\sqrt{q}\left(1 + \frac{d_L|B_{\mathrm{sys}}|}{d_R\Lambda} + \frac{1}{|U|}\right) \\ &\leq 3C\sqrt{q} \leq \frac{\sqrt{g}}{16} = \frac{(1-\theta)\ell}{d_R}. \end{aligned}$$

We obtain neighboring instances J_0, J_1 with $O(n)$ sets and set sizes and frequencies at most $\Lambda \le \exp(q^c)$, satisfying

$$\mathsf{EMD}\big(A(J_0), A(J_1)\big) \ge \frac{L-1}{d_L|\Sigma_U||B_{\mathrm{sys}}|(1+d_L/d_R)} \ge \frac{L-1}{\Lambda} = \Omega\left(\frac{s^{\delta_{\mathrm{LC}}}}{\Lambda}\right) = \Omega(n^{\delta}),$$

where the last equality uses $s = \Theta(n/\Lambda^2)$ and $\Lambda = n^{o(1)}$. This proves the theorem. □

13 PCP-Based Lower Bounds for Dominating Set

This section proves two sensitivity lower bounds for Dominating Set on graphs of maximum degree at most Δ and derives locality lower bounds from them. The IKW-based bound in Theorem 13.3 concerns approximation ratio $(1-\varepsilon)\ln\Delta$ for every constant $\varepsilon \in (0,1)$. The Dinur–Harsha-based bound in Theorem 13.5 gives a polynomial sensitivity lower bound for approximation ratios that are a small positive power of $\log\Delta$. We first isolate the common reduction from Set Cover to Dominating Set and then specialize it in the two approximation regimes. We then compare the resulting locality bounds with prior distributed results. Table 4 summarizes the sensitivity bounds and their degree ranges.

Table 4: Sensitivity bounds for Dominating Set on n vertices with maximum degree at most $\Delta \le n-1$. The first lower bound uses DH, with a sufficiently small absolute constant $\beta > 0$, and requires sufficiently large $\Delta \le \exp(C_{\mathrm{range}} \log^{\kappa} n)$ for some $\kappa \in (0,1)$. For every constant $\varepsilon \in (0,1)$, the second uses IKW and requires $\Delta \ge \Delta_0(\varepsilon)$ and $\log\Delta(\log\log\Delta)^2 \le c_0(\varepsilon)\log n$. The upper bound holds for every constant $\varepsilon > 0$ and sufficiently large Δ depending on ε.

Approximation factor	Sensitivity bound	Reference
	Lower bounds	
$\log^{\beta}\Delta$	$n^{\Omega(1)}$	Theorem 13.5
$(1-\varepsilon)\ln\Delta$	$n^{\Omega_\varepsilon(1/(\log\log\Delta)^2)}$	Theorem 13.3
	Upper bound	
$(1+\varepsilon)\ln\Delta$	$\exp\big(O_\varepsilon(\log^3\Delta)\big)$	Section 13.2, from [26]

Parameter guide. In the lower-bound proofs below, N denotes the size of the universe in the source Set Cover instance, $m := |\mathcal{F}|$ the number of sets, n the prescribed number of vertices, and Γ the parameter bounding both set size and element frequency in the Dominating Set reduction. Here and below, sensitivity for graph algorithms is measured under single-edge edits. For a Dominating Set algorithm A, we write $\mathsf{SwapSens}(A)$ for the maximum EMD between its output distributions on graphs in the stated input class that differ by one edge. The distance between output vertex sets is the size of their symmetric difference.

13.1 Reduction from Set Cover to Dominating Set

Fix an integer parameter $\Gamma \ge 1$. We use the classical reduction of [8]. We start with the element–set incidence graph. Choosing the set-vertices of a cover dominates all element-vertices. Helper vertices will dominate the remaining set-vertices, with at most Γ set-vertices per helper. For a Set Cover instance $I = (U = \{u_1, \dots, u_N\}, \mathcal{F} = (S_1, \dots, S_m))$ whose set sizes and element frequencies are at most Γ, let $G_I =$

$\mathcal{T}_{\mathrm{DS}}(I)$ be the graph on $U \sqcup \{S_i : i \in [m]\} \sqcup W$, where $W = \{w_1, \ldots, w_{\lceil m/\Gamma \rceil}\}$. In graph notation, S_i denotes a tagged vertex representing index i, whose identity is independent of the contents of the subset $S_i \subseteq U$. The three vertex classes are disjoint, and different indices give distinct vertices even when their subsets coincide. Its edges are the incidences uS_i with $u \in S_i$ and the helper edges $S_i w_{\lceil i/\Gamma \rceil}$. Thus

$$\Delta(G_I) \le \Gamma + 1, \qquad |V(G_I)| = N + m + \left\lceil \frac{m}{\Gamma} \right\rceil.$$

Recovery and sensitivity. Given a vertex set $D \subseteq V(G_I)$, retain the indices of its set-vertices and replace each element-vertex in D by a set containing that element. Ignore helper vertices. Formally, let $\sigma_I(u)$ be the smallest index $i \in [m]$ with $u \in S_i$, or $\perp$ if no such index exists. Define $\phi_I : V(G_I) \to [m] \cup \{\perp\}$ by

$$\phi_I(S_i) = i, \qquad \phi_I(u) = \sigma_I(u), \qquad \phi_I(w) = \perp \quad (w \in W),$$

and set $R_I(D) := \phi_I(D) \setminus \{\perp\}$ for every $D \subseteq V(G_I)$. Here R_I abbreviates $R_{\mathrm{DS},I}$. If I is feasible and D dominates G_I, then every element-vertex is either selected or adjacent to a selected set-vertex. In either case, $R_I(D)$ contains a set covering that element, and $|R_I(D)| \le |D|$. Adding or removing one vertex in D changes at most one recovered index. An incidence edit changes one graph edge and can change σ_I only at the edited element, replacing at most one recovered index by another. Thus $C_{\mathcal{T}} = C_{\mathcal{R}} = 1$ and $D_{\mathcal{R}} \le 2$.

Padding to a prescribed size. For each pair of integers $\Delta \ge 2$ and $t \ge 0$, fix a labeled star forest $\mathcal{P}_{\Delta,t}$ on a designated set of t vertices. Its components have size at most Δ and its domination number is $\lceil t/\Delta \rceil$: writing $t = a\Delta + b$ with $0 \le b < \Delta$, take a copies of $K_{1,\Delta-1}$ and, if $b > 0$, one copy of $K_{1,b-1}$, where $K_{1,0}$ is an isolated vertex.

Fix $C_{\mathcal{F}} \ge 1$, a positive integer target n, and an integer $\Gamma \ge 1$. We choose the source universe size to leave room for padding to exactly n vertices. The padding must increase the optimum by at most a constant factor, so that recovery preserves the approximation guarantee up to a constant loss. We use $C_{\mathcal{F}}$ to bound the ratio of the number of sets to the universe size. Define a padding factor C_{pad}, an approximation-loss bound C_{lift}, and a source universe size N by

$$C_{\mathrm{pad}} := 8(C_{\mathcal{F}} + 1), \qquad C_{\mathrm{lift}} := C_{\mathcal{F}} + 2C_{\mathrm{pad}} + 3, \qquad N := \left\lfloor \frac{n}{C_{\mathrm{pad}}} \right\rfloor.$$

Let $I = (U, \mathcal{F} = (S_1, \ldots, S_m))$ have $|U| = N$, $m \le C_{\mathcal{F}} N$, and set sizes and element frequencies at most Γ, and set $t_I := n - |V(G_I)|$. Whenever $t_I \ge 0$, define $H_I := G_I \sqcup \mathcal{P}_{\Gamma+1,t_I}$. Extend ϕ_I to $V(H_I)$ by mapping every padding vertex to $\perp$, and use the same formula $R_I(D) = \phi_I(D) \setminus \{\perp\}$ on all $D \subseteq V(H_I)$. Since the padding is disjoint from G_I and is unchanged under incidence edits, the recovery and sensitivity guarantees above continue to hold.

The next proposition shows that the padding is well-defined for all sufficiently large target sizes and records the approximation and perturbation guarantees of this construction.

Proposition 13.1 (Reduction to Dominating Set with a prescribed number of vertices)**.** For every $C_{\mathcal{F}} \ge 1$, there is a threshold $n_0 = n_0(C_{\mathcal{F}})$ such that, for every $n \ge n_0$ and every integer $\Gamma \ge 1$, each instance I in the bounded Set Cover family above has $t_I \ge 0$ and satisfies the following.

1. *Size, degree, and optimum.* We have

$$|V(H_I)| = n, \qquad \Delta(H_I) \le \Gamma + 1, \qquad n \le 2C_{\mathrm{pad}} N,$$
$$\mathsf{opt}(H_I) \le C_{\mathrm{lift}} \mathsf{opt}(I) \quad \text{if } I \text{ is feasible.}$$

2. *Recovery from a fixed instance.* If I is feasible and D dominates H_I, then $R_I(D)$ is a set cover of I and $|R_I(D)| \le |D|$. For all $D, \widetilde{D} \subseteq V(H_I)$,

$$|R_I(D) \triangle R_I(\widetilde{D})| \le |D \triangle \widetilde{D}|.$$

3. *One incidence edit.* If I and $\widetilde{I}$ belong to this bounded family, have the same universe and a common index set, and differ by one incidence edit, then H_I and $H_{\widetilde{I}}$ have a common vertex set and differ by exactly one edge. Moreover, every vertex subset D of this common vertex set satisfies

$$|R_I(D) \triangle R_{\widetilde{I}}(D)| \le 2.$$

Proof. Writing $m := |\mathcal{F}|$, for all sufficiently large n depending only on $C_{\mathcal{F}}$ and uniformly over $\Gamma \ge 1$,

$$|V(G_I)| = N + m + \left\lceil \frac{m}{\Gamma} \right\rceil \le (1 + 2C_{\mathcal{F}})N + 1 \le n/2, \qquad n \le 2C_{\mathrm{pad}}N.$$

This proves $t_I \ge 0$ and $|V(H_I)| = n$; the bounds for G_I and the star forest give the degree claim.

If I is feasible, bounded set size and $m \le C_{\mathcal{F}}N$ give

$$\mathsf{opt}(I) \ge N/\Gamma, \qquad m/\Gamma + 1 \le (C_{\mathcal{F}} + 1)\mathsf{opt}(I).$$

Moreover,

$$\mathsf{opt}(G_I) \le \mathsf{opt}(I) + m/\Gamma + 1,$$
$$\mathsf{opt}(\mathcal{P}_{\Gamma+1,t_I}) = \left\lceil \frac{t_I}{\Gamma+1} \right\rceil \le \frac{n}{\Gamma+1} + 1 \le (2C_{\mathrm{pad}} + 1)\mathsf{opt}(I).$$

The first bound is obtained by selecting the set-vertices of an optimal cover and all helper vertices. Since domination numbers add over components, these bounds give the claimed optimum comparison.

Suppose now that I is feasible and D dominates H_I. Since there are no edges between G_I and the padding forest, the set $D \cap V(G_I)$ dominates G_I. For each $u \in U$, if $u \in D$, then $\sigma_I(u) \in R_I(D)$ and $u \in S_{\sigma_I(u)}$. Otherwise, D contains a set-vertex S_i adjacent to u, so $i \in R_I(D)$ and $u \in S_i$. Thus $R_I(D)$ is a set cover of I. Since this recovery is the image of D under ϕ_I, with $\perp$ removed, it does not increase size. For arbitrary $D, \widetilde{D} \subseteq V(H_I)$,

$$R_I(D) \triangle R_I(\widetilde{D}) \subseteq \phi_I(D \triangle \widetilde{D}),$$

which proves the fixed-instance Lipschitz bound.

Finally, suppose $\widetilde{I}$ is obtained from I by one incidence edit at (u_0, S_j). Then m is unchanged, so all vertices, helper edges, and the fixed padding forest are common; only the edge u_0S_j changes. The maps ϕ_I and $\phi_{\widetilde{I}}$ agree outside u_0. For every $D \subseteq V(H_I)$, the set $R_{\widetilde{I}}(D)$ is therefore obtained from $R_I(D)$ by deleting at most one index and adding at most one index. This proves the bound on the change in recovery. □

Use the parameters of Proposition 13.1, with n sufficiently large. Let A_{DS} have approximation ratio r_{DS} on n-vertex graphs of maximum degree at most $\Gamma + 1$. For Set Cover instances I on N elements with at most $C_{\mathcal{F}}N$ sets, each of size at most Γ and each element occurring in at most Γ sets, define

$$A_{\mathrm{SC}}(I) := R_I(A_{\mathrm{DS}}(H_I)).$$

Extend A_{SC} to all instances on N elements by returning an optimal cover on feasible instances outside the family and an arbitrary output on infeasible instances outside the family.

Corollary 13.2 (Approximation and sensitivity transfer to Dominating Set)**.** The induced algorithm A_{SC} is a $C_{\mathrm{lift}} r_{\mathrm{DS}}$-approximation on all Set Cover instances of universe size N. On instances with at most $C_{\mathcal{F}} N$ sets, set sizes at most Γ, and element frequencies at most Γ, the reduction has transfer parameters $C_{\mathcal{T}} = C_{\mathcal{R}} = 1$ and $D_{\mathcal{R}} = 2$. For every neighboring pair $I, \widetilde{I}$ in that family with a common universe and index set,

$$\mathsf{EMD}(A_{\mathrm{SC}}(I), A_{\mathrm{SC}}(\widetilde{I})) \geq L \quad \Longrightarrow \quad \mathsf{SwapSens}(A_{\mathrm{DS}}) \geq L - 2.$$

Proof. The recovery and optimum bounds in Proposition 13.1 prove the approximation guarantee on instances satisfying these three bounds; outside this family, the extension is optimal whenever a cover exists. The same proposition gives $C_{\mathcal{T}} = C_{\mathcal{R}} = 1$ and $D_{\mathcal{R}} = 2$. For each neighboring pair in the statement, Lemma 3.3 and the triangle inequality give

$$\mathsf{EMD}(A_{\mathrm{SC}}(I), A_{\mathrm{SC}}(\widetilde{I})) \leq \mathsf{EMD}(A_{\mathrm{DS}}(H_I), A_{\mathrm{DS}}(H_{\widetilde{I}})) + 2.$$

Since H_I and $H_{\widetilde{I}}$ differ by one edge, this proves the sensitivity implication directly on the given pair. □

13.2 Logarithmic Approximation in the Maximum Degree

We prove a sensitivity lower bound for Dominating Set algorithms with approximation ratio $(1-\varepsilon)\ln\Delta$, for every constant $\varepsilon \in (0,1)$, where Δ bounds the maximum degree. Set $\Gamma := \lfloor \Delta \rfloor - 1$ and apply the IKW Set Cover lower bound of Theorem 12.5, with set size and element frequency bounded by a sufficiently small ε-dependent fraction of Γ. The common reduction above then gives Dominating Set instances of maximum degree at most Δ while preserving the sensitivity lower bound up to a constant factor.

Theorem 13.3 (Sensitivity lower bound for Dominating Set via IKW)**.** For every constant $\varepsilon \in (0,1)$, there exist an integer $\Delta_0 = \Delta_0(\varepsilon) \geq 2$ and a constant $c_0 = c_0(\varepsilon) > 0$ such that, for all sufficiently large n and every $\Delta_0 \leq \Delta \leq n-1$ satisfying

$$\log\Delta \cdot (\log\log\Delta)^2 \leq c_0 \log n,$$

any algorithm for the Dominating Set problem on graphs with n vertices and maximum degree at most Δ with approximation ratio

$$(1-\varepsilon)\ln\Delta$$

in expectation must have sensitivity

$$2^{\Omega_\varepsilon\left(\frac{\log n}{(\log\log\Delta)^2}\right)}.$$

Proof of Theorem 13.3. Let c_1, Γ_0 be the constants from Theorem 12.5 with error parameter $\varepsilon/4$, and fix $C_{\mathcal{F}} \geq 1$ bounding its number of sets by $C_{\mathcal{F}} N$. Use the corresponding C_{pad} from Proposition 13.1. For n, Δ in the stated range, put

$$N = \lfloor n / C_{\mathrm{pad}} \rfloor, \qquad \Gamma = \lfloor \Delta \rfloor - 1, \qquad \gamma = \left\lfloor \frac{\varepsilon\Gamma}{16 C_{\mathrm{pad}}} \right\rfloor.$$

Taking Δ_0 large enough and c_0 small enough ensures $\gamma \geq \Gamma_0$, $\Gamma + 1 \leq \Delta$, and

$$(1+\varepsilon/4)(1-\varepsilon)\ln\Delta \leq (1-\varepsilon/4)\ln\gamma, \qquad \log\gamma(\log\log\gamma)^2 \leq c_1 \log N.$$

Here $N = \Theta(n)$ and $\ln\gamma = \ln\Delta - O_\varepsilon(1)$. The stated degree range also gives $N \geq 16\gamma/\varepsilon$ for sufficiently large n.

Consider Set Cover instances I on N elements with at most $C_{\mathcal{F}}N$ sets, set sizes at most γ, and element frequencies at most γ. Apply the construction of Proposition 13.1 with parameter Γ, obtaining a graph H_I on exactly n vertices with maximum degree at most $\Gamma + 1 \le \Delta$. If I is feasible, then $\mathsf{opt}(I) \ge N/\gamma$ and

$$\begin{aligned}\mathsf{opt}(H_I) &\le \mathsf{opt}(I) + \left\lceil \frac{m}{\Gamma} \right\rceil + \left\lceil \frac{t_I}{\Gamma + 1} \right\rceil \\ &\le \mathsf{opt}(I) + \frac{n}{\Gamma} + 2 \\ &\le (1 + \varepsilon/4)\mathsf{opt}(I).\end{aligned}$$

The second inequality uses $m + t_I \le n$. For the last, $n \le 2C_{\mathrm{pad}}N$ and the choice of γ give $n/\Gamma \le (\varepsilon/8)\mathsf{opt}(I)$, while $N \ge 16\gamma/\varepsilon$ gives $2 \le (\varepsilon/8)\mathsf{opt}(I)$.

Let A' be an algorithm as in the statement. On the bounded family above, define $A(I) := R_I(A'(H_I))$ using the recovery from Proposition 13.1. Extend A to all Set Cover instances on N elements by returning an optimal cover on feasible instances outside this family and an arbitrary output on infeasible instances outside this family. The recovery does not increase size, so the preceding optimum bound and approximation comparison make A a $(1 - \varepsilon/4)\ln\gamma$-approximation algorithm in expectation. Theorem 12.5 gives neighboring instances $I, \widetilde{I}$ on the same universe and index set such that

$$\mathsf{EMD}(A(I), A(\widetilde{I})) \ge L = 2^{\Omega_\varepsilon(\log N/(\log\log\gamma)^2)}.$$

Each instance has at most $C_{\mathcal{F}}N$ sets, set sizes at most γ, and element frequencies at most γ. The sensitivity comparison from Corollary 13.2 therefore gives

$$\mathsf{SwapSens}(A') \ge L - 2 = 2^{\Omega_\varepsilon(\log n/(\log\log\Delta)^2)}. \qquad \square$$

For comparison, approximation ratio $(1 + \varepsilon)\ln\Delta$ permits sensitivity bounded independently of n for sufficiently large Δ. We use KMW's CONGEST algorithm to compute a fractional dominating set, then round its coordinates independently and add every undominated vertex. Let $G = (V, E)$ have maximum degree at most Δ, and let $N[v]$ denote the closed neighborhood of v. For an LP accuracy parameter $0 < \eta \le 1$, KMW's algorithm [26, Corollary 4.1] computes a nonnegative vector $x = (x_v)_{v \in V}$ in $O(\log^2(\Delta+1)/\eta^4)$ rounds, with

$$\sum_{u \in N[v]} x_u \ge 1 \quad (v \in V), \qquad \sum_{v \in V} x_v \le (1 + \eta)\mathsf{opt}(G).$$

For randomized rounding, select each vertex v independently with probability $\min\{1, \ln(\Delta + 1)x_v\}$. After one additional CONGEST round to exchange selection bits with their neighbors, all vertices left undominated join the selected set, producing a dominating set D. By LP feasibility, each vertex is left undominated by the rounding with probability at most $1/(\Delta + 1)$. Also, summing the LP constraints gives $n \le (\Delta + 1)\sum_v x_v$. Thus

$$\mathbf{E}[|D|] \le \ln(\Delta + 1)\sum_v x_v + \frac{n}{\Delta + 1} \le (1 + \eta)(1 + \ln(\Delta + 1))\mathsf{opt}(G).$$

For any constant $\varepsilon > 0$, choose $\eta = \min\{\varepsilon/2, 1/2\}$ so that the LP approximation leaves room for the additive term in $1 + \ln(\Delta + 1)$. For sufficiently large Δ depending on ε,

$$(1 + \eta)(1 + \ln(\Delta + 1)) \le (1 + \varepsilon)\ln\Delta.$$

The entire procedure therefore gives a $(1+\varepsilon)\ln\Delta$ approximation in expectation in $t = O_\varepsilon(\log^2\Delta)$ CONGEST rounds, including the final neighbor check.

To bound sensitivity, fix the vertex identifiers and the prescribed degree bound across inputs, and couple executions on graphs differing by one edge using the same independent random numbers at corresponding vertices. By locality, each vertex's output is determined by its radius-t input neighborhood and the random numbers in that neighborhood. As in the argument for (1), these input neighborhoods coincide outside distance $t+1$ from the edited edge's endpoints, with distance measured in the graph containing the edge. The coupled outputs therefore agree outside this region, which contains at most $2(\Delta+1)^{t+1}$ vertices. This bounds their symmetric difference under the coupling and hence gives sensitivity at most

$$\exp\bigl(O_\varepsilon(\log^3\Delta)\bigr). \tag{59}$$

For every constant $\varepsilon\in(0,1)$ and every sufficiently large constant Δ depending on ε, this upper bound is independent of n, whereas Theorem 13.3 requires sensitivity $n^{\Omega_{\varepsilon,\Delta}(1)}$ for $(1-\varepsilon)\ln\Delta$-approximation. The bounds do not give matching dependence on Δ for the sensitivity itself.

Applying the sensitivity-to-locality bound (1) of Fleming and Yoshida [18] to Theorem 13.3 gives the following distributed lower bound.

Corollary 13.4 (Locality lower bound via IKW)**.** For every constant $\varepsilon\in(0,1)$, there exist a constant $c_0 = c_0(\varepsilon) > 0$ and an integer $\Delta_0 = \Delta_0(\varepsilon) \geq 2$ such that, for all sufficiently large n and every $\Delta_0 \leq \Delta \leq n-1$ satisfying

$$\log\Delta\,(\log\log\Delta)^2 \leq c_0\log n,$$

any non-signaling algorithm for Dominating Set on n-vertex graphs of maximum degree at most Δ with approximation ratio $(1-\varepsilon)\ln\Delta$ in expectation has locality at least

$$\Omega_\varepsilon\left(\frac{\log n}{\log\Delta\,(\log\log\Delta)^2}\right).$$

Proof. Fix $\varepsilon\in(0,1)$ and use the constants from Theorem 13.3, decreasing c_0 as needed below. Apply that theorem and (1), using $\log(\Delta+1) = \Theta(\log\Delta)$. For some $a_\varepsilon > 0$ depending only on ε and an absolute constant $b > 0$, the locality t satisfies

$$t \geq a_\varepsilon\frac{\log n}{\log\Delta\,(\log\log\Delta)^2} - b.$$

The degree restriction makes the fraction at least $1/c_0$. Taking $c_0 \leq a_\varepsilon/(2b)$ absorbs the additive term uniformly over the stated degree range. □

For every constant $\varepsilon\in(0,1)$, the CONGEST algorithm described above gives a $(1+\varepsilon)\ln\Delta$ approximation in $O_\varepsilon(\log^2\Delta)$ CONGEST rounds for sufficiently large Δ depending on ε. The locality lower bound in Corollary 13.4 for $(1-\varepsilon)\ln\Delta$-approximation is $\omega(\log^2\Delta)$ whenever $\log^3\Delta\cdot(\log\log\Delta)^2 = o(\log n)$. In particular, for every sufficiently large constant Δ depending on ε, achieving a $(1-\varepsilon)\ln\Delta$ approximation on graphs of maximum degree at most Δ requires $\Omega_{\varepsilon,\Delta}(\log n)$ locality even in the non-signaling model, whereas a $(1+\varepsilon)\ln\Delta$ approximation takes a constant number of rounds independent of n.

13.3 Polylogarithmic Approximation in the Maximum Degree

We use the Dinur–Harsha Set Cover family to prove a polynomial sensitivity lower bound for Dominating Set algorithms whose approximation ratio is a constant positive power of $\log \Delta$. Starting from the instances in Theorem 12.6, whose universe size can be prescribed exactly, we choose the Set Cover parameter q from the degree bound Δ and set $\Gamma := \Delta - 1$. We then apply the common reduction and exact-size padding above. The choice of q ensures that set sizes and element frequencies are at most Γ, and that the induced Set Cover algorithm satisfies the approximation guarantee required by Theorem 12.6.

Theorem 13.5 (Sensitivity lower bound for Dominating Set via Dinur–Harsha)**.** There exist $\beta_0, \delta, \kappa \in (0,1)$, absolute constants $C_{\mathrm{DS}}, C_{\mathrm{range}} > 0$, and an integer $\Delta_0 \geq 2$ such that for all sufficiently large n and every integer Δ satisfying

$$\Delta_0 \leq \Delta \leq \exp(C_{\mathrm{range}} \log^{\kappa} n),$$

any algorithm that $C_{\mathrm{DS}} \log^{\beta_0} \Delta$-approximates Dominating Set on n-vertex graphs of maximum degree at most Δ must have sensitivity $\Omega(n^{\delta})$.

Proof of Theorem 13.5. Let $B_0, c_0, C_0, \delta_0 > 0$ be the constants from Theorem 12.6, decreasing B_0 so that $B_0 < 1$. Set $\delta = \min\{\delta_0, 1/2\}$, $C_{\mathrm{DS}} = 1$, and choose

$$0 < \beta_0 < \min\{1/2, 1/(4c_0), B_0/4\}, \qquad \kappa = B_0/2.$$

Fix $C_{\mathcal{F}} \geq 1$ bounding the number of sets in that theorem by $C_{\mathcal{F}} N$, and use the corresponding constants $C_{\mathrm{pad}}, C_{\mathrm{lift}}$ from Proposition 13.1. For n, Δ in the stated range, put

$$N = \lfloor n/C_{\mathrm{pad}} \rfloor, \qquad \Gamma = \Delta - 1, \qquad q = \max\left\{2, \left\lceil (64 C_{\mathrm{lift}}/C_0)^2 \log^{2\beta_0} \Delta \right\rceil \right\}.$$

Since $2\beta_0 c_0 < 1$, choosing Δ_0 sufficiently large gives $\exp(q^{c_0}) \leq \Gamma$. Also $2\beta_0 \kappa < B_0/2$, so the degree range implies $q \leq \log^{B_0/2} n \leq \log^{B_0} N$ for sufficiently large n, after fixing a sufficiently small $C_{\mathrm{range}} > 0$. Finally, the choice of q ensures

$$C_{\mathrm{lift}} \log^{\beta_0} \Delta \leq C_0 \sqrt{q}.$$

Consequently Corollary 13.2 converts an algorithm A' as in the statement into a $C_0\sqrt{q}$-approximation algorithm A for Set Cover on N elements. Theorem 12.6 gives neighboring instances $I, \widetilde{I}$ on the same universe and index set satisfying $\mathsf{EMD}(A(I), A(\widetilde{I})) = \Omega(N^{\delta_0})$. Each instance has at most $C_{\mathcal{F}} N$ sets, and set sizes and element frequencies at most $\exp(q^{c_0}) \leq \Gamma$. Corollary 13.2 therefore gives

$$\mathsf{SwapSens}(A') \geq \Omega(N^{\delta_0}) - 2 = \Omega(n^{\delta}). \qquad \square$$

Within the theorem's degree range, we apply the sensitivity-to-locality argument directly. For larger values of Δ, we first restrict the algorithm to graphs satisfying a smaller upper bound on maximum degree within that range. This gives the following consequence; extending to all $\Delta \leq n - 1$ uses a smaller approximation exponent.

Corollary 13.6 (Locality lower bound via Dinur–Harsha)**.** There exist absolute constants $C > 0$ and $\beta \in (0,1)$ and an integer $\Delta_0 \geq 2$ such that, for all sufficiently large n and every $\Delta_0 \leq \Delta \leq n-1$, any $C \log^{\beta} \Delta$-approximate non-signaling algorithm for Dominating Set on n-vertex graphs of maximum degree at most Δ has locality at least

$$\Omega\left(\frac{\log n}{\log \Delta}\right).$$

Table 5: Selected approximation and locality bounds for Dominating Set on n vertices with maximum degree at most $\Delta \le n-1$. For every constant $\varepsilon \in (0,1)$, IKW has the ε-dependent degree restrictions in Table 4. DH allows all sufficiently large $\Delta \le n-1$, with $\beta \in (0,1)$. Upper bounds are for constant $0 < \varepsilon \le 1$, and hidden constants may depend on ε. The $(1+\varepsilon)\ln\Delta$ rows require sufficiently large Δ depending on ε. The Chang–Li upper bound succeeds with high probability.

Approximation ratio	Locality lower bound or upper bound	Model	Reference
	Lower bounds		
$1+\varepsilon$, $0<\varepsilon\le\varepsilon_0$	$\Omega(\log n/\varepsilon)$	LOCAL	[7]
Polylogarithmic in Δ	$\Omega\left(\min\left\{\sqrt{\frac{\log n}{\log\log n}}, \frac{\log\Delta}{\log\log\Delta}\right\}\right)$	LOCAL	[11, 27]
$(1-\varepsilon)\ln\Delta$	$\Omega_\varepsilon\left(\frac{\log n}{\log\Delta\,(\log\log\Delta)^2}\right)$	Non-signaling	Corollary 13.4
$C\log^\beta\Delta$	$\Omega\left(\frac{\log n}{\log\Delta}\right)$	Non-signaling	Corollary 13.6
	Upper bounds		
$1+\varepsilon$	$\mathrm{poly}(\log n)$	LOCAL	[20]
$1+\varepsilon$	$O_\varepsilon(\log n(\log\log n)^3)$	LOCAL	[7]
$(1+\varepsilon)\ln\Delta$	$O_\varepsilon(\log n)$	LOCAL	[27]
$(1+\varepsilon)\ln\Delta$	$O_\varepsilon(\log^2\Delta)$	CONGEST	From [26]

Proof of Corollary 13.6. Use the constants of Theorem 13.5 and set $\beta = \kappa\beta_0$. For sufficiently large n, write $L = \log n$ and put

$$\widehat{\Delta} := \min\left\{\lfloor\Delta\rfloor, \left\lfloor \exp\left(\frac{C_{\mathrm{range}}}{2}L^\kappa\right)\right\rfloor\right\}.$$

This integer lies in the degree range of Theorem 13.5. After increasing Δ_0, there is an absolute $c_* > 0$ such that

$$\log\widehat{\Delta} \ge c_*\min\{\log\Delta, L^\kappa\} \ge c_*\log^\kappa\Delta,$$

where the second inequality uses $1 \le \log\Delta \le L$. Choose $C \le C_{\mathrm{DS}}c_*^{\beta_0}$. Then

$$C\log^\beta\Delta \le C_{\mathrm{DS}}\log^{\beta_0}\widehat{\Delta}.$$

Let A be a t-local algorithm with the stated approximation guarantee. Restrict A to graphs of degree at most $\widehat{\Delta}$. Theorem 13.5 gives sensitivity $\Omega(n^\delta)$ on this class. Since $\log\widehat{\Delta} = O(\log^\kappa n)$ with $\kappa < 1$, the ratio $\log n/\log\widehat{\Delta}$ tends to infinity. The sensitivity-to-locality bound (1) gives, after absorbing its additive constant,

$$t = \Omega\left(\frac{\log n}{\log\widehat{\Delta}}\right) = \Omega\left(\frac{\log n}{\log\Delta}\right),$$

since $\widehat{\Delta} \le \Delta$. □

13.4 Comparison with Existing Distributed Bounds

Table 5 compares our two locality lower bounds with prior upper and lower bounds for Dominating Set.

Chang and Li [7] establish logarithmic dependence on n for approximation ratios sufficiently close to one. For approximation factors growing polylogarithmically with Δ, the KMW lower bound [11, 27]

becomes independent of n when Δ is constant. For every sufficiently large constant Δ, our locality lower bounds retain an $\Omega(\log n)$ dependence on graphs of maximum degree at most Δ, even in the non-signaling model. For $\Delta = \log^a n$ with constant $a > 0$, Corollary 13.6 gives $\Omega(\log n/\log\log n)$ for $C\log^\beta \Delta$-approximation, whereas the KMW bound in the table is $\Omega(\log\log n/\log\log\log n)$. For every constant $\varepsilon \in (0,1)$, our IKW locality lower bound applies to $(1-\varepsilon)\ln\Delta$-approximation throughout its stated degree range, while a $(1+\varepsilon)\ln\Delta$ approximation can be attained in $O_\varepsilon(\log^2\Delta)$ CONGEST rounds for sufficiently large Δ depending on ε, as discussed after Corollary 13.4.

Part V
Representative Selection and Directed Spanners

This part proves sensitivity and distributed lower bounds for minimum unweighted directed 3-spanner: selecting as few arcs as possible so that every input arc has a directed path of length at most three in the selected subgraph. For every constant approximation factor, we obtain a polynomial sensitivity lower bound on weakly connected DAGs of bounded total degree, where the degree bound may depend on the approximation factor. This gives an $\Omega(\log n)$ lower bound on LOCAL and non-signaling locality for all sufficiently large graph sizes n.

Section 14 first converts the low-soundness Label Cover bounds into bounds for selecting representatives from groups. Section 15 encodes these selections by three-arc directed paths while controlling both input edits and maximum degree. It also gives a polynomial sensitivity lower bound for a $\log^\beta n$ approximation factor on DAGs of degree $\exp(O(\log^\kappa n))$, for absolute constants $\beta > 0$ and $\kappa \in (0, 1/2)$. Section 16 uses the degree bounds to convert sensitivity into LOCAL and non-signaling locality lower bounds.

14 Sensitivity Lower Bounds for Min-Rep

We give a second application of the low-soundness Label Cover theorems, independent of the covering reductions in Sections 12 and 13. The intermediate problem is *Min-Rep*: choose a small collection of representatives that contains a compatible pair for every pair of label groups joined by at least one compatibility edge. This is the standard representative-selection problem used in spanner hardness reductions; see, for example, [12]. We analyze both the value of a recovered Label Cover solution and the effects of input edits and changes to the supplied representative set. The resulting reduction gives sensitivity $\Omega(n^\delta)$ for $\log^\beta n$-approximation on instances with n representatives, for some absolute constants $\beta, \delta > 0$.

There are two issues with applying Min-Rep directly to Theorem 11.1. The Label Cover graph need not be regular, and it may contain parallel edges carrying different projections. We address the first issue by copying label groups according to the source degrees, and the second by requiring a compatible representative pair to satisfy every parallel constraint between its source vertices. Both operations allow a recovery rule independent of the input predicates and projections.

14.1 The Problem and the Input-Edit Convention

A Min-Rep instance J consists of disjoint groups $(A_i)_{i\in L}$ and $(B_j)_{j\in R}$ and a simple bipartite *compatibility graph* with edge set

$$E_J \subseteq \bigcup_{i\in L,\, j\in R} A_i \times B_j.$$

Its superedge set is

$$\mathcal{E}_J := \{(i,j) \in L \times R : E_J \cap (A_i \times B_j) \neq \varnothing\}.$$

The bipartite graph $(L, R, \mathcal{E}_J)$ is the *supergraph*; its vertices index groups, and its edges are called *superedges*. A representative set $S \subseteq \bigcup_i A_i \cup \bigcup_j B_j$ is feasible if, for every $(i,j) \in \mathcal{E}_J$, some compatibility edge between A_i and B_j has both endpoints in S. The objective is to minimize $|S|$. Groups with no incident superedge need not contribute a representative. Every instance is feasible, including the empty compatibility graph, whose optimum is zero.

Throughout a family the representative vertices, groups, and their names are fixed. An input edit adds or deletes one compatibility edge. In particular, deleting the last compatibility edge between two groups *removes* that superedge; it does not leave an unsatisfiable demand. This convention is sufficient for sensitivity transfer: the recovered Label Cover algorithm needs a value guarantee only on satisfiable source instances. On other source instances, it must remain defined and obey the perturbation bounds; the reduction need not preserve a gap in optimum. Outputs are subsets of the same representative universe and have Hamming distance $|S \triangle \widetilde{S}|$. Sensitivity is the maximum output EMD over such single-edge edits within the input class under consideration. Approximation guarantees follow the convention of Section 2.

14.2 Copies Proportional to Vertex Degrees

We replace each source vertex's label group by as many copies as its degree. If we choose a Label Cover edge uniformly, counting parallel edges separately, and then independently choose one group copy uniformly at each endpoint, every left group has the same probability of being chosen, as does every right group. This will let us bound the recovered Label Cover value using only the total expected number of selected representatives, even when source degrees vary.

Fix a swap-closed Label Cover family on a common bipartite multigraph (U, V, E) with $m = |E| \geq 1$. Degrees count parallel edges. Put

$$d_z := \deg(z), \qquad D := \max_z d_z, \qquad Q := \max\{|\Sigma_U|, |\Sigma_V|\}.$$

The bounds D, Q and all auxiliary choices below are common to the family. Isolated source vertices receive a fixed default label in recovery and create no group. The spanner reduction in Section 15.1 requires every group to have the same size. We therefore pad groups to size Q with isolated dummy representatives. For each nonisolated $u \in U$ create d_u groups $A_{u,h}$, $h \in [d_u]$. In each group, the vertex corresponding to $a \in \Sigma_U$ is named (u, h, a); add $Q - |\Sigma_U|$ distinct dummy vertices with fixed names. Similarly, for each nonisolated $v \in V$ and $h \in [d_v]$, create $B_{v,h}$ with vertices (v, h, b) for $b \in \Sigma_V$ and $Q - |\Sigma_V|$ dummy vertices. Dummy vertices have no compatibility edges and cannot help satisfy a superedge, so they leave the optimum unchanged. Thus every group has size Q, while the source alphabets, predicates, and projections retain their original domains. There are

$$r = 2m \quad \text{groups}, \qquad N_{\text{rep}} = 2mQ \quad \text{representatives}. \tag{60}$$

For a source instance I, each distinct adjacent pair u, v, and each $h \in [d_u]$, $h' \in [d_v]$, insert the compatibility edge $(u, h, a)\,(v, h', b)$ for $a \in \Sigma_U$, $b \in \Sigma_V$ exactly when

$$P_u(a) = 1 \quad \text{and} \quad f_e(a) = b \quad \text{for every edge } e \text{ joining } u, v. \tag{61}$$

Let $\mathcal{T}_{\text{MR}}(I)$ be the resulting Min-Rep instance.

Lemma 14.1 (Structure and completeness)**.** The construction has at most mD^2Q compatibility edges. Its supergraph has maximum degree at most D^2, and its compatibility graph has maximum degree at most D^2Q. If I is satisfiable, every group has an incident superedge and

$$\mathsf{opt}(\mathcal{T}_{\mathrm{MR}}(I)) = r = 2m.$$

Proof. For a fixed group $A_{u,h}$, the number of potentially adjacent right groups is at most $\sum_{v:\, uv\in E} d_v \le Dd_u \le D^2$, where the sum is over *distinct* neighbors. The same bound holds on the right. A left representative has at most one compatible label in each adjacent right group. A right representative has at most Q neighbors in each adjacent left group. These observations prove the two degree bounds. Summing $d_u d_v Q \le D^2Q$ over the at most m distinct adjacent source pairs proves the edge bound.

Let π satisfy I. Selecting the copy of $\pi_U(u)$ in every $A_{u,h}$ and the copy of $\pi_V(v)$ in every $B_{v,h'}$ covers every potential group pair by (61). This gives a feasible set of size r. Every group comes from a positive-degree source vertex, so it has an incident superedge on a satisfiable input and must contribute at least one representative. The lower bound r follows. □

14.3 Value Recovery and Perturbation Bounds

We next recover a Label Cover labeling from a representative set. On satisfiable source instances, a feasible representative set of expected size at most ar will give expected value at least a^{-2}, for $a \ge 1$, where r is the number of groups. We then bound the effects of changes to the supplied representative set and swaps of source predicates or projections to transfer a sensitivity lower bound.

For every representative set, the recovery R_{MR} returns a random source labeling as follows. Independently for each nonisolated source vertex z, choose $h \in [d_z]$ uniformly. If its chosen group contains selected representatives, choose one of them uniformly and return its source label, mapping dummy vertices to fixed default source labels. If the chosen group contains no selected representative, return a fixed default source label. Isolated vertices also receive fixed labels. The rule uses the source degrees and fixed label names, but no predicate or projection.

We use the recovery parameters of Definition 3.1: $C_{\mathcal{R}}$ bounds the change in recovered output EMD per unit of representative-set distance, and $D_{\mathcal{R}}$ bounds the change under one source swap with the supplied representative set held fixed.

Lemma 14.2 (Value and Lipschitz bounds for recovery)**.** Suppose I is satisfiable and a random set $\boldsymbol{S}$ is a feasible solution to $\mathcal{T}_{\mathrm{MR}}(I)$ almost surely. If $\mathbf{E}\,|\boldsymbol{S}| \le ar$ for a real $a \ge 1$, then

$$\mathbf{E}\big[\mathsf{val}_I(R_{\mathrm{MR}}(\boldsymbol{S}))\big] \ge a^{-2}. \tag{62}$$

Moreover, for arbitrary representative sets $S, \widetilde{S}$,

$$\mathsf{EMD}(R_{\mathrm{MR}}(S), R_{\mathrm{MR}}(\widetilde{S})) \le |S \triangle \widetilde{S}|. \tag{63}$$

Thus the recovery has $C_{\mathcal{R}} \le 1$ and $D_{\mathcal{R}} = 0$.

Proof. Condition first on a feasible set S. Let $\ell_{z,h}$ be the number of selected representatives in group (z,h). Every such number is positive, by Lemma 14.1. Write $\boldsymbol{\pi}$ for the labeling returned by $R_{\mathrm{MR}}(S)$, and choose an edge $e = (u,v)$ uniformly from E, counting parallel edges separately, independently of this recovery. Conditional on e, the copy indices $h \in [d_u]$ and $h' \in [d_v]$ used by the recovery at its endpoints are

independent and uniform. Feasibility supplies a selected compatible pair in these groups. By (61), that pair satisfies every edge joining u, v, including the sampled edge e. Hence

$$\Pr\big[\boldsymbol{\pi} \text{ satisfies } e \mid S, e, h, h'\big] \geq \frac{1}{\ell_{u,h}\ell_{v,h'}} \geq \frac{4}{(\ell_{u,h} + \ell_{v,h'})^2}.$$

Each left group $A_{u,h}$ is chosen with probability $(d_u/m)(1/d_u) = 1/m$: the sampled edge has left endpoint u with probability d_u/m, and the recovery chooses copy h with probability $1/d_u$. Each right group $B_{v,h'}$ is also chosen with probability $1/m$ by the same calculation. Consequently, now including the randomness of $\boldsymbol{S}$,

$$\mathbf{E}[\ell_{u,h} + \ell_{v,h'}] = \frac{\mathbf{E}\,|\boldsymbol{S}|}{m} \leq 2a.$$

Apply Jensen's inequality to the positive random variable $\ell_{u,h} + \ell_{v,h'}$ and the convex function $z \mapsto 4/z^2$. This proves (62) using only the expected size of $\boldsymbol{S}$.

If $S, \widetilde{S}$ differ at one representative in a group of z, share the copy indices and all recovery randomness outside z. Only the recovered label at z can differ. A path that adds or deletes one representative at a time and the triangle inequality prove (63). The fixed recovery rule proves $D_{\mathcal{R}} = 0$. The same bounds on distributions follow from Lemma 3.3. □

Lemma 14.3 (Input edits)**.** A single predicate or projection swap in the source changes at most

$$J_0 := 2D^3Q \tag{64}$$

compatibility edges. The two resulting Min-Rep instances admit a shortest single-edge-edit path along which the upper bounds on the number of edges and the two maximum degrees in Lemma 14.1 continue to hold.

Proof. A predicate swap at u changes only edges incident to copies of u; a projection swap on $e = (u, v)$ affects only edges between copies of u and v. In either instance, the number of edges incident to copies of u is at most

$$d_u Q \sum_{v:\, uv \in E} d_v \leq D^3 Q.$$

The two compatibility edge sets therefore differ in at most $2D^3Q$ edges. Delete edges present only in the first instance, then insert those present only in the second. Each intermediate compatibility graph is a subgraph of one endpoint, and its supergraph is also a subgraph of that endpoint's supergraph. The upper bounds on the number of edges and the degrees are therefore preserved. □

Theorem 14.4 (Label Cover to Min-Rep sensitivity)**.** Let $\mathcal{I}$ be a swap-closed Label Cover family on a common bipartite multigraph (U, V, E) and common alphabets, with $m = |E| \geq 1$. Let D be its maximum degree, $Q = \max\{|\Sigma_U|, |\Sigma_V|\}$, and $r = 2m$ the number of groups in $\mathcal{T}_{\mathrm{MR}}(I)$. Suppose every algorithm on $\mathcal{I}$ with expected Label Cover value at least $1/g$ on satisfiable members has swap sensitivity at least L, where $g > 1$. Let $\mathcal{J}$ consist of the images $\mathcal{T}_{\mathrm{MR}}(I)$, $I \in \mathcal{I}$, together with the instances on a path from Lemma 14.3 for each neighboring source pair. Any Min-Rep algorithm defined on $\mathcal{J}$ that, for every satisfiable $I \in \mathcal{I}$, returns an almost-surely feasible set of expected size at most $\sqrt{g}\, r$ on $\mathcal{T}_{\mathrm{MR}}(I)$ has single-compatibility-edge sensitivity on $\mathcal{J}$ at least

$$\frac{L}{2D^3Q}. \tag{65}$$

In particular this holds for any $\sqrt{g}$-approximation algorithm for Min-Rep on $\mathcal{J}$.

Proof. Let A be a Min-Rep algorithm satisfying the hypothesis, and let B be the Label Cover algorithm that, on input $I \in \mathcal{I}$, returns $R_{\mathrm{MR}}(A(\mathcal{T}_{\mathrm{MR}}(I)))$. By Lemma 14.2, B has expected value at least $1/g$ on every satisfiable $I \in \mathcal{I}$. For two neighboring source instances $I, \widetilde{I}$, follow the path in $\mathcal{J}$ from $\mathcal{T}_{\mathrm{MR}}(I)$ to $\mathcal{T}_{\mathrm{MR}}(\widetilde{I})$ supplied by Lemma 14.3; its length is at most $2D^3Q$. Couple the outputs of A along this path and apply (63). The swap sensitivity of B on $\mathcal{I}$ is therefore at most $2D^3Q$ times the sensitivity of A on $\mathcal{J}$. Since $\mathsf{SwapSens}(B, \mathcal{I}) \geq L$ by hypothesis, this proves the claimed lower bound on the sensitivity of A. The final assertion uses $\mathsf{opt}(\mathcal{T}_{\mathrm{MR}}(I)) = r$ on satisfiable inputs. □

The spanner reduction need not provide a multiplicative Min-Rep approximation on instances with many groups having no incident superedge. To apply Theorem 14.4, it suffices that, on the image of every satisfiable Label Cover instance, the spanner reduction returns an almost-surely feasible representative set of expected size at most $\sqrt{g}\, r$. The recovered labeling needs no value guarantee on the other source instances.

14.4 Consequences of the Low-Soundness Theorems

We first use the Dinur–Harsha theorem to obtain a polynomial sensitivity lower bound for every sufficiently large number of representative vertices. We then apply the IKW theorem, which trades a larger approximation factor against a smaller sensitivity exponent at the sizes of its construction.

Corollary 14.5 (Min-Rep with a prescribed number of representative vertices)**.** There are absolute constants $\beta, \delta > 0$, $\kappa \in (0, 1/2)$, and $C > 0$ such that, for all sufficiently large n, every $\log^{\beta} n$-approximation algorithm for Min-Rep with exactly n representative vertices has sensitivity $\Omega(n^{\delta})$. The bound is witnessed by two instances on the same n representative vertices, with the same partition into groups, that differ by one compatibility edge. In both instances, group sizes and the maximum degrees of the compatibility graph and the supergraph are at most $\exp(C \log^{\kappa} n)$.

Proof. Let $B, \delta_{\mathrm{LC}} > 0$ be from Theorem 11.1. For source parameters s, g in that theorem's range, let $\mathcal{I}_{s,g}$ be the supplied family on a common multigraph (U, V, E) with $|U| = s$. Write $m = |E|$, and let D, Q be its maximum degree and maximum alphabet size as in Section 14.2. Since $m \leq sD$, there are constants $a \geq 1$, $C_0 > 0$ such that, uniformly for these families with $g \geq 2$,

$$D^3Q \leq \exp(C_0 g^a), \qquad 2mQ \leq s \exp(C_0 g^a).$$

Choose $0 < \theta < B$ with $a\theta < 1/2$, put $s = \lfloor n^{1/2} \rfloor$, and set $g = \log^{\theta} s$. For sufficiently large n, the source theorem applies and the number of representative vertices is less than n. Pad to n by adding fixed isolated singleton groups on the left. Padding changes neither the optimum nor the edge-edit distance; recovery ignores the new groups. Choose any $0 < \beta < \theta/2$. Then $\log^{\beta} n \leq \sqrt{g}$ for all sufficiently large n. By Theorem 14.4, the sensitivity is at least

$$\frac{\Omega(s^{\delta_{\mathrm{LC}}})}{\exp(C_0 g^a)} = n^{\delta_{\mathrm{LC}}/2 - o(1)}.$$

Take a constant $\delta < \delta_{\mathrm{LC}}/2$ and $\kappa = a\theta < 1/2$. Every group has size at most Q, and Lemma 14.1 bounds the maximum degrees of the supergraph and the compatibility graph by D^2 and D^2Q, respectively. These three quantities are at most $D^3Q \leq \exp(C_0 g^a)$. Since $g^a = \log^{\kappa} s \leq \log^{\kappa} n$, they are at most $\exp(C_0 \log^{\kappa} n)$. The padding and all names are independent of the source predicates and projections, so the two instances witnessing the lower bound have the same n representative vertices and differ by one compatibility edge. □

For the IKW consequence, let N be the number of original 2-CSP variables and let $k = k(N)$ be sufficiently large with $k = o(\log N)$, as in Theorem 6.1. Write C_{IKW} for its constant C in the value guarantee, and η for its sensitivity exponent. For the resulting Label Cover family, let s be the total number of vertices and D, Q its maximum degree and alphabet size. With $g = \exp\left(C_{\mathrm{IKW}}\sqrt{k}\right)$ and $D, Q \le 2^{O(k)}$, Theorem 14.4 gives

$$\text{approximation } \exp\left(C_{\mathrm{IKW}}\sqrt{k}/2\right), \qquad \text{sensitivity at least } s^{\eta/k}/2^{O(k)}.$$

Since $s = N^{\Theta(k)}$ and $k = o(\log N)$ in that theorem, $k^2 = o(\log s)$, so the denominator can be absorbed in a bound $s^{\Omega(1/k)}$. We record the corresponding spanner consequence in Corollary 15.9.

15 Sensitivity Lower Bounds for Unweighted Directed 3-Spanner

We reduce Min-Rep to minimum unweighted directed 3-spanner and use the reduction to transfer the sensitivity lower bounds of Section 14. All graphs in this section are simple directed graphs; all arcs have unit length and unit cost. For $G = (V, A)$, a *directed 3-spanner* is an arc set $H \subseteq A$ such that every $(u, v) \in A$ has a directed u–v path of length at most three in (V, H). This is equivalent to $d_H(u, v) \le 3d_G(u, v)$ for every reachable ordered pair, by replacing each arc of a shortest path. Thus $\mathsf{opt}(G)$ is the minimum number of arcs in a directed 3-spanner of G.

We approximate the number of arcs while requiring stretch at most three. Outputs are embedded in the common universe of all ordered pairs of distinct vertex names, and their distance is $|H \triangle \widetilde{H}|$. Input sensitivity is under insertion or deletion of one arc. A degree bound means $\Delta(G) = \max_v(\deg^+_G(v) + \deg^-_G(v))$. The constructions below are directed acyclic graphs (DAGs), and we also preserve weak connectivity, meaning connectivity after forgetting arc directions.

For every constant approximation factor $\alpha \ge 1$, we obtain sensitivity $\Omega(n^\delta)$ on weakly connected n-vertex DAGs of degree at most a constant Δ_α, for an absolute $\delta > 0$ and all sufficiently large n. We also obtain a polynomial sensitivity lower bound for approximation factor $\log^\beta n$ with maximum degree $\exp(O(\log^\kappa n))$, for absolute $\beta > 0$ and $\kappa \in (0, 1/2)$. These degree bounds are essential for the next section: they limit how many output arcs can be affected by a change within a local neighborhood.

Classical reductions from Min-Rep to undirected spanners use large girth of the supergraph to exclude unwanted short routes [12]. We instead use directions and two-hop auxiliary routes to characterize every three-hop alternative explicitly. The auxiliary vertices are local to individual groups. There is no vertex adjacent to all representatives, and the maximum degree depends only on the degrees and alphabet sizes of the source Label Cover instance.

15.1 The Directed Construction

The construction represents a compatible pair by a three-arc path from a new source to a new sink. Selecting the two representatives corresponds to including the first and last arcs; the middle arc records their compatibility. Auxiliary two-arc routes span the first and last arcs when they are unselected. Using either auxiliary route in place of one of these arcs gives a path of length four, which cannot span the source-to-sink demand arc within stretch three. We repeat the sources and sinks so that the cost of selecting representatives can absorb the cost of the other arcs.

Consider a Min-Rep instance J with fixed groups $(A_i)_{i \in L}$ and $(B_j)_{j \in R}$, where $r = |L| + |R| \ge 2$, every group has size $Q \ge 1$, and $M = |E_J|$. Let $M_{\max}$ be a common upper bound on the number of compatibility

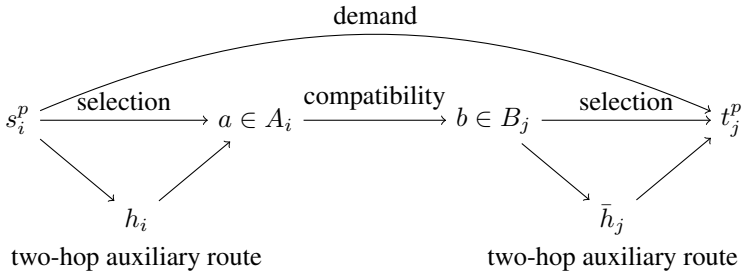


Figure 4: The arcs associated with a compatible pair $(a, b) \in E_J$ and a repetition index $p \in [x]$. All arcs have unit length and unit cost. Omitting a demand arc requires the three-arc route through a compatible pair and both selection arcs; using either auxiliary route would require at least four arcs. The four-arc paths connecting consecutive hubs are not shown.

edges in the family. The number of repetitions is an integer x satisfying

$$x \geq Q + \lceil M_{\max}/r \rceil + 4. \tag{66}$$

In particular, x does not change when a compatibility edge is edited. Retain all representative vertices. For every left group A_i create sources s_i^p, $p \in [x]$, and a hub h_i. For every right group B_j create sinks t_j^p, $p \in [x]$, and a hub $\bar{h}_j$. The representatives and hubs are shared by all x repetitions. Insert the following arcs:

$$\begin{aligned}
A_{\text{comp}} &:= \{a \to b : (a, b) \in E_J\},\\
A_{\text{sel}} &:= \bigcup_{i \in L} \{s_i^p \to a : a \in A_i,\ p \in [x]\} \ \cup \ \bigcup_{j \in R} \{b \to t_j^p : b \in B_j,\ p \in [x]\},\\
A_{\text{aux}} &:= \bigcup_{i \in L} \{s_i^p \to h_i,\ h_i \to a : a \in A_i,\ p \in [x]\}\\
&\quad \cup \bigcup_{j \in R} \{b \to \bar{h}_j,\ \bar{h}_j \to t_j^p : b \in B_j,\ p \in [x]\},\\
A_{\text{dem}} &:= \{s_i^p \to t_j^p : (i, j) \in \mathcal{E}_J,\ p \in [x]\}.
\end{aligned}$$

Each arc is included only once, even if it occurs for several values of p. The selection and auxiliary arcs are fixed throughout the family. Demand arcs are present precisely for actual superedges, including when J is obtained from an unsatisfiable Label Cover instance.

To ensure weak connectivity, order all r hubs with the left hubs first and the right hubs second. Between each consecutive pair insert a directed path of length four from the earlier hub to the later hub, using three private new vertices. Its length prevents it from participating in an alternative path of length at most three from a source to a sink. Let A_{con} be these $4(r-1)$ arcs, and let $\mathcal{T}_{\text{sp}}(J)$ be the resulting graph. The private vertices and paths are fixed across the family.

Lemma 15.1 (Structure and three-hop routes)**.** Suppose the compatibility graph has maximum degree at most Δ_{rep} and the supergraph has maximum degree at most Δ_{grp}. The graph $\mathcal{T}_{\text{sp}}(J)$ is a weakly connected DAG whose number of vertices n_0 and maximum total degree satisfy

$$n_0 = r(Q + x + 4) - 3, \qquad \Delta(\mathcal{T}_{\text{sp}}(J)) \leq x + Q + \Delta_{\text{rep}} + \Delta_{\text{grp}} + 4. \tag{67}$$

For each demand arc $s_i^p \to t_j^p$, any alternative directed path of length at most three is exactly

$$s_i^p \to a \to b \to t_j^p, \qquad a \in A_i,\ b \in B_j,\ (a, b) \in E_J. \tag{68}$$

Proof. There are rQ representatives, rx sources and sinks in total, r hubs, and $3(r-1)$ private connector vertices. Each group is connected to its hub after forgetting directions, and the connector paths join all hubs. A topological ordering puts sources first, then the left hubs in their fixed order, then left representatives, right representatives, the right hubs in their fixed order, and sinks. The internal vertices of each connector path are placed in path order between its endpoints. Thus no directed cycle is introduced.

Each source or sink has degree at most $Q+1+\Delta_{\mathrm{grp}}$, each representative at most $x+1+\Delta_{\mathrm{rep}}$, each hub at most $x+Q+2$, and each internal vertex of a connector path at most two. These give (67).

An alternative path of length at most three from a source to a sink cannot contain an internal vertex of a connector path, since traversing that connector path requires four arcs. Sinks have no outgoing arcs, so a path using a demand arc must end immediately at that demand's sink. A route using h_i or $\bar{h}_j$ in place of a selection arc has length at least four. The only remaining possibility of length at most three is (68). □

Let B_J be the number of compatibility, auxiliary, and connector arcs:

$$B_J := M + r(Q+x) + 4(r-1).$$

Our spanner will include all these arcs, with the remaining cost determined by the selected representatives. We also record bounds on the number of vertices and the optimum to control the cost of padding later.

Lemma 15.2 (A spanner from representatives)**.** If S is a feasible Min-Rep set, the graph has a 3-spanner of size

$$B_J + x|S|. \tag{69}$$

In particular, if $\mathsf{opt}(J) = r$, then

$$\mathsf{opt}(\mathcal{T}_{\mathrm{sp}}(J)) \le 3xr. \tag{70}$$

Also $n_0 \le 2rx$ and $\mathsf{opt}(\mathcal{T}_{\mathrm{sp}}(J)) \ge n_0 - 1$.

Proof. Include every compatibility, auxiliary, and connector arc. Include $s_i^p \to a$ exactly when $a \in S$, and $b \to t_j^p$ exactly when $b \in S$. Include no demand arc. Each omitted selection arc has its two-hop auxiliary route. Each demand has a three-hop route through a compatible pair in S. All other input arcs are included, so this is a 3-spanner of the stated size. By (66), $M + rQ + 4(r-1) \le rx$, and hence $B_J + xr \le 3xr$. The same inequality $x \ge Q+4$ gives $n_0 \le 2rx$. Finally, replacing each input arc by a directed path preserves weak connectivity, so every 3-spanner has at least $n_0 - 1$ arcs. □

15.2 Recovery by Sampling a Repetition

We recover representatives by choosing an index $p \in [x]$ uniformly and examining the selection and demand arcs with that index. A single compatibility edit can create or remove x demand arcs, but each changed selection or demand arc in the output is used with probability $1/x$. This averaging will cancel the factor x in the sensitivity transfer.

For every left group A_i and right group B_j, fix an ordering of $A_i \times B_j$ independently of the compatibility graph. Let $w_J(i,j) \subseteq A_i \cup B_j$ be the set of endpoints of the first edge in $E_J \cap (A_i \times B_j)$ under this ordering, or $\varnothing$ if this block is empty. For an arbitrary set H of ordered pairs on the construction's common vertex set and $p \in [x]$, define

$$S_p(J,H) := \bigcup_{i \in L} \{a \in A_i : s_i^p \to a \in H\} \cup \bigcup_{j \in R} \{b \in B_j : b \to t_j^p \in H\}$$

$$\cup \bigcup_{\substack{i\in L,\, j\in R:\\ s_i^p \to t_j^p \in H}} w_J(i,j).$$

Pairs not of these forms are ignored. In particular this rule is defined also for outputs infeasible for the current input. Let $R_{\mathrm{sp},J}(H)$ choose $p \in [x]$ uniformly and return $S_p(J,H)$.

Lemma 15.3 (Feasibility, size, and recovery sensitivity)**.** If H is a 3-spanner of $\mathcal{T}_{\mathrm{sp}}(J)$, then every $S_p(J,H)$ is feasible for Min-Rep, and

$$\mathbf{E}\,|R_{\mathrm{sp},J}(H)| \le \frac{2|H|}{x}. \tag{71}$$

For arbitrary output sets $H, \widetilde{H}$ on the common domain,

$$\mathsf{EMD}(R_{\mathrm{sp},J}(H), R_{\mathrm{sp},J}(\widetilde{H})) \le \frac{2}{x}|H \triangle \widetilde{H}|. \tag{72}$$

If $J, \widetilde{J}$ differ by one compatibility-edge edit, then for every fixed H,

$$\mathsf{EMD}(R_{\mathrm{sp},J}(H), R_{\mathrm{sp},\widetilde{J}}(H)) \le 4. \tag{73}$$

Proof. Fix a superedge (i,j) and an index $p \in [x]$. If its demand arc is in H, the rule inserts a compatible pair. Otherwise Lemma 15.1 supplies the route (68) in H, and both its representatives are selected by the rule. This proves feasibility. Every selection arc is charged once when summing $|S_p|$ over p; every demand arc is charged at most twice. Thus $\sum_p |S_p(J,H)| \le 2|H|$, proving (71).

For (72), use the same p in both recoveries. A change of one selection arc affects at most one representative for one value of p, and a demand arc affects at most two for one value of p. Every other arc has no effect. Summing over the arcs in $H \triangle \widetilde{H}$ proves the bound, also for infeasible outputs.

An input edit changes $w_J(i,j)$ for at most one group pair. The old and new sets each contain at most two representatives. Under the same choice of p, at most four representatives can therefore change. This includes deletion of the last compatibility edge, when one of the two sets is empty. Averaging proves (73). □

For this reduction, $C_{\mathcal{T}}$ bounds the number of arc edits caused by one edit to the compatibility graph. The parameter $C_{\mathcal{R}}$ bounds the change in recovered output EMD per unit of arc-set distance. The parameter $D_{\mathcal{R}}$ bounds the effect on recovery of one compatibility-edge edit with the output arc set held fixed, as in Definition 3.1.

Lemma 15.4 (Sensitivity-transfer parameters)**.** A compatibility-edge edit changes at most $x+1$ arcs of the constructed graph. The transformation and recovery have parameters

$$C_{\mathcal{T}} \le x+1, \qquad C_{\mathcal{R}} \le 2/x, \qquad D_{\mathcal{R}} \le 4, \qquad C_{\mathcal{T}} C_{\mathcal{R}} \le 4.$$

If two instances $J, \widetilde{J}$ in the Min-Rep family have compatibility graphs of maximum degree at most Δ_{rep} and supergraphs of maximum degree at most Δ_{grp}, then $G = \mathcal{T}_{\mathrm{sp}}(J)$ and $\widetilde{G} = \mathcal{T}_{\mathrm{sp}}(\widetilde{J})$ can be joined by a shortest arc-edit path of weakly connected DAGs satisfying the degree bound in (67).

Proof. The compatibility arc itself changes. Only when its block becomes empty or nonempty do the x demand arcs change as well. All other arcs and all vertices are fixed. This proves the bound on the number of arc edits; the recovery bounds follow from Lemma 15.3. To obtain a shortest arc-edit path from G to $\widetilde{G}$, first delete the arcs present only in G, then insert those present only in $\widetilde{G}$. Each intermediate graph is a subgraph of G or $\widetilde{G}$ and contains the weakly connected spanning subgraph with arc set $A_{\mathrm{aux}} \cup A_{\mathrm{con}}$. All intermediate graphs respect the same acyclic orientation and satisfy the common degree bound. □

Thus the product $C_{\mathcal{T}} C_{\mathcal{R}}$ stays bounded even when a superedge appears or disappears. The additive term $D_{\mathcal{R}}$ accounts for the input dependence of the compatible pair $w_J(i, j)$ selected for a demand arc in H.

15.3 Connected Padding to a Prescribed Number of Vertices

To obtain a lower bound at every prescribed graph size, we must add vertices while controlling their contribution to the optimum. Repeating the graph before adding a short tail keeps this extra cost within a constant factor and allows recovery by sampling one copy.

Let G be a weakly connected DAG on a fixed set of $n_0 \geq 2$ vertices, and fix an anchor vertex. For every $n \geq n_0$, write $n = qn_0 + h$, where $q \geq 1$ and $0 \leq h < n_0$. Construct $P_n(G)$ from q disjoint named copies of G by joining their anchors with arcs oriented from copy j to copy $j+1$ for $j \in [q-1]$. Append a directed path of h new vertices starting at the last anchor. All names, anchors, and added arcs are fixed independently of the input arcs of G.

For any set H of ordered pairs of distinct vertices of $P_n(G)$, let $U_n(H)$ choose one of the q copies uniformly, retain only pairs whose endpoints both lie in that copy, and return them under the original vertex names of G. Thus U_n is defined on all arc sets on the common vertex universe and is independent of the input arcs.

Lemma 15.5 (Connected replication and padding)**.** For the construction and recovery above, the following properties hold.

1. $P_n(G)$ is a weakly connected DAG on exactly n vertices of maximum total degree at most $\Delta(G) + 2$.
2. Every 3-spanner of $P_n(G)$ restricts to a 3-spanner on each of the q fixed copies of G, and
$$\mathsf{opt}(P_n(G)) = q\mathsf{opt}(G) + (q-1) + h \leq 2q\mathsf{opt}(G).$$
3. For every arc set H on the common vertex universe, $\mathbf{E}\,|U_n(H)| \leq |H|/q$. The recovery U_n is $1/q$-Lipschitz in output EMD on this entire domain and is independent of the input arcs. An edit of one arc of G changes exactly q arcs of $P_n(G)$.

Proof. The arcs joining copies and forming the tail are bridges in the underlying undirected graph, and every one must be included in a 3-spanner. A directed path that leaves a copy cannot return to it. Hence an input arc inside that copy can only be spanned by arcs of that copy. This proves the optimum identity and the restriction property. Conversely, the union of spanners of the copies and all added arcs spans the padded graph.

Since $\mathsf{opt}(G) \geq n_0 - 1$,
$$(q-1) + h \leq q + n_0 - 2 \leq q(n_0 - 1) \leq q\mathsf{opt}(G),$$
proving the factor-two bound. The degree and DAG assertions follow from the construction. To compare $U_n(H)$ and $U_n(\widetilde{H})$, choose the same copy in both recoveries. Each changed pair internal to a copy is seen with probability $1/q$, while all other pairs are ignored. This proves the Lipschitz bound; the same counting applied to H proves the size bound. A change to one input arc of G changes that arc in each of the q copies, proving the final claim. □

For the graph family in Section 15.1, the subgraph with arc set $A_{\mathrm{aux}} \cup A_{\mathrm{con}}$ inside each copy remains present throughout every deletion-then-insertion path. Thus the preceding padding also preserves weak connectivity, acyclicity, and the degree bound along the paths needed for sensitivity transfer. One need not assume that intermediate graphs are themselves images of Min-Rep instances.

15.4 Sensitivity Transfer from Label Cover

We now compose the three recoveries to obtain a Label Cover algorithm from a spanner approximation algorithm. To guarantee the required expected Label Cover value, it suffices to bound the expected size of the recovered representative set on images of satisfiable Label Cover instances, as in Theorem 14.4. Fix a common source graph and alphabets as in Section 14.2: m is its number of edges, D its maximum degree, Q the larger alphabet size, and $r = 2m$ the number of Min-Rep groups. Choose

$$x = (D^2+1)Q + 4, \qquad n_0 = r(Q+x+4) - 3, \qquad \Delta_* := x + Q + D^2Q + D^2 + 6. \tag{74}$$

The choice of x satisfies (66), since $M_{\max} = mD^2Q$ is a common upper bound. All graphs below have maximum total degree at most Δ_*, including padding and intermediate arc-edit paths.

Proposition 15.6 (Sensitivity transfer to directed 3-spanners with a prescribed number of vertices)**.** Let $\mathcal{I}$ be a swap-closed Label Cover family on the common source graph above, and suppose expected value $1/g$ on its satisfiable members requires swap sensitivity at least L. Fix any $n \geq n_0$ and any $\alpha \geq 1$ with

$$144\alpha^2 \leq g. \tag{75}$$

Any α-approximation algorithm for minimum directed 3-spanner on all weakly connected n-vertex DAGs of maximum total degree at most Δ_* has single-arc sensitivity at least

$$\frac{L}{4J_0} - 1, \qquad J_0 = 2D^3Q. \tag{76}$$

The witnessing pair is in this same graph class.

Proof. Fix an α-approximation algorithm A on the stated graph class. For each $I \in \mathcal{I}$, construct

$$J_I = \mathcal{T}_{\mathrm{MR}}(I), \qquad G_I = \mathcal{T}_{\mathrm{sp}}(J_I), \qquad \widehat{G}_I = P_n(G_I).$$

All vertex sets, the number q of padding copies, and the choices of names and anchors are the same for every $I \in \mathcal{I}$. Define a Label Cover algorithm B by

$$B(I) := R_{\mathrm{MR}}\big(R_{\mathrm{sp},J_I}(U_n(A(\widehat{G}_I)))\big).$$

Here U_n restricts the spanner output to a uniformly chosen copy of G_I and restores its original vertex names. Writing H for this restricted arc set, R_{sp,J_I} chooses $p \in [x]$ uniformly and returns $S_p(J_I, H)$, and R_{MR} recovers a source labeling. Each recovery uses fresh randomness conditional on its supplied output.

If I is satisfiable, Lemma 14.1 gives a Min-Rep solution of size r, and Lemmas 15.2 and 15.5 give $\mathsf{opt}(\widehat{G}_I) \leq 6qxr$. The composition $R_{\mathrm{sp},J_I} \circ U_n$ produces an almost-surely feasible Min-Rep set $\boldsymbol{S}$ with

$$\mathbf{E}\,|\boldsymbol{S}| \leq \frac{2}{qx}\,\mathbf{E}\,|A(\widehat{G}_I)| \leq 12\alpha r.$$

By Lemma 14.2, the expected value of the recovered labeling $B(I)$ is at least $(12\alpha)^{-2} \geq 1/g$. On other source instances B remains defined, but we require no bound on the expected value of $B(I)$. The approximation guarantee for A continues to hold on the entire stated graph class.

For one compatibility-edge edit, the padded graphs differ in at most $q(x+1)$ arcs. The composition $R_{\mathrm{sp},J_I} \circ U_n$ has Lipschitz constant $2/(qx)$. At a fixed supplied arc set, this input edit changes the recovered

distribution by EMD at most four. The subsequent map R_{MR} is 1-Lipschitz and input-independent. A source swap can be realized by at most J_0 compatibility-edge edits. If S_A is the spanner algorithm's sensitivity on the stated graph class, triangle inequalities along the corresponding valid graph paths therefore give

$$\mathsf{SwapSens}(B, \mathcal{I}) \le 2J_0(1 + 1/x)S_A + 4J_0 \le 4J_0S_A + 4J_0.$$

This is the sensitivity-transfer argument of Theorem 3.2, with paths confined to the graph class just verified. Its left-hand side is at least L. Rearranging proves (76). □

15.5 Consequences of the Low-Soundness Theorems

Our Label Cover lower bound via Dinur–Harsha (Theorem 11.1) gives a constant positive sensitivity exponent at every sufficiently large graph size, including for approximation factors that are small powers of $\log n$. Our lower bound via IKW (Theorem 6.1) gives a different trade-off for the graph sizes produced by the construction: approximation factor $\exp\left(c_1\sqrt{k}\right)$ requires sensitivity $n^{c_2/k}$ for absolute constants $c_1, c_2 > 0$, where k is the IKW direct-product parameter. Table 6 summarizes the sensitivity and degree guarantees.

Table 6: Sensitivity lower bounds for minimum unweighted directed 3-spanner on weakly connected n-vertex DAGs. The first two rows use DH and hold for every sufficiently large n, with a constant sensitivity exponent and $\kappa \in (0, 1/2)$. The third row uses IKW and holds at constructed sizes $n = N^{\Theta(k)}$, with k sufficiently large and $k = o(\log N)$ as in Theorem 6.1. Here $c_1 > 0$ is an absolute constant.

Approximation factor	Maximum total degree	Sensitivity lower bound	Reference
Any constant $\alpha \ge 1$	$O_\alpha(1)$	$n^{\Omega(1)}$	Corollary 15.8
$\log^\beta n$	$\exp(O(\log^\kappa n))$	$n^{\Omega(1)}$	Corollary 15.8
$\exp\left(c_1\sqrt{k}\right)$	$2^{O(k)}$	$n^{\Omega(1/k)}$	Corollary 15.9

Theorem 15.7 (Directed 3-spanner sensitivity via Dinur–Harsha)**.** There are absolute constants $a \ge 1$, $C > 0$, $\delta > 0$, and $\theta > 0$ with $a\theta < 1/2$ such that the following holds uniformly for all sufficiently large n and all real

$$144 \le g \le \log^\theta n.$$

Every $(\sqrt{g}/12)$-approximation algorithm for minimum unweighted directed 3-spanner on weakly connected n-vertex DAGs of maximum total degree at most

$$\Delta_g := \lceil \exp(Cg^a) \rceil \tag{77}$$

has single-arc sensitivity $\Omega(n^\delta)$. Both input graphs witnessing the sensitivity lie in the stated class.

Proof. Let $B, \delta_{\mathrm{LC}} > 0$ be from Theorem 11.1. For source parameters s, g in its range, let $\mathcal{I}_{s,g}$ be the supplied family on a common graph (U, V, E) with $|U| = s$. Write $m = |E|$ and let D, Q be its maximum degree and alphabet size. Use these parameters to define J_0, Δ_*, n_0 as in Proposition 15.6 and (74). Since $m \le sD$, the source theorem's uniform structural bounds allow constants $a \ge 1, C_1 > 0$ such that, for these families with $g \ge 144$,

$$J_0 \le \exp(C_1 g^a), \qquad \Delta_* \le \exp(C_1 g^a), \qquad n_0 \le s \exp(C_1 g^a).$$

Increase a, C_1 once to absorb all absolute constants. Choose $0 < \theta < B$ with $a\theta < 1/2$ and put $s = \lfloor n^{1/2} \rfloor$. For all sufficiently large n, uniformly over the claimed g-range, $g \le \log^B s$ and $\exp(C_1 g^a) = n^{o(1)}$. Thus $n_0 \le n$ and the source theorem supplies $L = \Omega(s^{\delta_{\mathrm{LC}}})$. Choose $C \ge C_1$, so that the degree bound Δ_* of Proposition 15.6 is at most Δ_g. Fix an algorithm A with approximation factor $\alpha = \sqrt{g}/12$ on the graph class in the statement, and let S_A be its sensitivity on that class. Applying the proposition to the restriction of A to degree at most Δ_* gives

$$S_A \ge \frac{\Omega(s^{\delta_{\mathrm{LC}}})}{4\exp(C_1 g^a)} - 1 = n^{\delta_{\mathrm{LC}}/2 - o(1)}.$$

Choose any constant $0 < \delta < \delta_{\mathrm{LC}}/2$. The graph paths in the proposition stay within degree $\Delta_* \le \Delta_g$, which proves the claimed witness assertion and uniformity. □

Corollary 15.8 (Growing and constant approximation factors)**.** There are absolute constants $\beta \in (0,1)$, $\kappa \in (0, 1/2)$, $\delta > 0$, and $C > 0$ such that $\log^\beta n$-approximation for minimum unweighted directed 3-spanner has sensitivity $\Omega(n^\delta)$ already on weakly connected DAGs of maximum total degree at most $\exp(C \log^\kappa n)$. Moreover, for every constant $\alpha \ge 1$ there is a constant degree bound Δ_α such that α-approximation on weakly connected n-vertex DAGs of degree at most Δ_α has sensitivity $\Omega(n^\delta)$ for all sufficiently large n.

Proof. For the first assertion set $g = \log^\theta n$ in Theorem 15.7, take $\kappa = a\theta < 1/2$, and choose any $0 < \beta < \theta/2$. For large n, $\log^\beta n \le \frac{\log^{\theta/2} n}{12}$, so the same lower bound applies to this stronger approximation guarantee. For the second assertion take $g = (12\alpha)^2$ and $\Delta_\alpha = \Delta_g$. This constant g is in the theorem's range once n is sufficiently large. The exponent δ can be chosen uniformly; the size threshold and degree bound may depend on α. □

Corollary 15.9 (The IKW trade-off)**.** Let N denote the number of variables in the source 2-CSP. There are absolute constants $c_1, c_2, C_2, a_1, a_2 > 0$ with the following property. Let $k = k(N)$ be integer-valued, with k sufficiently large and $k = o(\log N)$. For all sufficiently large N, the IKW construction yields a common graph size n satisfying

$$N^{a_1 k} \le n \le N^{a_2 k}$$

for which every $\exp\left(c_1\sqrt{k}\right)$-approximation algorithm for minimum unweighted directed 3-spanner has sensitivity at least $n^{c_2/k}$ on weakly connected DAGs of maximum total degree at most $2^{C_2 k}$.

Proof. Take the Label Cover family used in the proof of Theorem 6.1, constructed in Section 7. Its parameter N counts the original 2-CSP variables, and k is the direct-product parameter specified in the statement. Let s be its total number of vertices, m its number of edges, and D, Q its maximum degree and alphabet size. This family has positive degrees, so $s/2 \le m \le sD$. Write C_{IKW} for the source theorem's constant C in the value guarantee and η for its sensitivity exponent. With L denoting the source sensitivity lower bound and g the reciprocal value guarantee, its parameters satisfy

$$s = N^{\Theta(k)}, \quad D, Q \le 2^{O(k)}, \quad L \ge s^{\eta/k}, \quad g = \exp\left(C_{\mathrm{IKW}}\sqrt{k}\right).$$

Take $n = n_0$ in Proposition 15.6. Then $s \le n \le s2^{O(k)}$, $J_0 \le 2^{O(k)}$, and $\Delta_* \le 2^{O(k)}$. Choose $c_1 < C_{\mathrm{IKW}}/2$; for sufficiently large k, $144\exp\left(2c_1\sqrt{k}\right) \le g$. The resulting sensitivity is at least $s^{\eta/k}/2^{O(k)} - 1$. Since $k^2 = o(\log s)$, the denominator and additive constant can be absorbed, giving $n^{c_2/k}$ for some $c_2 > 0$. The source size bounds and $n \le s2^{O(k)}$ imply the stated bounds on n after changing a_1, a_2. This assertion concerns the sizes generated by the IKW construction. The bound for every sufficiently large graph size n is given in Theorem 15.7. □

Role of the degree bound. To derive locality lower bounds from sensitivity, we also bound the maximum degree. A single arc deletion can force a large increase in the optimum when degrees are large: take disjoint nonempty vertex sets A, B and two fresh vertices u, v. Include all arcs from A to B, together with $a \to u$, $u \to v$, and $v \to b$ for all $a \in A, b \in B$. With $u \to v$, the $|A| + |B| + 1$ arcs through u, v form a 3-spanner. After its deletion, all $|A||B|$ arcs in $A \times B$ are necessary. This elementary example has large degree. In contrast, Theorem 15.7 controls the degree uniformly, and Corollary 15.8 keeps it constant for each constant approximation factor. The next section uses precisely this additional control.

16 Distributed Consequences for Directed 3-Spanner

We convert the arc-set sensitivity bounds into distributed lower bounds. For every constant approximation factor, we obtain an $\Omega(\log n)$ lower bound on LOCAL rounds and non-signaling locality on DAGs of constant maximum total degree. The degree bound and size threshold may depend on the approximation factor. We also derive bounds for approximation factors that grow as a small power of $\log n$.

The communication network is the underlying undirected graph: vertices can communicate in both directions, independently of input-arc directions, as in the distributed spanner formulation of [6]. Each vertex initially knows its incident input arcs and their endpoints. Identifiers and global parameters such as n and a prescribed degree bound are fixed across compared inputs. Local computation and message size are unrestricted.

Each vertex may mark a subset of its incident input arcs, and the union of all marked arcs is the output spanner. This includes the convention that an arc is owned by its tail, as well as the convention that either endpoint may select it. All unmarked or absent arcs have output bit zero.

16.1 From Arc Sensitivity to Non-signaling Locality

At each vertex, the output is the vector of indicators specifying which incident arcs it selects. We apply the definition of non-signaling locality in Section 2 to the joint distribution of these vertex outputs. The following bound accounts for the fact that one affected vertex may change several output arcs.

Lemma 16.1 (Locality bound for arc-set outputs)**.** For a t-local non-signaling algorithm whose outputs are subsets of the input arcs, its single-arc sensitivity on graphs of maximum total degree at most $\Delta \geq 2$ is at most

$$4\Delta(\Delta + 1)^{t+1}. \tag{78}$$

Proof. Let $G, \widetilde{G}$ be graphs on a common vertex set V that differ by one arc with endpoint set X, $|X| = 2$. In the underlying undirected graph of $G \cup \widetilde{G}$, let W be the vertices at distance at most $t + 1$ from X. The union is one of the two inputs, so its degree is at most Δ, and $|W| \leq 2(\Delta + 1)^{t+1}$.

The radius-t input neighborhood of $V \setminus W$ is identical in G and $\widetilde{G}$. The extra layer in the definition of W ensures this equality even when the input labels at boundary vertices specify all their incident arcs. Non-signaling therefore gives identical joint output marginals on $V \setminus W$. Couple those vectors to agree, and extend the coupling to full outputs by conditional sampling. Every differing selected arc then has at least one endpoint in W. Each vertex in W is incident to at most 2Δ arcs present in either G or $\widetilde{G}$. The Hamming distance between the coupled arc sets is at most $2\Delta|W|$, proving (78). □

16.2 Locality and Round Lower Bounds

We combine the polynomial sensitivity lower bound of Theorem 15.7 with the sensitivity upper bound of Lemma 16.1. Taking logarithms gives the following lower bound on non-signaling locality and LOCAL rounds.

Theorem 16.2 (Parameterized distributed lower bounds)**.** Use a, C, θ, δ from Theorem 15.7. Let t denote the locality radius in the non-signaling model or the number of communication rounds in LOCAL. For every sufficiently large n and every $144 \le g \le \log^{\theta} n$, a $(\sqrt{g}/12)$-approximation for minimum unweighted directed 3-spanner on weakly connected n-vertex DAGs of maximum total degree at most $\Delta_g = \lceil \exp(Cg^a) \rceil$ requires

$$t = \Omega\left(\frac{\log n}{g^a}\right) \quad \text{in the non-signaling and LOCAL models.} \tag{79}$$

The implicit constant is absolute.

Proof. The sensitivity lower bound is $\Omega(n^{\delta})$, while $\log \Delta_g = O(g^a) = o(\log n)$ uniformly in the stated range. Taking logarithms in Lemma 16.1 gives

$$\delta \log n - O(1) \le \log(4\Delta_g) + (t+1)\log(\Delta_g + 1).$$

The additive $O(g^a)$ terms can be absorbed because $g^a = o(\log n)$. This proves (79). □

Corollary 16.3 (Polylogarithmic approximation factors)**.** There are constants $\beta \in (0,1)$, $\kappa \in (0, 1/2)$, and $\tau = 1 - \kappa \in (1/2, 1)$ such that every $\log^{\beta} n$-approximation for minimum unweighted directed 3-spanner requires non-signaling locality at least

$$\Omega(\log^{\tau} n)$$

and the same lower bound holds for the number of LOCAL rounds. The lower bound already holds on weakly connected DAGs of maximum total degree $\exp(O(\log^{\kappa} n))$, with no bound on local computation.

Proof. Set $g = \log^{\theta} n$ and $\kappa = a\theta < 1/2$ in Theorem 16.2, and choose $0 < \beta < \theta/2$ as in Corollary 15.8. Since $g^a = \log^{\kappa} n$, this gives the stated bounds. □

Corollary 16.4 (Every constant approximation factor)**.** For every constant $\alpha \ge 1$, there is a constant Δ_α such that α-approximation for minimum unweighted directed 3-spanner on weakly connected n-vertex DAGs of maximum total degree at most Δ_α requires non-signaling locality $\Omega_\alpha(\log n)$, and the same lower bound holds for the number of LOCAL rounds.

Proof. Use $g = (12\alpha)^2$ in Theorem 16.2. □

For the IKW family, take N, k, n and the constant c_1 from Corollary 15.9, with k sufficiently large and $k = o(\log N)$. At the graph sizes $n = N^{\Theta(k)}$ supplied by that construction, let t_{LOCAL} denote the number of LOCAL rounds and t_{NS} the non-signaling locality. The same conversion gives

$$t_{\mathrm{LOCAL}}, t_{\mathrm{NS}} = \Omega\left(\frac{\log n}{k^2}\right) \tag{80}$$

for $\exp\left(c_1\sqrt{k}\right)$-approximation algorithms. The maximum total degree bound Δ from Corollary 15.9 satisfies $\log \Delta = O(k)$. The prefactor 4Δ in (78) therefore contributes $O(k)$ to the logarithm of the sensitivity upper bound. The logarithm of the sensitivity lower bound is $\Omega(\log n/k)$, which dominates $O(k)$ because $k^2 = o(\log n)$.

Table 7: Selected approximation and locality bounds for minimum unweighted directed 3-spanner on n vertices. The three lower-bound rows use the DAG families and parameter ranges of Table 6, with $\tau = 1 - \kappa \in (1/2, 1)$. All three lower bounds also allow failure probability at most n^{-3}. The upper bound holds on general directed graphs with high probability, for $\varepsilon > 0$.

Approximation ratio	Maximum total degree	Locality bound	Reference
	Lower bounds (LOCAL and non-signaling)		
Any constant $\alpha \geq 1$	$O_\alpha(1)$	$\Omega_\alpha(\log n)$	Corollary 16.4
$\log^\beta n$	$\exp(O(\log^\kappa n))$	$\Omega(\log^\tau n)$	Corollary 16.3
$\exp\left(c_1\sqrt{k}\right)$	$2^{O(k)}$	$\Omega(\log n/k^2)$	(80)
	Upper bound (LOCAL)		
$1+\varepsilon$	Unrestricted	$\mathrm{poly}(\log n/\varepsilon)$	[6]

16.3 Algorithms Allowed to Fail with Small Probability

The preceding results use almost-sure feasibility and an expected-cost approximation guarantee. We now consider algorithms that, with high probability, output a directed 3-spanner of G with at most $\alpha\mathsf{opt}(G)$ arcs, where α is the required approximation factor. We show that the same locality and round lower bounds hold when the failure probability is at most n^{-3}.

Lemma 16.5 (Sensitivity with a small failure probability)**.** Suppose that, on every graph G in an n-vertex input class, a randomized algorithm A outputs a directed 3-spanner with at most $\alpha\mathsf{opt}(G)$ arcs with probability at least $1 - p$. There is an α-approximation algorithm B on the same input class such that, for every graph,

$$\mathsf{EMD}(A(G), B(G)) \leq p\, n(n-1).$$

Consequently, if all such B have sensitivity at least L, then A has sensitivity at least $L - 2pn(n-1)$.

Proof. Define B by retaining A's output when it satisfies both conditions and replacing it otherwise by a fixed optimal spanner of G. The algorithm B is used only in the analysis and need not be distributed. Under this coupling, $A(G)$ and $B(G)$ differ in at most $n(n-1)$ output coordinates on an event of probability at most p. The output $B(G)$ always has at most $\alpha\cdot\mathsf{opt}(G)$ arcs. Apply the triangle inequality to the neighboring pair witnessing B's sensitivity. □

Thus Theorem 16.2 and corollaries 16.3 and 16.4 and the IKW bound (80) remain valid for algorithms with failure probability at most n^{-3}: the sensitivity loss is at most $2/n$, which is negligible compared with both the DH bound $n^{\Omega(1)}$ and the IKW bound $n^{\Omega(1/k)}$. The locality and round conclusions use a worst-case round bound for LOCAL and the stated locality bound for non-signaling output families. The argument does not by itself extend the statements to constant failure probability or to expected termination time. Outputs may first be restricted locally to actual input arcs, so this convention causes no communication overhead.

Comparison with existing distributed results. Table 7 summarizes selected upper and lower bounds.

Censor-Hillel and Dory [6] obtain a $(1+\varepsilon)$-approximation in $\mathrm{poly}(\log n/\varepsilon)$ LOCAL rounds with high probability for constant stretch, allowing unrestricted local computation. Their directed unweighted CONGEST lower bound is stated for stretch $k \geq 5$. Dinitz and Nazari [13] give a randomized algorithm that,

with high probability, computes an $O(\sqrt{n}\log n)$-approximation for directed k-spanner in $O(k\log n)$ LOCAL rounds using polynomial local computation. For any constant approximation ratio $\alpha \geq 1$, our lower bound requires logarithmic locality even on DAGs whose maximum degree is bounded by a constant depending only on α. For approximation ratio $\log^{\beta} n$, our lower bound requires $\Omega(\log^{\tau} n)$ locality and thus rules out $(\log\log n)^{O(1)}$ locality. The CONGEST lower bound of Censor-Hillel and Dory for directed unweighted k-spanner with $k \geq 5$ and their LOCAL lower bound for weighted 2-spanner [6] do not establish these conclusions for unweighted directed 3-spanner. Our bounds hold with unrestricted message size and local computation and extend to non-signaling algorithms.

A direction for future work is to extend these sensitivity-based arguments to the HYBRID model [3], which augments LOCAL with limited global communication.

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Statement of AI Usage

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